

Inkstain Blog - Colorado River Analysis

Section: Cracks in the Upper Colorado River Basin wall

Source: <https://www.inkstain.net/2026/07/cracks-in-the-upper-colorado-river-basin-wall/>

People are finally starting to say the quiet parts, the parts so many Upper Basin people have been grumbling about in private, out loud. Via an incredibly important Heather Sackett story:

Colorado Sen. Dylan Roberts, a District 8 Democrat who represents several Western Slope counties, including Eagle, Grand, Garfield, Routt and Summit, asked Colorado's lead negotiator, Becky Mitchell, whether the people of Colorado should have confidence that negotiations among the seven states that share the Colorado River have put the state in the best possible position. The states have been at an impasse for more than two years without a deal for future management as reservoirs continue to decline to record-low levels.

"My constituents just see fighting and intransigence," Roberts said. "And it's concerning to me, especially as a Western Slope lawmaker ... that the strategy is just 'Let's hire more lawyers; we're going to court no matter what.' That doesn't give me confidence, because I don't think Colorado fares well when we go to court against Arizona and California and Nevada, throwing our fate to the nine justices on the U.S. Supreme Court."

Section: Some good Colorado River news, some bad news, and a request for help

Source: <https://www.inkstain.net/2026/06/some-good-colorado-river-news-some-bad-news-and-a-request-for-help/>

A grab bag from my friends and colleagues working on Colorado River issues....

The good news

From friend of Inkstain Karl Flessa (the guy who helped get me started thinking about the Colorado River Delta), a new analysis concluding that despite the terrible hydrology and political difficulties, environmental restoration work in the delta is working:

The Pulse Flow of 2014 demonstrated the feasibility of the restoration effort. Subsequent creation, irrigation and maintenance of 559 hectares (1,381 acres) of riparian vegetation attracted birds and other wildlife. Deliveries to the river channel raised water tables, supported existing vegetation and increased the length of the flowing river. Local communities have benefitted from recreational, educational and job opportunities.

The bad news

In 2026 and for only the third time in the 21st century, there was no accumulation, and no recovery, of total Basin live storage during the snowmelt season. Nor was there any accumulation or recovery of total live storage in Lake Powell and Lake Mead during the 2026 snowmelt season.

A request for help

And from friends of Inkstain Jason Robison, Matt McKinney, and Doug Kenney, a request for your input on a survey of folks attitudes toward the Colorado River Post-2026 management process.

Section: Colorado River Basin – new report from my colleagues on the implications of running on empty

Source: <https://www.inkstain.net/2026/06/colorado-river-basin-new-report-from-my-colleagues-on-the-implications-of-running-on-empty/>

I've been on a "Colorado River sabbatical" of late, but I took a peek last week at Reclamation's latest 24-month study. Holy moly things have gotten bad since the last time I looked!

Those not on sabbatical already know all of this, but to keep Lake Powell above a surface elevation of 3,500 feet, Reclamation is:

- increasing releases out of Flaming Gorge on the Wyoming-Utah border
- dropping releases out of Lake Powell to 6 million acre feet this year

Even with those two "hail Mary" moves, Lake Mead is projected in the "most probable" scenario to drop to elevation 1,020 by summer 2027. Under the "minimum probable" forecast, Mead drops all the way to elevation 1,008 in 2027.

We are on the brink, as a group of my colleagues explains in a new analysis out this morning (Monday June 1, 2026), of a system crash:

If the Colorado River Basin (Basin) experiences another dry year, similar to Water Year 2025, it is likely that reasonably accessible storage in Lake Powell and Lake Mead would be mostly depleted, even if consumptive uses and losses are at or near historic lows. Run-of-the-river operations would shortly ensue. This would be an outcome with devastating consequences.

That's from the latest report from the team of Castle-Schmidt-Kuhn-Sorensen-Tara, the Traveling Wilburys of the Colorado River. I've been on "sabbatical", so I didn't work on this one with my friends. (The joke is that I'm busy catching up on old movies, which is at least partly true, did you know Billy Wilder made, like, 50 movies?)

Even a wet year, my friends conclude, would only provide a short reprieve from the need to significantly reduce consumptive use.

Building on a similar analysis done last September (I was a co-author on that one), the authors attempt to overcome one of the shortcomings of the traditional Colorado River accounting systems, which is to treat any water above "dead pool" as usable storage. This is not the case, with clear do-not-cross lines in the reservoirs that are maintained for technical reasons well above the bottom, defined by my colleagues as...

"realistically accessible storage" (hereafter, "RAS") in the major reservoirs. RAS is the water available above protected elevations determined by Reclamation. These levels are 3500 feet in Lake Powell, and 975 feet in Lake Mead. "Active storage" is a term widely used in water resource engineering and refers to all the stored water that is above "dead pool" that could theoretically be released. This is the metric of storage reported by Reclamation in the 24-Month Study. Forecasts of active storage may not be fully illuminating, however, because Reclamation currently intends to protect higher levels in the reservoirs.

One of the reasons for my "sabbatical" is, frankly, an agonized frustration with the abject failure of Colorado River governance at the basin scale, and a desire to turn my attention to the local level, which is where the problem solving responsibility seems to rest right now. Each community needs to be having a serious conversation right now about the specifics of its Colorado River water supply, and how it intends to go about using less. Blaming other people for using too much isn't particularly useful at this point, we seem to have chosen to hand that set of questions (the rule-based part of "who is entitled to how much") over to the courts, and who knows what that process holds. We know the answer for everyone is "use less water", and each community needs to be getting on with that conversation.

Section: 2026-05-19: Federal managers increase release for the silvery minnow

Source: <https://www.inkstain.net/2026/05/2026-05-19-federal-managers-increase-release-for-the-silvery-minnow/>

Federal water managers yesterday (May 18, 2026) began pushing a pulse of water through New Mexico's rapidly drying Middle Rio Grande to try to encourage the endangered Rio Grande silvery minnow to spawn.

From a note sent 'round to the Bureau's water management list yesterday by Carolyn Donnelly, water operations supervisor for the bureau's Albuquerque Operations Office:

Due to the record low snowpack and well above normal temperatures in March, the Rio Grande did not have a typical spring flow increase. That type of spring flow increase is what triggers the Rio Grande silvery minnow to spawn, and without it, the silvery minnow spawn has been minimal this year. Because 2026 is the third poor hydrologic year in a row, silvery minnow numbers are concerningly low.

The U.S. Fish and Wildlife Service therefore asked the Bureau of Reclamation to use its leased San Juan – Chama water to create a brief higher flow through Albuquerque, hoping to trigger a more robust silvery minnow spawn. Crews will be out in the river this week collecting silvery minnow eggs to provide to hatcheries for later augmentation and broodstock.

Flows at Albuquerque's Central Avenue Bridge have been hovering around 10 cubic feet per second. The river's managers bumped the release out of Cochiti Dam at the head of the valley by 250 cfs yesterday. (The increase actually started upstream at Abiquiu Reservoir Sunday.) The water being released was imported from the Colorado River Basin via the San Juan-Chama diversions in the mountains of southern Colorado, then leased by the USBR from San Juan-Chama contractors not using their full allotment.

The inordinate mental cost of writing the preceding paragraph at 5:37 a.m. reminded me of the complexity. "The Bureau increased"? "The Army Corps of Engineers increase"? USFWS? SJC? And what happens as the water passes through the system? It's passing a bunch of Middle Rio Grande Conservancy District (a local agency) diversions, which will have to be managed to keep water in the main channel for the fish. Stunningly polycentric, this system.

Or maybe I just need to finish my second cup of coffee. The 2026 water year is testing a lot of our mental and emotional faculties, and the chemicals we use to manage them. It's a good thing weed is now legal in New Mexico.

Section: The Utility of Operationally Neutral and Flexible Conservation Pools in the Colorado River Basin

Source: <https://www.inkstain.net/2026/05/the-utility-of-operationally-neutral-and-flexible-conservation-pools-in-the-colorado-river-basin/>

Nota bene: A guest post from friends of Inkstain John Berggren and Kevin Wheeler

John Berggren (Regional Policy Manager, Western Resource Advocates)

Kevin Wheeler (Principle, Water Balance Consulting)

5/5/2026

As everyone is well aware, the snowpack and associated runoff this year are truly awful. It will be one of the worst, if not worst, on record. The Bureau of Reclamation is using unprecedented policy levers to keep Lake Powell from reaching levels so low that the safety of the dam may come into question. And it

is unclear if those policy levers will be enough, especially if the rest of the spring and summer are warm and dry. This got us thinking—are there other tools in the toolbox we are not considering?

While current guidelines may not contain much else, you only have to look as far as Reclamation's Post-2026 Draft Environmental Impact Statement to see there are options being proposed. We explore how one of those—flexible and operationally neutral conservation pools—can actually provide system protection without the political, legal, and hydrological risk that comes with currently available options. The current Intentionally Created Surplus program in the Lower Basin is an excellent framework to explore how conserved water can be creatively moved between reservoirs, wherever it is needed most.

No need for new infrastructure, complicated agreements, or shifting of risk—just smart reservoir accounting and operations. We hope this level of flexibility and creativity is being considered in new Post-2026 Guidelines and in the Basin State's negotiations.

Water Year 2026 Challenges

The Bureau of Reclamation's April 24-month study indicates that under the 'most probable' scenario, Lake Powell will fall below 3,500 ft by August of this year if no further action is taken. Reclamation has indicated, however, they intend to plan for the 'minimum probable' scenario which has the reservoir crossing this critical threshold in July. In response, Reclamation will be releasing additional water from the Upper Initial Units (UIU) under the 2019 Drought Response Operations Agreement (DROA) in the range of 660 kaf to 1 maf. Reclamation has indicated these releases would come from Flaming Gorge and will help keep Lake Powell from dropping to critical levels.

Concurrently, Reclamation has announced they intend to reduce Lake Powell releases from 7.48 maf to 6 maf during Water Year 2026 under Section 6E of the 2024 Record of Decision from the Supplemental Environmental Impact Statement to the 2007 Interim Guidelines (SEIS). Reducing Powell releases has implications for elevations in Lake Mead as well as significant Compact implications. In the absence of any operational adaptations, the 10-year cumulative flow at Lee Ferry will fall to 82.7 maf by the end of the current water year. If releases are reduced to 6 maf and 1.48 maf of water is withheld in Lake Powell between April and September 2026, as is currently planned, the 10-year cumulative flow will be approximately 81.2 maf (Figure 1). This would risk triggering legal action by the Lower Basin.

Figure 1. Lee Ferry Cumulative 10-year Flow. The blue line shows the estimated decline of the cumulative 10-year Lee Ferry flow before any adjustments and the orange line shows an accelerated decline following a 1.48 maf reduction to Lake Powell releases made under 6E SEIS authorities.

Despite this immediate legal risk, Reclamation's modeling demonstrates that both mechanisms will be required to protect Lake Powell this year, but just barely. Under 'minimum probable' hydrology, these actions would still result in Lake Powell dropping to 3,500 ft—dangerously close the hydropower intakes.

The Upper Basin has reluctantly accepted the proposed DROA releases which will significantly impact Flaming Gorge, but harbors significant concerns over whether water releases from the UIUs will be used to benefit the Lower Basin and how storage of the UIU reservoirs will be recovered. Similarly, the Lower Basin has issues with the reduction of Lake Powell releases to 6 maf, citing concerns over violations of the Compact and suggesting that more water should be released from the UIUs. Neither Basin is happy with Reclamation's response to the dire hydrology this water year.

How could Water Year 2026 have been different?

The challenges discussed above present an opportunity to examine exactly how a primary tool developed for several of the alternatives in the Post-2026 Draft Environmental Impact Statement (DEIS) — operationally neutral and flexible conservation pools — could provide another means to handle extremely dry years. While current guidelines do not allow for this flexibility, Water Year 2026 does provide a concrete example for how moving conservation pool water could mitigate low runoff years and protect critical infrastructure without many of the challenges discussed above. In this

hypothetical, we demonstrate how if conserved water currently stored in Lake Mead were to be treated as an operationally neutral conservation pool with flexibility in where it is located, Reclamation could more easily protect Lake Powell.

As of EOY 2024 (the most recent Reclamation Colorado River Accounting and Water User Report), there was 3.32 maf of conserved water, called Intentionally Created Surplus (ICS), being stored in Lake Mead. If this current conserved volume would be considered operationally neutral, as in some of the proposals in the DEIS alternatives, then Reclamation could move a portion of this conserved water up to Lake Powell. ICS water can be “moved” to Lake Powell by reducing releases from Glen Canyon Dam. Importantly, though, these reductions in Powell releases would not occur under the 2024 SEIS 6E authority and therefore would have no impact on Compact compliance.

In the current case, the proposed 1.48 maf reduction of releases from Lake Powell could be considered a transfer of ICS water from Lake Mead to Lake Powell. The reduction of releases would be added to the physical volume crossing the Lee Ferry Compact point, as though 7.48 maf were released from Powell, even though only 6 maf were physically released, so there would be no risk of a compact violation. That water would help to prevent Lake Powell from reaching critical elevation levels but would remain under the ownership of the Lower Basin contributors. Similarly, instead of needing to release 1 maf from Flaming Gorge, Reclamation could release only 500 kaf from the UIUs as part of DROA, and move an additional 500 kaf of ICS water up to Lake Powell by making a physical release of 5.5 maf for Water Year 2026 from Lake Powell (again, as though 7.48 maf were released). This would keep more water in Flaming Gorge which serves as a potential buffer if Water Year 2027 is similarly dry.

Figure 2. Options for Protecting the Elevation of Lake Powell. Under the minimum probable hydrologic condition from the April 24-month study, the orange line shows combinations of contributions from Upper Initial Units (UIUs) and Water Year 2026 releases from Lake Powell to maintain 3,500 ft elevation. The blue line shows the resulting minimum elevation of Lake Mead during Water Year 2026 on the right axis.

As shown in Figure 2, several options exist to maintain Lake Powell at 3,500 ft during the projected lowest month of March 2027. The options towards the left side of the graph indicate lower releases from UIUs and reduced releases from Lake Powell (i.e. increases in release reductions relative to the initially assumed release of 7.48 maf). The options on the right side of the graph rely on larger contributions from UIUs and greater releases from Lake Powell (i.e., smaller release reductions).

What if additional water is needed later in Water Year 2026?

Concerningly, the minimum probable hydrology is not the worst-case scenario and therefore additional flexibility may be required. The April to July runoff of Water Year 2026 is tracking very closely to Water Year 2002, which then resulted in the historically lowest inflow to Lake Powell. With the starting elevation today over 120 ft lower than it was in 2002, such a sharp decline would have far worse implications for Lake Powell. Using both levers to the extent currently planned may not be enough to keep Lake Powell at a safe elevation. Building upon the April 24-month study, if April to September inflow is equivalent to 2002, the “hole” in Lake Powell that Reclamation would need to fill might require an additional 530 kaf from currently projected volumes to maintain Lake Powell at 3,500 ft. To fill this hole, Reclamation may need to release additional water from the UIUs, but that presents significant physical risks because we cannot be certain what 2027 will be like and that would certainly face significant Upper Basin opposition. On the other hand, additional reductions from Lake Powell below 6.0 maf faces mounting legal risks, including exceeding Reclamation’s authorities under the SEIS and further exacerbating Compact concerns along with Lower Basin opposition. Having the flexibility to, again, pull from an operationally neutral conservation pool to respond to changing conditions and fill the inherently uncertain hole, provides a solution that avoids many of the legal, hydrologic, and political uncertainties.

What happens to Lake Mead?

The other benefit to having a flexible conservation pool is not just in the short term to move water up to protect Powell, but in the medium term as it provides the mechanism to move that water back down in future years. The water might have been needed in Water Year 2026 to protect Powell, but in Water Year 2027 it could be moved back down to Mead, depending on system conditions and priorities. Reducing Powell releases under Reclamation's 6E authority provides no similar mechanism to recover that water in Mead in future years.

What if Water Year 2027 is dry?

In the absence of institutional barriers, prudent extended drought management operations in a generalized multi-reservoir context suggest keeping as much water as possible higher in the system, bringing water down only when it is needed. This approach keeps more options open for unknown future conditions. Furthermore, it minimizes evaporative losses due to cooler temperatures at higher elevations. In the context of the current Colorado River crisis, this would equate to minimizing additional releases from the UIUs while maximizing the use of adaptations to releases from Lake Powell, but only to the extent possible without impairing the operations of Lake Mead. If this mechanism were possible, more water could be retained in the UIUs while less water is released from Lake Powell. This approach increases the overall system protection if Water Year 2027 were to be dry.

Section: The Bard

Source: <https://www.inkstain.net/2026/04/the-bard/>

Neil Innes's bard in Holy Grail is one of my favorite film characters.

Bravely Bold Sir Robin, rode forth from Camelot

He was not afraid to die, O' Brave Sir Robin!

He was not at all afraid to be killed in nasty ways

Brave, Brave, Brave, Brave Sir Robin

He was not the least bit scared to be mashed into a pulp,

Or to have his eyes gouged out, and his elbows broken!

To have his knee caps split, and his body burned away!

Innes, who wrote the song, goes on, but you get the point. Sir Robin was brave. It's a little over the top, but that's the point! To be fair, "over the top" is the point of the whole film, but in this particular case Innes's bard is playing a role – to not only document, but to support and encourage Sir Robin's bravery.

But then, you know, three-headed giant knight stuff happens. The bard keeps trying to sing hero stuff – the "ought" in moral philosophy's "is/ought" dichotomy – and Robin glares nervously and tries to shut him down.

And then Sir Robin – we'll let Innes tell it:

Bard: Brave Sir Robin Ran Away!

Sir Robin: "NO!"

Bard: Bravely Ran Away Away!

Sir Robin: "I didn't!"

Bard: When danger reared its ugly head, he bravely turned his tail and fled!

Sir Robin: "I never!"

Bard: Yes, brave Sir Robin turned about, And gallantly, he chickened out!

Whether Sir Robin should have stayed and fought the three-headed giant knight is beside the point. I never wanted to be Sir Robin. I wanted to be the bard. But the “is/ought” dichotomy is tough terrain. There is danger when the bard thinks they know what ought to be done.

Section: The Colorado River and the Tragedy of the Anti-Commons

Source: <https://www.inkstain.net/2026/03/the-colorado-river-and-the-tragedy-of-the-anti-commons/>

Some notes on the current state of the Colorado River...

I'm preparing for a panel discussion this evening in Albuquerque. I promised – three-finger promise, Scout's honor, which still means something to me – that I wouldn't use any swear words., either in the blog post or the panel discussion.

The state of the water

- Per the latest numbers from my colleague/collaborator/friend Jack Schmidt, Lake Powell currently holds 1.57 million acre feet of water above the protect-the-infrastructure no-go line of elevation 3,500.

- Storage at this point in the year is similar to 2022, when we began a hair-about-to-be-on-fire drill as Interior raced to figure out how to protect Glen Canyon Dam because of newly understood (or newly publicly understood) risks of dropping below minimum power pool and using the dam's outlook works. That constraint still holds.

- The forecast this year is a catastrophe compared to 2022: 1.75 million acre feet for the 2026 runoff season, compared to 3.8 maf in the 2022 runoff season.

- The result, according to the most probable forecast from Reclamation, is that absent some sort of action (see governance below) Powell will drop below 3,500 in September, and stay that way until the spring runoff in 2027.

- According to the min probable forecast, which is realistic given the looming heat-pocalypse, we hit 3,500 by July and stay there forever (by which I mean as far as the current 24-month forecast runs – as the late Jim Morison wrote, the future's uncertain and the end is always near).

The state of the governance

The state of the governance nests two separate by closely linked problems: near term actions and long term rules.

Near term actions

Protecting Glen Canyon Dam from that 3,500 no-go line requires coming up with a least 2 million acre feet of water over the next two years – to get us past that spring 2027 problem described above. There are two ways to do this. The first is to release a bunch of water from upstream, primarily Flaming Gorge Reservoir. How much? Dunno. The second is to cut releases from Glen Canyon Dam, reducing flows through the Grand Canyon and into Lake Mead. How much? Dunno, though we may find out soon.

The current rules, adopted in response to the challenges of 2022-23, allows releases from Glen Canyon Dam to drop this year to 6 million acre feet, which effectively gets 1.5 million of the needed 2 million feet from Lake Mead by reducing releases thereto. Another 500,000 in releases from upstream reservoir gets you 2 million acre feet, with room to do more if the hydrology gets even worse – which it might.

Longer term actions

The longer term stuff is where, as a student of governance, this gets really interesting for me. As a citizen of the basin, I am inclined to swear words at the dysfunction that has left us with no long term plan beyond the end of this year. But I Scout's honor promised, so shifting to the "student of governance" schtick gives me a view from nowhere way to approach this dispassionately, without the, y'know, words that would have made Mr. Vinatieri, my Scoutmaster, disappointed in me.

Others have chronicled the failure of the seven U.S. Colorado River Basin states to come to a consensus agreement on a set of river operating rules, we need not repeat that here, other than to note that what we have here is a classic case of what has been called the tragedy of the anticommons. This is a situation where many people or entities – in this case the states of the Colorado River Basin – each have the power to block a solution that might be to the benefit of the community as a whole. In this case, each of the seven states of the Colorado River Basin have blocking power over solutions that would prevent the reservoirs from crashing.

See above: the reservoirs are crashing and we have no plan to prevent it because any proposal that might prevent it has been blocked by one or more states that object.

The reason behind this is a set of rules written beginning in the 1920s governing the river – the Colorado River Compact and a series of ad hoc additions that followed – that attempted to lay out rules for managing the river but failed to include functional processes for modifying the rules when they proved inadequate to changing the situation. We're now stuck with a system under which each of the seven basin states has blocking power over any attempt to change the rules.

This violates one of the fundamental institutional design principles identified by the late Elinor Ostrom, who taught us so much about how we succeed or fail in overcoming the tragedy of the commons: "How will the rules ... be changed over time with changes in the performance of the resource system, the strategies of participants, and external opportunities and constraints?" We have to have rules about how we rewrite the rules. We lack that.

Despite this, we have succeeded in the past, in a series of rule-writing exercises that began in the late 1990s, by depending on principled actors at the state level recognizing that they needed to balance their need to protect their own community's water supplies against the need to solve problems at the scale of the basin as a whole.

My personal values on this question are both instrumental (things that I think are in the best interests of myself and my community) and deontological (things that I think are fundamental moral principles). The second first: I think we have ethical obligations to those upstream and downstream of us in shared river basins. This is, for me, fundamental. The second is instrumental – I think compromise is in the best interests of my community's water supply and therefore its future, because if we end up in litigation and the system crashes, we stand to lose a lot more than if we compromise, are willing to act on our obligations to our downstream neighbors by using less ourselves.

The last two years of increasingly hostile negotiations among the states make clear that behavior that recognizes those principles is gone, replaced by interpersonal bickering and a game of chicken driving the basin toward litigation (effectively hoping to manage the basin by convincing a judge of our preferred interpretation of ambiguous rules written a century ago) and reservoir collapse.

Thar be dragons.

Section: Quoting Jeff Kightlinger and Jim Lochhead

Source: <https://www.inkstain.net/2026/03/quoting-jeff-kightlinger-and-jim-lochhead/>

As the former CEOs of two of the largest water utilities using water from the Colorado River, we have been deeply engaged in interstate and federal negotiations on the river for over 30 years. Those negotiations were tough, but the basin states ultimately reached agreement, including reducing

California's use of water by 800,000 acre-feet and adopting the current set of operating rules for the federal reservoirs.

Although we fought hard to protect our individual interests, we understood that the public interest is not always served by "winner-take-all" negotiating tactics. Agreements benefiting the river as a whole also benefit every individual water user, and the environment. We recognized the responsibility of the states, tribal nations, water users and conservation organizations in the basin to determine our own future, and not to have it dictated by interstate litigation or federally mandated solutions.

The clock is running out. Unfortunately, the current negotiations remain stuck, characterized by brinksmanship and talking points.

8 Comments

-

Having watched this slow motion train wreck for most of the 30 years Jeff and Jim describe above, I concur that negotiations during this period (2007 Interim Guidelines, DCP) have been characterized by preservation of self interest above all. However, the river, and the hydrology that supports it, does not care about water rights established over a century ago. This is a watershed like any other watershed, and successful adjudications of watersheds are based on what the river actually delivers, not some ancient, inaccurate estimate. To that end, the Supply Driven Alternative in the DEIS is the only truly logical approach here. All others, it would seem to me, begin the negotiation with hydrological fiction and will lead us right back to where we are today.

-

It's really interesting to hear about the long history of these negotiations. Reducing California's use by that much highlights the scale of the challenge.

-

"In short, if the states fail to reach agreement, the federal government will be forced to impose an alternative that invites lawsuits and imposes economically and environmentally disastrous emergency operations. No one, in any part of the basin, will be spared from the consequences."

-

Just sent this to the Sentinel: In reply to "Time for states to reach consensus on the future of the Colorado River" Colorado comes to the meetings with 1) refusal to take mandatory cuts, 2) the lie that the Lower Basin states are still taking more than their share, and 3) some sort of red herring, the most recent of which is that Arizona farmers are selling alfalfa to Saudi Arabia. (That seriously annoys me, an Arizona resident. My bet is that Upper Basin farmers are selling alfalfa to Saudi Arabia as well. You can't control who buys your product. We are, essentially, sending Colorado River water to Saudi Arabia.) The Upper Basin states have consistently taken 4 maf (million acre-feet). That's why the Compact worked for 80 years and Lake Mead and Lake Powell were full in 2000. The Lower Basin States have reduced usage to 6 maf. Here are my questions: 1) What is the graph of usage by state since 2000? It's mainly California, Colorado and Arizona. 2) How accurate are the numbers in the Upper Basin for stream discharge, rating curves, withdrawals, and return flows? Acoustic Doppler instruments in use for 40 years approach 1%. 3) Why hasn't the Upper Basin lost its appropriative right by non-use? and 4) Is Denver building infrastructure to take more Colorado River water? The Law of the River is that the Upper Basin will deliver 75 maf/10 years to Lee Ferry. That's the law, ratified by all the states. It seems to me that they could reach an agreement based on current usage of the Lower Basin taking 6 maf and the Upper Basin taking 4 maf with a base of 4.5 maf and 3 maf, respectively, and equal proportional additional cuts, if necessary, based on the run of the river, considering that the water comes from the Upper Basin. Michael Carpenter, hydrologist, USGS retired

[home address and phone for ID to newspaper]worked on Glen Canyon Environmental Studies, Upper Colorado River Endangered Fish Recovery Program, and lower Colorado River

-

I have to reply to Michael Carpenter's comment above, as it is indicative of the messaging (brainwashing?) that Arizona has perpetrated on its citizens. Fallacy #1: Arizona and the Lower Basin (LB) states would like us to believe that they have "conserved" great quantities of water over the past several years. IN FACT, What they have (almost) achieved is bringing their uses into line with a sustainable level of consumption that accounts for the evaporative losses incurred in storing and delivering water to the LB States. The LB can yell until they're blue in the face about AZ vs CA and how much water is "allocated" to each state through that decision, but the simple math is that unless the LB states account for evaporative losses which are on the order of 1.3-1.5 Maf/yr., Lake Mead will continue to drop EVEN WITH the UB in full compliance with the compact. Any rational conversation about LB consumption must include an accounting for these losses. When these losses are accounted for, the LB states have just barely squeaked back into the black over the past few years with respect to their 7.5 Maf allotment under the compact. That 6.0 MAF of "use" that is cited above equates to ~7.5 Maf of actual consumption. And let's not forget, a large chunk of this "conservation" in the LB was paid for by ~\$3Billion dollars of taxpayer money. Corollary to Fallacy #1: The "Sweet Spot". Not only does Arizona want us to believe they are suffering through so-called "cuts", they have actually acted in ways to guarantee increased releases of water from Lake Powell and have hastened the critically low storage levels we have today (see <https://www.inkstain.net/2018/04/is-the-central-arizona-project-gaming-reservoir-levels-to-take-more-water-from-the-upper-basin/>). The ironic thing about the "Sweet Spot" slide is it vividly illustrates the fact that even with the Upper Basin delivering 8.23 MAF, the Lower Basin is over-consuming by not assigning evaporation to the states as part of their allocation. "Releases of 8.23 MAF will drop the lake level (Mead) more quickly and drive the system into shortage more quickly" – Central Arizona Project. The chickens are coming home to roost. Fallacy #2: The Upper Basin (UB) states "refuse" to take mandatory cuts. UB water users take mandatory cuts ALL THE TIME. Its called prior appropriation and in Colorado was ratified in the State Constitution in 1876. If you look at John Fleck's previous post that includes a graph of UB consumptive use, you see a lot of variability from year to year. that is not a change in DEMAND, it is a change in AVAILABILITY. those years of low consumptive use occur when there is insufficient water to meet all the demands on the system. Even during the years of higher consumptive use many water users in the UB states are still forced to curtail their usage. Water users in the UB are subject to shortages on a regular basis (oh, and they don't get any of that \$3B to mitigate their losses...). What about mandatory cuts in Federal projects you say? Well, Reclamation's Colorado Big Thompson project regularly allocates water to its users at less than a full quota. in fact, since the millennial drought began, there has been exactly one year with a 100% quota. Since 2000, the annual quota has hovered around 75%. We certainly don't see any news coming out of the Arizona spin machine acknowledging involuntary and uncompensated reductions and curtailments of water to users in the Upper Basin do we? Meanwhile, the LB did not declare an official shortage until 2021. And that shortage reduced the allocation by less than 10%. While it may seem counter-productive to revisit the past instead of looking for solutions, it IS important to understand how "we" got into this situation, and look at the facts instead of political spin. As for an equitable solution? perhaps we should go back to the foundational ideas of the compact itself. The compact negotiators arrived at a 50/50 split of the waters in the basin (with a bit "extra" for the LB – read Science be Damned <https://www.indiebound.org/book/9780816540051?aff=jfleck>). Seems to me that a reasonable starting point given the original intent would be a 50/50 split of what is actually available now.

-

"How Ancestral Puebloans Thrived After The 536-541-AD Climate Catastrophe"
<https://www.ancient-origins.net/news-history-archaeology/climate-catastrophe-0016105>

Full report: <https://doi.org/10.15184/aqy.2021.19>

-

Here is a graph of discharge at Lee's Ferry since 2000 with 10-year average in red. Do the Upper Basin states owe any additional for Mexico's 1.5 maf than the 75 maf/10 years they owe Lee Ferry? As I see it, the Upper Basin has done its job. I guess by Lake Powell being depleted. But if the Upper Basin has delivered, why is Lake Mead depleted? IS THERE A PLOT OF TOTAL DISCHARGE OF THE COLORADO RIVER AND EACH STATE'S WITHDRAWAL SINCE 2000. Well, the graph won't post but it shows a minimum of more than 8 maf, 10-year average at Lee's Ferry gage. In reply to John Carron. I believe my statements are correct, Lower Basin use is down to 6 maf and Upper Basin use has been consistent at 4 maf. I agree with him that arguments coming from Arizona don't deal with Arizona taking a lower right than other Lower Basin states in order to get the CAP. I did not make those arguments. My main question deals with the fact, as I see it, that things do not add up. Even with evaporative loss from Lake Powell, the Upper Basin delivered. The reservoirs were full in 2000. Is there a graph of total discharge in the Upper Basin, delivery to Lee Ferry (Lee's Ferry is good enough), evaporative losses from Powell and Mead, depletion of Powell and Mead, and usage by each Lower Basin state.

-

One of the ugly truths of water management is that measuring water diversions is frequently not accurate at all. My guess is that diversions could be way more than the reported use in many places. If you stand at the Imperial Dam watching most of what is left of the river disappear into the All American Canal the sense you get is not "wow, that looks like a really accurate flow measurement system". From there, the use of water is sometimes measured based on acreage of crops rather than actual metering – slide gate positions are considered meters in some places. The error bars in measurement are rather large. The phrase "what gets measured gets managed" comes to mind and IMHO the measurement systems stink. Heck, there are still a lot of quantified water rights in California that are defined in miners inches!

Section: In which my colleagues and I share thoughts on the future of Colorado River governance

Source: <https://www.inkstain.net/2026/02/in-which-my-colleagues-and-i-share-thoughts-on-the-future-of-colorado-river-governance/>

It is hard to know where to begin. The Department of the Interior's Post-2026 Colorado River draft environmental impact statement, and the deep questions it raises, is an "everything including the kitchen sink" sort of process.

But at its root, the question it raises is simple: Tell us what you're going to do.

It is easiest to quote the Draft EIS itself on the central question: "In critically dry periods, all alternatives have unacceptable performance." Roger that. Tell us what you're going to do.

In the short run, with a meager snowpack and no clear explanation of how federal and state managers are going to operate the reservoir system, the basin's dams and diversions now, we have no clear picture of what will happen in 2026. Tell us what you're going to do.

Some specifics

The ad-hoc collective that Allen Best dubbed "the Traveling Wilburys of the Colorado River" – Anne Castle, Eric Kuhn, Jack Schmidt, Kathryn Sorensen, Rin Tara, and me – took yet another stab at offering our suggestions in comments we submitted yesterday to Interior's P26 EIS process.

I've been mentally largely elsewhere lately, finishing up the book and managing my health (I'm doing great! Thanks for asking!), so I mostly show up at the last minute on these things to dub my vocals, but my friends are very kind and inclusive, and they do good work. In particular, stuff like this full of both useful NEPA-speak and also substance:

The analyzed alternatives do not meet the purpose and need of the proposed federal action. "In critically dry periods, all alternatives have unacceptable performance." Therefore, the alternatives analyzed do not meet the purpose and need for the federal action as described in the DEIS. It is incumbent on Reclamation and Interior to provide at least one alternative that meets the purpose and need.

As I said: Tell us what you're going to do.

The Final EIS and Record of Decision must specify how short-term operations will address the current vulnerability of the system and its critical infrastructure. The new operating regime for the Colorado River will be implemented at a time of unusually low storage volumes and unprecedented low flows. The vulnerability of the system in providing an adequate water supply is unprecedented. Regardless of the duration of the next set of guidelines, it is imperative that Reclamation provide a clear picture of what actions will be implemented in the near term (i.e., next year, next 3 years, next 5 years) to protect critical infrastructure, and to protect public health and safety.

As I said: Tell us what you're going to do.

The preferred alternative should be based on the broadest possible interpretation of federal authority to ensure a sustainable water supply provided by the river. The DEIS states that, "The Basic Coordination Alternative is designed to be implementable without agreements among Basin water users regarding distributions of lower Colorado River mainstream shortages, storage and delivery of conserved water from system reservoirs, or other voluntary agreements...." Because the Basic Coordination alternative does not meet key performance objectives for a large percentage of the Dry and Critically Dry futures, it is incumbent on the Agency to explore and propose using the outer limits of their authority to achieve a system that provides a sustainable water supply. Reclamation and Interior should not default to a "safe" legal approach when those operations are insufficient to balance supply and demand and provide a prudent operating regime. Lawsuits over the operating guidelines adopted by the Department of the Interior are inevitable, regardless of which alternative is adopted, even a theoretically "safe" one. Therefore, the preferred alternative should utilize the broadest possible interpretation of Reclamation's and Interior's authority to provide a predictable and resilient Colorado River so that the system can continue to operate in a reasonable manner while the lawsuits proceed.

This one's on you, Secretary Burgum. We do offer suggestions, but if you don't like ours, then tell us your alternative: Tell us what you're going to do.

This one has seemed for a long time like a no-brainer to me:

The Upper Basin demand scenarios utilized in the modeling are unrealistically high.... [T]he Upper Basin demand scenario utilized in modeling the impacts of the various alternatives is much higher than historical use and does not reflect likely future actual use. The 2016 UCRC demand scenario embedded in the DEIS modeling is approximately 40% to 50% higher than recent average demand. Public statements by representatives of the Upper Division states emphasize the limitations imposed on Upper Basin water users by annual hydrology, but the UCRC demand scenario fails to take those hydrologic limits into account.

This one's on my Upper Basin leaders: Really?

Reclamation should specifically identify what additional authority is needed to implement an operating regime that meets the purpose and need for the EIS over the long term. There are no alternatives identified in the DEIS that meet the purpose and need and that Reclamation can implement without additional legal authority. This creates significant uncertainty around river management, despite the intended purpose of providing water users a greater degree of predictability about water availability in the future. Reclamation should describe specifically what additional legal authorities it believes would be required to implement desired operating provisions. Furthermore, Reclamation has indicated the intention to "seek additional authorities" to protect critical reservoir infrastructure. These additional authorities should be explained.

Again, Secretary Burgum: Tell us what you're going to do.

The colossal failure of the Colorado River Basin leadership is on display in this simple sentence from the draft EIS: "In critically dry periods, all alternatives have unacceptable performance."

You've spent the last few years telling us what you can't do. It's on you now: Tell us what you're going to do.

(Lots more in the full comments, it's a useful primer on the current state of play.)

Section: Colorado River news isn't all bad!

Source: <https://www.inkstain.net/2026/02/colorado-river-news-isnt-all-bad/>

I've been pretty successfully checked out of Colorado River work while I put the finishing touches on the new book (pre-order now!) but my colleagues are on it with a new post looking at the over-winter storage at the big reservoirs behind Hoover and Glen Canyon dams:

Key Points

- The rules that control releases from Lake Powell and Lake Mead are very different. Lake Powell's releases are determined by an Annual Operating Plan that has little flexibility during the year. Lake Mead's releases change each month in response to changing delivery requirements to Lower Basin users. The impact of these different release rules on each reservoir's storage was illustrated this autumn and early winter when Lake Powell steadily declined and Lake Mead steadily increased. The magnitude of Powell's decline and Mead's increase compensated for one another, and the total combined storage in Lake Powell and Lake Mead did not change.
- During the four months between October 1 and February 1, Lake Mead's releases were reduced in response to decreasing Lower Basin demands, but Lake Powell's releases were not similarly reduced. Lake Powell lost 615,000 af during the four-month study period, and Lake Mead gained the same amount.
- On February 1, Lake Mead had 2,714,000 af more water than Lake Powell, the largest difference between the two reservoirs since April 2022.
- Modest flood inflows in early October delayed drawdown of Lake Powell by six weeks. Releases during the four-month study period were the second smallest since at least 2010^[1]. Releases from Lake Mead were the smallest since at least 2010. Despite the small inflows to Lake Mead, the increase in storage in Lake Mead during the study period was the largest since 2019.
- The four-month delay in depletion of the combined storage in Lake Powell and Lake Mead saved between 400,000 to 900,000 af.
- Forecasts for spring snowmelt inflow to Lake Powell are not encouraging and have been declining all winter, because Rocky Mountain snowpack remains meager.

[1]We compared the inflows, outflows, changes in storage, and Lower Basin consumptive uses between 2010 and 2026.

This feels like good news, or least not bad news, except for that last bullet. The Upper Basin median forecast is just 40 percent of the median.

Section: Colorado River Assigned Water: Quoting Sorensen et al

Source: <https://www.inkstain.net/2026/02/colorado-river-assigned-water-quoting-sorensen-et-al/>

For more than a century of development, Colorado River governance has lived under a tension between individual communities' desires to use more water and the collective need to balance basin-scale supply and use for the benefit of the region as a whole. Incentives favoring individual communities at the expense of the collective good have brought us to the edge of the current crisis.

The latest project from our little Colorado River working group takes a deep dive into the arcane rules governing what Arizona State University's Kathryn Sorensen has dubbed "assigned water" – water that results of conservation efforts, set aside in the Colorado River Basin's reservoir storage accounts for the future use of whoever did the conserving. Sort of.

While most of the attention right now is focused on the top line conflict among the basin states, there remains a need to get the down-in-the-weeds details right.

Originally developed as a tool called "Intentionally Created Surplus" in the 2007 Colorado River operating guidelines (who names these things?), Assigned Water has become a critical tool for managing water use reductions in a way that helps ensure reliability for the water users to whom the tool is available. It's extremely valuable, but is not without its drawbacks – crowding out conservation efforts that might benefit the basin as a whole, inadvertently shifting system water that again might benefit the basin as a whole into an individual water agency's savings account, and delaying shortage declarations that might have enforced deeper cuts, sooner.

Section: A truly awful Colorado River snowpack so far in 2026

Source: <https://www.inkstain.net/2026/02/a-truly-awful-colorado-river-snowpack-so-far-in-2026/>

Via Eric Kuhn, the snowpack above Lake Powell is miserable. 2002 was the lowest year on record for unregulated April-July inflow into Lake Powell. This year is starting off even worse.

4 Comments

-

Snowpack and runoff are two different things

24 years ago, there were way more fires and way less trees

How many months this year will have zero natural flow into Lake Powell?

-

Thank you for these posts. Could you provide a link to the source for this graph?

-

2002 snowpack and runoff were terrible on the Rio Grande and San Juan. Think the San Juan Chama Project diverted about 6,000 acre-feet into the Rio Grande that year (normal would be between 80,000 and 120,000 acre-feet. Heck, for Colorado on the mainstem Rio Grande, we had to negotiate a new delivery by Colorado to New Mexico because the actual inflow that year was lower than any contemplated in the Rio Grande Compact. Only thing that allowed us to manage thru the year was releases of water stored in previous years. If this dryness continues, with no storage to speak of, 2026 will be a very difficult year of water management/flows and, unfortunately, possibly fires.

Section: Quoting John Entsminger

Source: <https://www.inkstain.net/2025/12/quoting-john-entsminger/>

If you distill down what my six partners just said, I believe there's three common things: Here's all the great things my state has done. Here's how hard/impossible it is to do any more. And here are all the reasons why other people should have to do more.

As long as we keep polishing those arguments and repeating them to each other, we are going nowhere.

10 Comments

-

Well... He's right! I thought your state representative from New Mexico really said it better which was to avoid litigation at all cost... so it appears we're gonna let the Federal Government hand down from interior the rules of the river? What I thought also was insightful was the presentation given regarding local governorships Senate races and representative races who in the 2026 upcoming election in relation to the timeline for approving new operating rules. Given how polarized politics is these days and with Arizona and Nevada as swing states let's hope the Federal Government will not battle in water politics as we really need a strong, fair hand to understand the complexities of the Colorado river and the intricacies of the politics involved.

-

Like his idea of a 5 year plan. Also, Rhett's analogy of bankruptcy. Look for more next week

-

Who knows! If Trump gets briefed on the Compact and how the water is distributed- Or Not- he might find a way to put some common sense into the whole thing. Especially if he sees how much water his buddy Gavin gets annually.

From Dick Allison, the Old Canadian Hydrologist!

-

Good man John Entsminger.

-

Lower Basin States have cut to about 6 MAF, so recent comments by Upper Basin States that the lower basin is taking more than its share are way, way behind the times or just a bald-faced lie. Can we see a BOR plot of total flow, flow delivered to Lee Ferry, and flow delivered to Mexico? If Upper Basin is using 4 MAF and Lower Basin is using 6 MAF, Lower Basin is going to have to cut more, but Upper Basin is going to have to cut some. Or, make them deliver 75 MAF/10 years. Is Denver Water Board building infrastructure to take more?

-

Water, water everywhere. Nor any drop to drink. Tucson receives more rain in an average year, than we traditionally received from CAP. And we can harvest about 60% of the amount we consume every year, at a fraction of the cost of any of the augmented supplies that are being considered. TUCSON doesn't need CAP. GROWTH needs CAP. And we are being forced to pay for it. Example: Tucson Water provides permanent access to our shared supply, for a one-time charge of \$1500/unit. New development isn't required to bring its own water, so we now have 1/2 as much water per person as 40 years ago, because we doubled population but didn't require it bring more water, as we add people. If we get super efficient, 1M people can prosper on our rain. We don't need outside water sources. All the debate over water, is really a debate over how current population will subsidize the cost of bringing more water to allow more people to move here.

-

Empty faucets are always an impetus to water agreements. Note: “distill down”, a bit of redundancy.

-

/ “distill down” is a correct and commonly used phrase in written English.

It means to condense or simplify information or ideas into their most essential and important elements. It can be used in a variety of contexts, such as summarizing a complex topic, getting to the core of a problem, or simplifying instructions. Example: “In order to effectively communicate the key points of the presentation, we need to distill down the information and focus on the most important aspects.” Thanks for the language lesson schoolmarm Peter.

-

I have a few shot glasses left over from CRWUA if anyone is interested in other types of distillery

Section: Colorado River Deadlines & Incentives

Source: <https://www.inkstain.net/2025/12/colorado-river-deadlines-incentives/>

A guest post by Friend of Inkstain Michael Cohen....

By Michael Cohen, Senior Fellow, Pacific Institute

December 15, 2025

Key Takeaways

- The consensus-based effort to develop new rules to manage the Colorado River system hasn't worked – it's time for a new approach
- Federal leadership and the credible threat of managing reservoirs to protect the system is that new approach

Missing Deadlines

Way back at the end of the last century, at the annual Colorado River conference in Vegas, Marc Reisner repeated the Margaret Thatcher quote that consensus is the absence of leadership. On Veterans Day, the seven Colorado River basin states missed yet another deadline to reach consensus on a conceptual plan for managing the shrinking Colorado River after the current rules expire in 2026. Valentine's Day marks the next holiday deadline, this time for a detailed plan, but multiple missed deadlines give no indication that the states will reach consensus then, either.

The basin states can't agree on the substance of a new agreement. They also disagree on the process to get there. While Arizona has called for the federal government to break the negotiation logjam, Colorado opposes federal intervention and continues to call for consensus. Each basin-state negotiator acts to protect their state's interests, often at the expense of the short and long-term resilience of the Colorado River system as a whole and the 35 million people who rely on it. The continued failure to negotiate a plan challenges the efforts of irrigators, cities, businesses, and river runners throughout the basin to plan for 2027 and beyond.

Meanwhile, river runoff and reservoir storage get lower and lower and snowpack lags well below average. This is not a zero-sum game, with winners and losers. The more appropriate metaphor here is a shrinking pie, with smaller and smaller pieces.

Leadership

The basin state negotiators have met for years behind closed doors, without success. It's time for a new approach. Aggressive federal intervention and the credible threat of a federally-imposed Colorado River management plan would offer political cover – or a political imperative – for the negotiators. The

credible threat of a federal plan would give the negotiators the space to compromise without having to do so unilaterally and then being accused of not protecting their state's interests.

But federal leadership alone is not enough – it must be coupled with a plausible federal plan that compels the states to act and can meet the magnitude of the ongoing crisis. As the Department of the Interior announced in its 6/15/2023 press release, the purpose of and need for the post-2026 guidelines is “to develop future operating guidelines and strategies to protect the stability and sustainability of the Colorado River.” To date, the development of the post-2026 guidelines has prioritized routine operations of Glen Canyon and Hoover dams over the system as a whole, a focus inconsistent with the magnitude and urgency of the problem. Prioritizing routine dam operations and hydropower generation over water delivery and environmental protection elevates the tool over the task. Seeking to preserve routine operations of the dams while imposing draconian cuts on water users is not a path to resilience and precludes alternatives that would help stabilize the system.

The Plan

Instead, by early next year, the Secretary should announce that Interior will implement a federal plan incorporating the following elements:

- Grant Tribal Nations the legal certainty and the ability to access, develop, or lease their water.
- Make accessible (“recover”) the roughly 5.6 million acre-feet (MAF) of water stored in Lake Powell below the minimum power pool elevation and avoid the additional ~0.25 MAF of annual evaporative losses from Powell by storing such water in Lake Mead and using Powell as auxiliary storage.
- As a condition precedent, the Lower Basin states agree not to place a “compact call” for the duration of the agreement.
- Implement annual Lower Basin water use reductions for the following calendar year based on total system contents on August 1:- 75% – 60%: cuts to Lower Basin water uses increasing from 0 to 1.5 MAF<60% – 38%: static cut to Lower Basin water uses of 1.5 MAF<38% – 23%: increasing cuts to Lower Basin water uses of up to 3.0 MAF total
- below 23% of total system contents – cut Lower Basin water uses to the minimum required to protect human health and safety and satisfy present perfected rights
- If the Lower Basin states do not satisfy the condition precedent in #3 above, Reclamation limits Lower Basin deliveries to the minimum required to satisfy present perfected rights when total system contents are <75%.
- Recover water stored in federal Upper Basin reservoirs unless the Upper Basin states reduce annual water use based on total system contents:- <34% – 23%: Assuming the first 0.25 MAF “reduction” would be contributed by the elimination of Powell’s evaporative losses and gains from Glen Canyon bank storage, reduce Upper Basin water uses up to 0.65 MAF
- below 23% of total system contents – limit total Upper Basin water uses to 3.56 MAF (the minimum volume reported this century)
- Expand the pool of parties eligible to create Intentionally Created Surplus (ICS) beyond existing Colorado River contractors, to include water agencies and other entities with agreements to use Colorado River water.
- Eliminate the existing limits on the total quantity of Extraordinary Conservation ICS and DCP ICS that may be accumulated in ICS and DCP ICS accounts, while maintaining existing limits on delivery of such water.
- Fully mitigate the on-stream and off-stream community and environmental impacts of the water use reductions identified above.

- After a three-year phase-in period, condition Colorado River diversions on a clear “reasonable and beneficial use” standard predicated on existing best practices for water efficiency, including but not limited to the examples listed below (state(s) that already have such standards):
- Require removal of non-functional turf grass (California, Nevada)
- Incentivize landscape conversion and turf removal statewide (California, Colorado, Utah)
- Adopt stronger efficiency standards for plumbing and equipment (Colorado, California, and Nevada)
- Require urban utilities to report distribution system leakage, and to meet standards for reducing water losses (California)
- Require all new urban landscapes to be water-efficient (California)
- Require metering of landscape irrigation turnouts (Utah)
- Ensure that existing buildings are water-efficient when they are sold or leased (Los Angeles, San Diego)
- Require agricultural water deliveries to be metered and priced at least in part by volume (California)

Many of the elements listed above raise important questions about federal authorities, accounting and data challenges, the roles and obligations of state water officials to implement coordinated actions in-state, water access for disadvantaged communities, environmental compliance, and potential economic and social costs, among others. For each item listed, many details will need to be refined. Similarly, the plan’s duration will need to be determined. But as temperatures again climb into the high 40s in the Rockies near the Colorado River’s headwaters (in mid-December!), drying soils and reducing next year’s runoff, and the National Weather Service issues red flag fire warnings for Colorado’s Front Range, the need for bold action is clear.

The Dominy Bypass

Recovering water stored in Lake Powell will require the construction of new bypass tunnels around Glen Canyon Dam. Former Reclamation Commissioner Floyd Dominy sketched the design of such tunnels almost thirty years ago (see image). Such tunnels would enable the recovery of about 5.6 MAF of water stored below the minimum power pool elevation – more water than the Upper Basin states consume each year. Current operating rules and the scope of the current planning process effectively treat this massive volume of water as “dead storage” – a luxury the system can no longer afford. After Reclamation constructs the bypass tunnels, water recovery should be timed to maximize environmental and recreational benefits in the Grand Canyon.

Avoiding a Worse Outcome

Last year’s Colorado River conference featured a panel on the risks of litigation. Unfortunately, the continued failure to reach a deal, growing litigation funds, and the preference for repeating the same action that’s led to the continuing impasse suggest that some believe litigation could generate a better outcome (for them). Both sides have attorneys who assure their clients of victory. Yet, as Arizona learned in 1968, winning in the Supreme Court doesn’t ensure a better outcome and certainly won’t increase Colorado River flows. Placing faith in Congress could entangle this basin with challenges in other basins and other political considerations.

Running the River

Almost 160 years ago, John Wesley Powell – the reservoir’s namesake – demonstrated bold leadership, going where no (white) man had gone before. With leadership and a clear goal, he charted a route through the Colorado River’s iconic canyons. Now is the time for more bold leadership, a clear goal, and a plan to get there.

Section: Quoting Katie Hobbs on Colorado River Negotiations

Source: <https://www.inkstain.net/2025/12/quoting-katie-hobbs-on-colorado-river-negotiations/>

I don't envy Secretary Burgum the position that he's in, but we do need federal leadership. Secretary Burgum already reached out to convene a meeting. He invited the governors and the chief negotiators to DC earlier [last] week. That meeting didn't happen because three of the four Upper Basin governors weren't willing to go. [Nevada Gov. Joe] Lombardo reached out again ... So I'm hopeful that will happen. And I think it just is an indication that the federal government does recognize the need for them to come in and help negotiate a solution.

– Arizona Gov. Katie Hobbs, in conversation with Politico's Annie Snyder (prolly behind a paywall)

One Comment

-

The Upper Basin owes the lower Basin 75 MAF/10 years. The Compact is law whether there is drought/global warming or not. If the Upper Basin takes only 4 MAF and the Lower Basin is down to 6 MAF, the Upper Basin wouldn't have to have such a drastic reduction as the Lower Basin.

Section: Water is For Fighting Over, ten years on

Source: <https://www.inkstain.net/2025/12/water-is-for-fighting-over-ten-years-on/>

Ten years after I handed in the manuscript for my book *Water is For Fighting Over*, about the governance past, present, and future of the Colorado River, my Utton Center colleague Rin Tara and I took a look back at the book on our *Water Matters* podcast. Writing it changed my life, reading it changed Rin's, and we now spend a good chunk of our time in the office we share working together on Colorado River issues.

What of my optimism (because it was nothing if not an optimistic book) has held up? What do the current failures in Colorado River management tell us about the function of the basin's governance network? Does my deeper optimism about communities' abilities to thrive with less water remain? What can a new generation of water leaders (looking at you, Rin) bring to bear?

Special thanks to Rin for what was the most fun interview I can remember doing.

You can find it on Buzzsprout or, as they say, "wherever you get your podcasts."

Section: Equity in the Colorado River Basin

Source: <https://www.inkstain.net/2025/12/equity-in-the-colorado-river-basin/>

Bonnie Colby and Zoey Reed-Spitzer:

Access to water, a concern around the globe, is constrained in many areas of the Colorado River Basin (CRB) by water scarcity, pollution and lack of delivery and treatment infrastructure. These constraints disproportionately affect the basin's Native American, Hispanic and Black populations, groups which often have lower access to resources for mitigation and recovery. Moreover, these groups historically have not been represented in policy processes where key decision affecting water access and water quality are made. A number of tribal governments, in recent decades, have been at the negotiation tables due to their status as sovereign governments with significant water rights. However, there is no consistent process for inclusion of low-income communities of color in the basin. The urgent high-level dialogue among tribal governments, federal interests, states, cities and agricultural water users (and, to

some extent, environmental NGOs) focuses on allocating the ever-scarcer waters of the CRB, with little emphasis on disproportionate access.

From the excellent new report *Dancing With Deadpool*, a broad effort by a broad group of contributors to the Colorado River Research Group.

Section: Autumn Rains Delay Colorado River Basin-wide Reservoir Depletion

Source: <https://www.inkstain.net/2025/12/autumn-rains-delay-colorado-river-basin-wide-reservoir-depletion/>

By Jack Schmidt | December 3, 2025

Autumn Rains Delay Basin-wide Reservoir Depletion

In Brief

Unusually wet conditions in the Basin in October and November 2025, combined with reduced releases from some reservoirs, led to a basin-wide increase in storage for the two-month period. The combined contents of Lake Powell and Lake Mead increased during the two months for only the second time since 2010, and storage in the San Juan River basin increased by 19%, especially in Vallecito and Navajo Reservoirs. These changes were a welcome respite from the relentless depletion of storage that has dominated the last few years. Nevertheless, the upcoming winter snow season is predicted to be below average, and total active storage in the Basin is less than a 2 year supply when compared with recent Basin-wide consumptive uses and losses.

The Details

The rains of October and November 2025 slowed depletion of the Colorado River's reservoirs due to increases in stream flow and reduced reservoir releases in some places. Water levels rose in a few reservoirs, and autumn's rains provided a small bit of flexibility for water managers at the beginning of what is likely to be a below-average winter snow season.

As of November 30, the Basin's 46 reservoirs held 24.63 million af (acre feet) of active storage[1], of which 90% was in 12 federal reservoirs,[2] including 15.00 million af in Lake Powell and Lake Mead (hereafter, Powell+Mead) and 4.88 million af in 8 federal reservoirs upstream from Lake Powell (Fig.1). This amount of storage is similar to conditions in early 2022, a situation that was described at that time as a crisis. If we divide the total active storage in the Basin's 46 reservoirs by the basin-wide total annual rate of consumptive use and loss that was 12.7 million af in 2024, the basin-wide reservoir water supply would sustain Basin-wide use for less than 2 years. We continue to live at the doorstep of crisis.

Figure 1. Graph showing active storage in Colorado River basin reservoirs between January 1, 2021, and November 30, 2025.

Basin-wide reservoir storage stabilized in October and November, because Powell+Mead storage stabilized and storage in the San Juan River basin increased. Total Inflow to Lake Powell exceeded releases for more than one week between October 11 and October 18, when Lake Powell increased by 105,000 af[3] which is a 1.6% gain (Fig. 2). Approximately 40% of the total inflow came from the San Juan River, and the monthly October inflows were the largest since 2015. The gain in storage in Lake Powell during this weeklong period exceeded depletions during the rest of the month, and Lake Powell gained approximately 52,000 af during the month. Lake Powell lost 147,000 af in November.

Figure 2. Graph showing inflow and outflow from Lake Powell and active storage between October 1 and November 30, 2025. Total monthly flow at Lees Ferry, representing the total releases from Lake Powell, were 490,000 af in October and 501,000 af in November.

In contrast, the autumn rains did not significantly increase inflow to Lake Mead, because most of the inflows come from scheduled releases from Lake Powell. These reservoir releases were supplemented by 101,000 af of inflows downstream from Lees Ferry[4] and 8000 af from the Virgin River.[5] The most significant changes in Lake Mead occurred at the end of November when releases from Hoover Dam were significantly reduced (Fig. 3).

Figure 3. Graph showing inflow and outflow from Lake Mead and active storage between October 1 and November 30, 2025. Total monthly flow inflow of the Colorado River, representing the total releases from Lake Powell and inflows within Grand Canyon, were 574,000 af in October and 550,000 af in November. Reservoir releases from Hoover Dam were 485,000 af in October and 415,000 af in November. Withdrawals and return flows of the Southern Nevada Water Authority were not included in these data.

Together, total active storage in Powell+Mead increased by 63,000 af during October,[6] and decreased by only 38,000 af in November (Fig. 4).[7] More significant than the gains, however, was that the the pace of reservoir depletion was significantly slowed. Storage in Powell+Mead increased by approximately 25,000 af in October and November, only the second time since 2010 that total storage in these two reservoirs increased during these two months.[8]

Figure 4. Graph showing active storage in Lake Powell, Lake Mead, and in Powell+Mead between January 1, 2023, and November 30, 2025.

Reservoir storage in the San Juan River basin increased more than in any other part of the Colorado River Basin. Five San Juan basin reservoirs increased by 197,000 af in October and November, mostly in Navajo and Vallecito Reservoirs.[9] Not much happened elsewhere, however. The 21 reservoirs of the upper Colorado River watershed lost 57,000 af during October and November, and 16 reservoirs in the Green River watershed lost 10,000 af during the same period.

- [1] Active storage in 46 reservoirs are reported by Reclamation at https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html.

- [2] Taylor Park, Blue Mesa, Morrow Point, Crystal, Fontenelle, Flaming Gorge, Vallecito, Navajo, Lake Powell, Lake Mead, Lake Mohave, and Lake Havasu.

- [3] Inflow to Lake Powell was computed as the sum of mean daily discharge of the Colorado River at Gypsum Canyon near Hite (gage 09328960), Dirty Devil River above Poison Springs near Hanksville (09333500), Escalante River near Escalante (09337500), and San Juan River near Bluff (09379500), as reported by the U.S. Geological Survey. Outflow from Lake Powell was computed as the mean daily discharge of the Colorado River at Lees Ferry (09380000), because stream flow is measured 15 miles downstream from the dam and includes ground-water seepage around the dam. Lake Powell storage increased between October 10 and October 20, as reported by Reclamation.

- [4] Inflows within Grand Canyon were calculated as the difference between measurements of the Colorado River at Lees Ferry (09380000), Colorado River above Diamond Creek near Peach Springs (09420000), and Diamond Creek nr Peach Springs (09404208).

- [5] Virgin River below confluence Muddy River near Overton (09419530)

- [6] Between October 1 and November 1, 2025, active storage in Lake Powell increased 52,000 af and 11,000 af in Lake Mead.

- [7] Between November 1 and November 30, active storage in Lake Powell decreased by 147,000 af and increased by 109,000 af in Lake Mead.

- [8] During the previous 15 years between 2010 and 2024, total storage in Powell+Mead increased by 36,000 af in 2011. During the other 14 years of that period, the median depletion of Powell+Mead was 436,000 af.

- [9] Storage in Navajo Reservoir increased 126,000 af between October 9 and November 8 and increased by 114,000 af in October and November. Active storage in Vallecito Reservoir gained 68,000 af in October and November. At the end of November, Navajo Reservoir was 60% of its 1.65 million af capacity. Vallecito Reservoir was 77% of its 125,400 af capacity.

Section: California's 2025 use of Colorado River water is on track to be the lowest since 1949

Source: <https://www.inkstain.net/2025/10/californias-2025-use-of-colorado-river-water-is-on-track-to-be-the-lowest-since-1949/>

California's projected use of Colorado River water this year, 3.76 million acre feet as of Reclamation's Oct. 29 modeling runs, would be, as near as I can tell, the state's lowest use since 1949.

Also notable:

- Nevada's 197,280 acre feet would be the lowest since 1992.
- The two lowest years in Imperial Irrigation District's history (my dataset goes back to 1941) were last year and this year.
- This will be the third year in a row that Arizona's main stem use has been below 2 million acre feet. The last time that happened (three consecutive years below 2maf) was in the 1980s.

Total take by the US Lower Basin states is projected to be 5.917 million acre feet, the lowest total US main stem use since 1983.

A few things to note.

First, the tenuous fabric of the Basin States negotiations is predicated right now, in part, on the Lower Basin cutting 1.5 million acre feet of annual use. They've already done that.

Second, the current cuts are enabled by significant federal payments to compensate the water agencies for their cuts. As my colleagues and I wrote back in September, counting on that money in the future would be unwise.

Third, the economies of Arizona, southern Nevada, and southern California are chugging along just fine right now. As I have written in the past, having less water does not mean scary doom. We can do this.

A note on the data:

The projection of total 2025 use by Lower Basin water users is based on model runs done by the Bureau of Reclamation every few days. It's a rich source of data, with detailed accounting of the various conservation programs being run by the Lower Basin agencies. PDF [here](#).

The comparison with prior years is based in part on the Lower Basin accounting reports, prepared each year since 1964. For prior years, I have a dataset I got years ago from the technical staff at the Metropolitan Water District of California, who had pieced together California numbers back to 1941. (Thanks, Met!)

We have no comparable data for the Upper Basin that might allow us to track water use at this level of detail, and with this kind of immediacy. This is a problem.

Section: Quoting Kevin Wheeler

Source: <https://www.inkstain.net/2025/10/quoting-kevin-wheeler/>

The utility of a model is seldom in its ability to deliver a precise value but rather in whether it provides an improved understanding of a system that would not have existed in its absence.

– Kevin Wheeler, Modeling to Support Water Management Negotiations

Section: “the nearest thing I have seen to being true”

Source: <https://www.inkstain.net/2025/10/the-nearest-thing-i-have-seen-to-being-true/>

A bunch of odds and ends cluttering my brain, blog posts that are half written in my mind that are in the way:

Quoting Luis Villa on accessing the open data commons

We’ve been talking about open data for a long time, but since using data is hard to consume and manipulate, open data has never been as big as open code. But if open data + vibecoding = powerful, does that make open data way more relevant?

Yes.

See graph above.

I always have had more questions (sometimes ill-posed, sometimes well-thought-through) than my coding abilities can execute. (See also domestic wells below.)

The Commons

I pay for a subscription to Newspapers.com in order to have access to a large portion of my written work. I view what I have written over the course of my life – newspapers, books, blogs – as a mindful and intentional contribution to the information commons. But this aligns poorly with the formal economic and legal structures – “institutions” as we might define them for our water resources students, the rules that serve as the foundation for the more common-language definitions of “institutions” that might apply here, the organizations of publishing – newspapers and book publishers and Inkstain.

The newspaper paid me well (it wasn’t a lot of money, but I viewed it as a fair transaction) and owned what I produced. I pay now for the privilege of reading it. The books are more complicated. I choose to make Inkstain freely available.

Derrida and Adorno, two philosophers I have been poking at of late, are helping me think about the definitional challenges – not “the commons” in particular, but what we’re doing when we attach words/concepts to things, the cultural quicksand beneath our linguistic feet.

That Postcard

Found this in a stack of old Dad stuff. It is my origin story, my father as a young artist in a moment of profound change. In laying the groundwork for his life, it laid the groundwork for mine.

Domestic Wells

See Luis’s comment above about vibe coding and open data.

I am not sure what to do with this. I can’t unsee it.

I’m out on the epistemological thin ice here, but as a journalist I spent much of my life working in areas where that ice is thin, it’s where the interesting stuff happens.

Ostrom and the Colorado River

I’ve mostly been grabbing the handrail and trying not to fall off as my Wilburys friends, in what we see as a discourse vacuum, charge ahead with our critique of Colorado River governance:

- Tripwires and the Lee Ferry delivery numbers
- The need for near term actions
- Essential Pillars for Colorado River Governance

In a 2011 paper, Elinor Ostrom laid out one of the final versions of her “design principles,” characteristics of successful institutional arrangements for collective action around natural resource systems. We spend a lot of time on this in the class I teach with Bob Berrens each fall for UNM graduate students. It was at the heart of my book *Water is For Fighting Over*, and it is at the heart of *Ribbons of Green*, the book Bob and I wrote that UNM Press will be publishing next year.

(Did I mention how much I love teaching?)

There are two design principles in particular that are at the heart of the current Colorado River challenges. Quoting from Ostrom 2011:

- How are conflicts over harvesting and maintenance to be resolved?
- How will the rules affecting the above be changed over time with changes in the performance of the resource system, the strategies of participants, and external opportunities and constraints?

There is an additional principle from Ostrom that shows up over and over in her work, that's embedded in her explicit principles: a need for a shared understanding of the quantification of the resource.

I am thinking through how these ideas relate to the current Colorado River challenges. Those challenges suggest what I had thought was a functional system lacks these three things. I am thinking a lot about what I described in 2015 when I was writing *Water is For Fighting Over*, versus what I see happening in 2025. What has changed, or what did I miss?

In which I get my first ambulance ride

Burying the lead here (I always hated the artifice of the journalistic jargon-spelling “lede”), but I had occasion recently to spend a few days in the bubble of the medical-industrial complex. I'm fine, I think, but the identification of a “new” life-changing risk is in actuality the identification of a risk that has probably been there all along. It's just that now I know about it.

Which means I can do some stuff to reduce that risk, including magical pharmacology (“If I crash,” I told my bike-riding buddy Sunday, “be sure to tell the EMT's!”) and also saying more “nos” to the stresses of my life of public engagement. My contributions to the commons are not without personal cost, as well as the personal benefits I derive. (Sorry, J.)

It also means that I spend a lot of time thinking about this (new?) risk. This is subtext to all the rest of what I just wrote.

Section: The 1922 Colorado River Compact is Now the Obvious Elephant in the Negotiating Room

Source: <https://www.inkstain.net/2025/10/the-1922-colorado-river-compact-is-now-the-obvious-elephant-in-the-negotiating-room/>

By Eric Kuhn, Anne Castle, John Fleck, Kathryn Sorensen, Jack Schmidt, and Katherine Tara

As negotiators for the seven Colorado River Basin states rapidly approach Reclamation's November deadline for providing a framework for a seven-state agreement for the Post-2026 Operating Guidelines for Lakes Powell and Mead, a larger threat looms. Reclamation's recently released September 24-Month study minimum probable projection is consistent with our mass balance analysis of storage in the next year, solidifying the likelihood of critical conditions if the coming winter is dry. Reclamation's

latest analysis predicts that storage at Lake Powell would fall below the 3500-ft elevation as early August 2026 and might continue to be below this critical elevation until March 2028. As we noted in our recent white paper, Reclamation has committed to protecting Lake Powell from going below 3500 ft.

This projection of future conditions in the event of persistent dry conditions poses a conundrum—Reclamation could reduce releases from Powell to protect the 3500-ft reservoir elevation, but in doing so, low releases would most likely trigger the dreaded 1922 Colorado River Compact tripwire—the amount of water delivered from Lake Powell to Lake Mead during a 10-year period that is less than the threshold. The Lower Division states are likely to litigate if the 10-yr average wire is tripped. Under one prevailing interpretation of the Compact, Upper Basin states must not cause the 10-yr flow at Lee Ferry to be depleted to less than 82.5 MAF to deliver water to the Lower Basin and Mexico. As explained in a new white paper, there is a very real chance that the 10-yr running average will be 82.78 MAF, just a hair above the tripwire, one year from now. In alternate scenarios, the 10-yr running average would hit the tripwire in 2027 or 2028. If Reclamation exercises its authority to reduce Lake Powell deliveries to as low as 6 MAF, the tripwire is triggered even earlier. In the face of this imminent possibility, Basin States and the Federal Government must commit to an enforceable agreement to reduce their total consumptive Colorado River uses with an equitable sharing of the burden sufficient to justify a waiver of claims under the Compact for the duration of the agreement. The alternative is a deeply uncertain future for the Basin.

Section: Analysis of Colorado River Basin Storage Suggests Need For Immediate Action

Source:

<https://www.inkstain.net/2025/09/analysis-of-colorado-river-basin-storage-suggests-need-for-immediate-action/>

By Jack Schmidt,¹ Anne Castle,² John Fleck,³ Eric Kuhn,⁴ Kathryn Sorensen,⁵ Katherine Tara,⁶

While Colorado River Basin attention is focused on negotiating post-2026 operating rules, a near term crisis is unfolding before our eyes. If no immediate action is taken to reduce water use, our already-thin buffer of storage in Lake Powell and Lake Mead could drop to just 9 percent of the levels with which we started the 21st century.

Water consumption in the Basin continues to outpace the natural supply, further drawing down reservoir levels. While Basin State representatives pursue the elusive goal of a workable and mutually acceptable set of post-2026 operating rules, our review of the latest Bureau of Reclamation data shows that the gap between ongoing water use and the reality of how much water actually flows in the Colorado River poses a serious near term threat. Another year like the one we just had on the Colorado River would nearly exhaust our dwindling reserves.

In a report being issued today, we look at total mass balance in the system – reservoir storage, inflow, and water use – to help clarify how much water the Basin actually has to work with if next year's snowmelt runoff is similar to 2025, and the risks if we do not take near term action to reduce our use. The findings are stark.

Today, Lake Mead and Lake Powell hold 6.3 million acre feet of realistically accessible water – water stored above critical thresholds identified by Reclamation as necessary to protect the infrastructure and ensure reliable supplies. That is down from 39 million acre feet above those thresholds at the end of the 20th century. A below average monsoon, a dry warm fall, and depleted upstream reservoirs means that next year's snowmelt inflow to Powell could be well below average even with a decent winter. If 2026 is as dry as 2025 and water use continues at the rate of the recent past, something current operating rules allow, our analysis shows the reserve of usable water could drop next year to just 3.6 million acre feet – less than the lowest previous storage content that occurred in March 2023. Such a situation occurring on Sept. 1, 2026 would necessitate drastic reductions in use during the subsequent reservoir

depletion season, which typically results in another 1.4 million acre feet of drawdown before significant inflows begin. To be clear, we are not predicting what will happen next year. But a repeat of last year's low inflows is a very real possibility. If we wait to see if 2026 is dry, the actions needed to save the system will be far more unpleasant than if we act prudently now.

A solution can't wait for a long term agreement among the states. It may be difficult, if not impossible, for the Basin States to take such short term action. That reality puts the onus on the Department of the Interior to act.

1 Director, Center for Colorado River Studies, Utah State University, former Chief, Grand Canyon Monitoring and Research Center.

2 Getches-Wilkinson Center, University of Colorado Law School, former US Commissioner, Upper Colorado River Commission, former Assistant Secretary for Water and Science, US Dept. of the Interior.

3 Writer in Residence, Utton Transboundary Resources Center, University of New Mexico.

4 Retired General Manager, Colorado River Water Conservation District.

5 Kyl Center for Water Policy, Arizona State University, former Director, Phoenix Water Services.

6 Staff Attorney, Utton Transboundary Resources Center, University of New Mexico

Section: Awaiting the Colorado River 24-Month Study

Source: <https://www.inkstain.net/2025/08/awaiting-the-colorado-river-24-month-study/>

By John Fleck, Anne Castle, Eric Kuhn, Jack Schmidt, Kathryn Sorensen, and Katherine Tara

As we await Friday's (Aug. 15, 2025) release of the Bureau of Reclamation's Colorado River 24-Month Study, we need to remember a painful lesson of the last five years of crisis management: whatever you see in Reclamation's report of the "Most Probable" reservoir levels for the next two years, we must prepare for things to be much worse.

A year ago, Reclamation's "Most Probable" forecast told us to expect Lake Powell to hold 10.36 million acre feet of water at the end of July 2025, with a surface elevation 3,593 feet above sea level. Actual storage in Powell at the end of July was 7.46 maf, 2.9 million acre feet less, and the reservoir is 38 feet lower, than the "Most Probable" forecast.

Four years ago, one of us (Eric Kuhn) wrote this, which is helpful in understanding what is happening:

The problem: the assumptions underlying the study do not fully capture the climate-change driven aridification of the Colorado River Basin.

In 2022, a Utah State Center for Colorado River Studies team led by Jian Wang (including one of us, Schmidt) took this on in more technical detail – Evaluating the Accuracy of Reclamation's 24-Month Study of Lake Powell Projections. The finding provided technical support for an intuition water managers already had: the 24-Month Study has an optimistic bias.

It is a practical demonstration of the problem U.S. Geological Survey scientist Paul Milly and colleagues famously warned us about nearly two decades ago – in water management, climate change means the past is increasingly unhelpful in projecting the future.

The 24-Month Study: A Brief Primer

Produced monthly, Reclamation's 24-Month Study includes three scenarios: Most Probable, Minimum Probable, and Maximum Probable. The Study includes 18 pages of data and forecasts for twelve

Colorado River system reservoirs, from Fontenelle and Flaming Gorge in the north to Mohave and Havasu in the south, projecting things like elevation, storage, inflows, releases, evaporation, and hydropower production each month for the next two years.

Here is Wang et al's explanation of how it works:

Projections for reservoir elevations during the next few months are based on predictions of reservoir inflow using a widely accepted watershed hydrologic model run by the Colorado Basin River Forecast Center. The input data for that model are observed snowpack in the watershed, soil moisture, and anticipated precipitation and temperature. Projections for reservoir elevations beyond the immediately proximate winter, a year or more in the future ('second year projections'), are based on statistical probabilities calculated using analyses of past inflows during a 30-year reference period.

The resulting model runs represent a wide range of uncertainties, which are captured in three resulting scenarios:

- Most Probable: the middle of the range
- Maximum Probable: the 90th percentile scenario, meaning that 10% of the model runs predict even wetter hydrology and 90% predict drier.
- Minimum Probable: the 10th percentile scenario, meaning that 10% of the model runs predict even drier hydrology and 90% predict wetter.

The problem, implicit in the argument Milly et al. made nearly two decades ago, is that a 30-year reference period is no longer a reliable indicator of what we should expect in the future. It represents a river we no longer have. This is not to suggest any bias or partiality on the part of Reclamation, but merely that the algorithms and modeling used to produce the 24-Month Study have proven in recent years to be skewed more toward the the past than the true-to-life. Our response needs to reflect that reality.

Because of the changing conditions in the Colorado River Basin, the Minimum Probable scenario has become the most valuable in providing a reliable indicator of the future. Actual flows and reservoir levels have been tracking the minimum probable forecast since March of this year. As we enter the fall of 2025, with the weak summer monsoon for most of the Upper Basin coupled with weak La Niña conditions persisting through the fall and early winter, and NOAA's seasonal outlook pointing to a warmer and drier than average fall, it's a good bet that this trend will continue at least through mid-winter. The Basin should be prepared for minimum probable conditions, with a clear possibility that actual conditions could be worse than the 10th percentile scenario. The basin community needs to be ready to respond with the necessary water use reductions now to protect the Colorado River system on which we all depend.

Sources:

- August 2024 24-month studies: Most probable, Minimum probable
- July 31, 2025 reservoir levels: USBR hydro data

Section: Return of the Deadpool Diaries: The Colorado River news keeps getting worse

Source:

<https://www.inkstain.net/2025/07/return-of-the-deadpool-diaries-the-colorado-river-news-keeps-getting-worse/>

With the latest Bureau of Reclamation model runs highlighting the serious risks posed by the declining reservoir levels that Utah State's Jack Schmidt has been warning about, there are signs that the closed-room discussions among the seven basin states, after brief glimmers of hope last month, are

once again not going well.

The Reservoirs

The latest Bureau of Reclamation 24-month studies show a clear risk of Lake Powell dropping below minimum power pool in late 2026, with Lake Mead dropping to elevation 1,025 by the summer of 2027. This should be hair on fire stuff.

The “clear risk” here is based on Reclamation’s monthly “minimum probable” model runs – what happens if we have bad snowpacks next year, and the year after? These are probabilistic estimates, not predictions. But the whole point of Reclamation doing this is so that we can be prepared. We need a robust public discussion about what our plan is if we end up on this fork in the hydrologic road.

The warning signs are clearly there in Jack’s analyses. Frustrated by the delay in the traditional metrics we use for measuring and monitoring the Colorado River, Jack’s been doing routine updates on reservoir storage contents. The traditional metrics we use – the Upper Basin Consumptive Uses and Losses Reports, the Lower Basin Decree Accounting Reports, the Natural Flow Database – have significant lags. The reservoir data is there in real time, integrating how much the climate system provides and how much humans use. The data here are all public. Jack’s value add is to sum them up and slice and dice the resulting data structures.

The somewhat arcane but incredibly useful framework he’s been using in his recent analyses is the period of accumulation, when reservoirs rise as river flows exceed human uses above them and extractions below them, following by the period of decline, when we’re drawing down the reservoirs. This is a tool, or a way of thinking, that we could use in real time to adjust our behavior, noting bad reservoir conditions and reducing our use. This is not something our water allocation framework is well suited to do.

The Negotiations

For more than a year, those involved in the delicate interstate negotiations over future Colorado River water allocation rules have repeatedly asked that we give them space to have the hard conversations they need to have in private. The results, or lack thereof, have done nothing to earn our trust.

When Arizona’s Tom Buschatzke moved the up-until-then super secret “supply driven” allocation concept into public view a month ago, it seemed like a good sign along two dimensions. First, the idea of basing the amount of water delivered from Upper Basin to Lower Basin past Lee Ferry on actual hydrology, on a percentage of how much water the climate is actually providing, seemed like an eminently reasonable approach. Second, Buschatzke was talking about this in public.

Folks from the Upper Basin followed suit, and a round of positive press followed.

Talking to Alex Hager, I called it “a glimmer of hope.”

But as this shifts from the brief sunshine of public statements back to the closed door negotiations, any glimmer appears dim indeed.

The problems were already visible in that brief, glorious bit of sunshine of public discussion last month.

There are two critical questions that need to be settled to make this work. The obvious one is the number – what percentage of the three year natural flow are we talking about shepherding down past Lee Ferry? The second is more subtle: What happens if the Lee Ferry flow falls short of that number?

Speaking to the Arizona Reconsultation Committee, Buschatzke was clear that whatever percentage number they settled on would be an Upper Basin “delivery obligation” at Lee Ferry. Becky Mitchell, speaking on behalf of Colorado, (but effectively as the de-facto Upper Basin voice, the role the other Upper Basin states seem to have for all practical purposes ceded to her) said (per Heather Sackett’s excellent reporting) it was in no way to be considered a delivery obligation.

When I suggested in a blog post that Upper Basin states might need to curtail water users in order to ensure the agreed-upon-percentage (whatever that is) is met, I got an angry call informing me that the Upper Basin was considering no such thing.

What this makes clear is that the same disagreement over the irreducibly ambiguous legal question in Article III of the Colorado River Compact – does the Upper Basin have a Lee Ferry delivery obligation or not? – is simply being shifted to a new modeling framework.

Never mind the equally intractable question of what the Lee Ferry don't-call-it-a-delivery-obligation percentage might be. I don't know anything more than gossip, but the gossip suggests the attempt to settle on a number, or even a range of numbers that Reclamation might model as part of its NEPA analysis, also is not going well.

If I was talking to Alex Hager today, I would no longer describe a glimmer of hope.

The Failure Mode

One of the most useful questions I learned to ask as a reporter covering water involved drilling down to the question of what happens when scarcity finally bites. What is the failure mode? Who actually doesn't get water? How does that work?

The combination of Jack's analysis and Reclamation's latest 24-month study suggests that we need to be asking that question in the near term. When Powell approaches minimum power pool, and Mead drops below 1030, whose water use will be curtailed to protect the system? If your answer involves a defense of why your own water supply should not be reduced, you're doing this wrong. Everyone needs to be realistic about their risk of a legal outcome different from their agency lawyer's position. But we also need to recognize moral obligations here, to find ways to share in this shrinking river. How are we going to come together, as a community, to respond?

The longer term argument also needs to begin to take this form.

Let us imagine going to the Supreme Court to settle the question of whether the Upper Basin does or does not have a legal delivery obligation under Article III of the Colorado River Compact to deliver 75 million or 82.5 million acre feet per year past Lee Ferry. If you lose that litigation, what is the failure mode? Who actually doesn't get water? If your groupthink has convinced you that this is not a meaningful question, that you're sure to win, and the other basin is the one that needs to be thinking about failure modes, you need a second opinion, to get out of your groupthink bubble.

Whatever "bring it on" enthusiasm for litigation you're hearing from your groupthinkers needs to be tempered by an honest discussion about what happens to your communities' water supplies if you lose.

I'll also make a modest pitch here for a need to recognize moral obligations, to find ways to share this shrinking river.

Section: Summer Update on the Colorado River Water Supply

Source: <https://www.inkstain.net/2025/07/summer-update-on-the-colorado-river-water-supply/>

Jack Schmidt

Center for Colorado River Studies, Utah State University

14 July 2025

Water stored in the reservoirs of the Colorado River represents the account balance from which we draw water for use. The amount in the account is especially important during dry times when the demand by water users throughout the Basin exceeds income to the account, primarily snowmelt runoff, and is met by account withdrawals.

The annual cycle of reservoir hydrology includes two seasons – a relatively short season when reservoir storage increases and a relatively long season when storage decreases. In wet years, the season when storage increases typically begins in March or early April and may last until late July. In dry years, this season might not begin until May and end in mid-June. During the rest of the year, the Basin’s reservoirs are progressively depleted.

Snowmelt in 2025 was low, similar to what it was in 2012 and 2013; in early June, the Colorado Basin River Forecast Center predicted that this year’s unregulated snowmelt inflow to Lake Powell will end up being 54% of the recent 30-yr average. In the 21st century, only 2002, 2018, and 2021 had lower inflows to Powell. Not surprisingly, the amount of water that accumulated in the Basin’s reservoirs during the 2025 snowmelt season was also unusually low. There are a few ways to consider the Basin’s reservoirs. We can consider every reservoir for which data are readily available[1]; we can consider the major reservoirs actively managed by Reclamation[2]; or, we can consider just Lake Powell and Lake Mead (hereafter, Powell+Mead). Considering only Lake Powell or only Lake Mead doesn’t tell us much, because all of the Rocky Mountain snowmelt is first stored in Lake Powell and subsequently transferred to Lake Mead. In 2025, the 46 Basin reservoirs gained only 0.55 million af (acre feet) of water, of which only 0.28 million af accumulated in the 12 federal reservoirs and only 0.11 million af accumulated in Powell+Mead. That is a very small amount, especially compared to 2023 and 2024 (Fig. 1). That accumulation is being quickly consumed. By 1 July 2025, all of the 2025 accumulation in Powell+Mead had been released downstream or evaporated.

In contrast to previous dry years, however, today’s account balance is unusually low, about the same as in late July 2021 (Fig. 2). Depending on how you think about the reservoir system, today’s contents are between 34 and 45% full in relation to their condition at the beginning of the 21st century (Table 1).

Table 1. Present storage contents of reservoirs in the Colorado River Basin in relation to past conditions.

Storage contents, in million acre feet			
on 30 June 2025 Last time storage was as low Present storage as a percentage of storage in late July 1999			
entire Basin (n=46)	26.8	25-Jul-21	45%
federal reservoirs (n=12)	23.64	4-Sep-21	42%
Powell + Mead	15.93	20-Nov-21	34%

The implications for Lake Powell depend on whether Reclamation decides to emphasize water storage in Lake Powell or in Lake Mead, and whether water presently in Flaming Gorge reservoir will be released to supplement storage in Lake Powell. As of June 30, 32% of the reservoir storage in the Basin was in 42 reservoirs upstream from Powell, 30% was in Mead, and 29% was in Powell (Fig. 1). if past management practices prevail, storage upstream from Powell will be quickly reduced, and storage in Powell and Mead will be reduced more slowly. If Reclamation emphasizes storage in Lake Powell by reducing releases to Lake Mead through the Grand Canyon, hydropower production at Glen Canyon Dam will be maintained and the risk of entrainment of smallmouth bass through the turbines will be reduced. But this management approach will cause Lake Mead to fall more quickly, thereby reducing hydropower production at Hoover Dam and perhaps the quality of water withdrawn to southern Nevada. Water storage can’t be maximized in both reservoirs at the same time. Indeed, we are living in dry times!

[1] There are 46, https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html.

[2] There are 12 included in Reclamation’s monthly 12-month study reports (Taylor Park, Blue Mesa, Morrow Point, Crystal, Fontenelle, Flaming Gorge, Navajo, Vallecito, Lake Powell, Lake Mead, Lake Mohave, and Lake Havasu).

Section: The Colorado River “psst psst” scheme emerges into public view: the “Supply Driven” concept

Source: <https://www.inkstain.net/2025/06/the-colorado-river-psst-psst-scheme-emerges-into-public-view-the-supply-driven-concept/>

See note of correction/clarification below:

Arizona yesterday finally moved the super-secret idea at the heart of current Colorado River negotiations out of the shadows.

The idea is deceptively simple: base Lake Powell releases on a percentage of the three-year rolling average of the Colorado River’s estimated “natural flow” at Lee Ferry. Allocate water based not on a century-old hydrologic mistake, but rather based on what the river actually has to offer. It presents an attractive alternative to the increasingly baroque and unproductive shitshow that had taken over interstate negotiations.

Correction/clarification: It has the great virtue of each basin getting out of the other basin’s business – one clean, simple number. But establishing the right percentage remains the hard part. Make the percentage too high and the Upper Basin would have to cut users with pre post-Compact water rights, something that has forcefully been asserted to me since this blog was first posted the Upper Basin argues it has no ability to do. Make the percentage too low and Lake Powell fills up while Central Arizona goes dry.

But some of the early modeling suggests that there may be a sweet spot where a combination of Lower Basin cuts along the lines of what the Lower Basin has already been willing to offer, combined with modest Upper Basin system conservation programs, might thread a needle that could allow the crafting of a compromise. This is very good news if the negotiators and the folks back home who have been egging them on can seize this opportunity to set aside parochial smallness and think at the basin scale.

The possibility of a new approach was hinted at a CU Boulder’s Colorado River conference two weeks ago (I spent most of the conference hidden away watching and listening on Zoom through a covid haze, so it might have just been a fever dream, but I thought I heard the hints), and I’m told was a topic of some of the hallway conversations. But Tom Buschatzke’s reveal at yesterday’s meeting of the Arizona Reconsultation Committee (the closest thing we have to the much-needed C-SPAN for the Colorado River Basin) was the first public discussion of the hush-hush stuff that shouldn’t be quite so hush-hush given, y’know, 40 million of us stakeholders.

The full slide deck from the Colorado River C-SPAN Arizona Reconsultation Committee is useful. Reclamation’s Dan Bunk, for example, shared a slide showing the latest “min probable” forecast (hilarious typo – “min problem” now corrected) showing the system tanking – dropping below minimum power pool at Powell – in winter 2026. The min probable forecast has been a useful guide lately, frankly, and the latest version is horrifying. (On any other day this would be the lead, and probably deserves its own post, but I try not to work on Wednesdays.)

We don’t have a lot of time here.

Section: Brad Udall on climate change and the Colorado River

Source: <https://www.inkstain.net/2025/06/brad-udall-on-climate-change-and-the-colorado-river/>

Via Allen Best, Brad Udall’s critically important comments at last week’s Getches-Wilkinson Colorado River conference:

Within this basin, we can and have worked together to deal with a really sticky, difficult issue like climate change, to inform decision-making given the right partners, including the federal government at

the table. Point two is our current climate trajectory is beyond awful, and that makes our challenge even worse.

Section: The Colorado River Conclave

Source: <https://www.inkstain.net/2025/06/the-colorado-river-conclave/>

Fascinating observation from Jim Lochhead this morning at the Getches-Wilkinson Center Colorado River Conference about the nature of the current negotiations and the role of the federal government. It came during a panel moderated by Anne Castle focused on what we learned from the expiring 2007 river management guidelines, which are the subject of intense renegotiation among the seven basin states.

From the perspective of the panel's charge – what have we learned since the 2007 agreements – the way I phrased that, the way the current process is going, should seem weird to us: “intense negotiation among the seven basin states.”

According to Lochhead, a Coloradan who was in the room for the '07 negotiations, the current cloistered seven-state process is very different from what happened leading up to the '07 agreement. In 2007, Lochhead explained, the states weren't the decision maker, the federal government was the decision maker, playing a much more active role as facilitator compared to the current process, which has deferred to the states to come up with a deal.

This is not going well. At least I think it's not going well. Who knows? Lochhead likened it to the selection of a pope, as we all await the puff of smoke. “The current process seems to me to be like the conclave.”

In my gossip network, I've heard good things about the current role being played by Scott Cameron, the Trump Administration's point person on this stuff. We will hear from him tomorrow. I look forward to that.

Other stuff from the morning sessions

Weirdly, after driving all the way to Colorado for the meeting, I spent the morning in my hotel room on Zoom – a bit under the weather, not feeling up to the social battery drain of all those people, saving energy for tomorrow when I'm moderating the closing panel. But what I lost in social capital construction and maintenance, I made up for in being able to focus on the talks. Among them.

Brad Udall, our modern-day E.C. LaRue, was pretty frank about the climate change trajectory, arguing that we need to prepare for a 10 million acre foot river. For those not steeped in the numbers, that's not very much water. The current climate trajectory, Brad said, is “beyond awful.”

Gov. Stephen Roe Lewis from the Gila River Indian Community argued that enduring solutions to the Colorado River's problems will require federal financial help.

A couple of useful nuggets from my Bill Hasencamp of the Metropolitan Water District of Southern California. One: Bill talked about a really interesting analysis his team has done of the Intentionally Created Surplus Program, which concludes that there is a lot more water in the reservoirs right now, including in Lake Powell, than would otherwise be the case. They've briefed me on their analysis and shared the report with me, I just haven't had the time to write about it yet, it's super interesting.

Bill also talked about the weird state of the current state negotiations. One on one, people say they're interested in compromise, in finding an agreement. In the negotiating room, they stick to hard line positions. This circles back nicely to Lochhead's point that last time around, this was a federal process, not a state-run process.

Anne Castle made an incredibly important point about the challenges face by the state's negotiators. They are sent into the room to advocate for their state's water supplies. They need permission from their constituents to compromise, to be able and willing to give up some water in order in the interests of the good of the basin.

That's on all of us.

Section: Colorado River Basin Reservoir Storage: where do we stand?

Source: <https://www.inkstain.net/2025/06/colorado-river-basin-reservoir-storage-where-do-we-stand/>

Jack Schmidt* and John Fleck**

*Center for Colorado River Studies, Utah State University

**Utton Transboundary Resources Center, University of New Mexico School of Law

1 June 2025

We now begin June, when the Colorado River's two largest reservoirs, Lake Mead and Lake Powell, should be swelling with melting snow for use later this year and beyond, but that is not happening. Although Lake Powell is our reservoir and Lake Mead is theirs (or vice versa), the two reservoirs are effectively one very large facility located downstream from Upper Basin consumptive users and upstream from Lower Basin users. At least 60% of the total storage in 46 reservoirs tracked by Reclamation is in Lake Powell and Lake Mead. The total contents of the two reservoirs have been steadily declining since early July 2024 and continued to decline through at least 31 May 2025. Never in the past 15 years has the decline in total storage of Powell and Mead extended so late into spring. Current reservoir storage data are showing us, in real time, an ominous pattern familiar from past dry years: upstream use of water before it has a chance to get to Lake Powell combined with releases from Lake Mead to users further downstream is outpacing the melting snowpack's ability to replenish the two reservoirs.

While the normal tools we use for measuring and managing use of Colorado River water – the Consumptive Uses and Losses Reports and the Lower Basin decree accounting reports – lag by weeks or even years, reservoir storage, which is the net difference between stream flow into reservoirs and what is released downstream or is lost to evaporation, provides the closest thing we have to an accurate, real-time measure of the Colorado River basin's water budget. Right now, we are not doing well.

- The duration of time this year during which total storage in Lake Powell and Lake Mead has declined is unprecedented in the past 15 years. In a typical year, the steady decrease in the combined contents of Powell and Mead that begins the preceding summer ends in early May when Rocky Mountain snowmelt becomes significant. However, inflows to Lake Powell this year have yet to exceed releases from Lake Mead, and the total contents continue to decline, suggesting that this year's recovery in storage will be minimal.

- Data from other years also suggests that reservoir recovery this year will be relatively small. This year, total unregulated inflow to Lake Powell is predicted to be 55% of normal. Based on past trends, net increase in total reservoir storage of the 46 reservoirs tracked by Reclamation will be ~1.2 million acre feet (af). By July, we are likely to resume draw down the basin's reservoirs until the 2026 snowmelt season begins.

- Presently, storage in the watershed's reservoirs is comparable to conditions in late summer and fall 2021 when water managers expressed significant concern. The very wet conditions of 2023 averted a major crisis, but the system remains depleted. In 2024, total basin reservoir storage climbed by 2.5

million af, but subsequent drawdown of those reservoirs was 3.6 million af during the following 10 months. Although the net difference between reservoir gain and subsequent drawdown of 1.1 million af might be considered “balanced” in the context of the last 15 years, there is no question that we have begun to mine the bounty of 2023, and we are likely to continue to do so until at least spring 2026 unless we greatly reduce consumptive uses.

For too long, we have hoped that big wet years will occur with sufficient frequency to avert true crisis, but there have been too few of those wet years during the 21st century. Only three of the last 15 years have been sufficiently wet to result in a significant increase in reservoir storage given the magnitude of the basin’s consumptive uses. We can’t continue with a water management policy that hopes for another wet year. The basin’s water managers have no choice but to further reduce consumptive uses to sustainably manage the dwindling water supply.

In response to a previously posted mini-white paper on reservoir storage, a supportive friend commented, “Nobody cares.” Another friend said, “I don’t see how we can get agreement about recovering storage. Let’s hope for more wet years.” We should care, and we need to try harder.

These mini-white papers seek to demonstrate that reservoir storage data, analyzed in aggregate, provide timely and accurate data relevant to understanding the reliability and security of the Colorado River’s water supply. These data are more precise, accurate, and timely than estimates of natural runoff, reservoir inflow, consumptive uses, or evaporation. Reservoir storage data provided by Reclamation are a significant contribution to transparency in water management. However, these data are under-utilized and under-analyzed and are typically reported without long-term context. We can do better.

These data can be used to develop an excellent correlation between April-July unregulated inflow to Lake Powell, forecast by the Colorado Basin River Forecast Center, and anticipated increase in basin-wide storage. Such an analysis strongly indicates that the 2025 snowmelt runoff will yield only a small increase in basin storage and necessitate greater reductions in consumptive use so as to better position the basin’s water users should next year also be dry.

Section: What the Mexican Treaty negotiations of the 1940s can tell us about 21st century Colorado River governance

Source: <https://www.inkstain.net/2025/05/what-the-mexican-treaty-negotiations-of-the-1940s-can-tell-us-about-21st-century-colorado-river-governance/>

Eric Kuhn, with some help from Anne Castle and myself, has taken a useful dive into what was known in the 1940s as Congress was considering the treaty between the United States and Mexico regarding how to share the waters of the Colorado River.

Drawing on the analysis of Colorado’s Royce Tipton, Eric draws out a theme we explored in our book *Science Be Dammed* – the gap between the scientific understanding of the day and the hydrologic reality we must now face as we try to manage a shrinking river.

Tipton, representing the state of Colorado, made a compelling case that there was enough water, including return flows from Lower Colorado River water use, to meet a 1.5 million acre foot allocation to Mexico without impairing US users.

California, already growing accustomed to living off the surplus of other water users’ unused apportionments, opposed the deal. Any water allocated to Mexico would cut into that surplus.

(California’s) major Colorado River projects were either complete, like the All-American Canal, or nearing completion, such as Metropolitan Water District’s Colorado River Aqueduct. In the 1930s, California agencies had signed contracts with the United States for 5.36 maf of Hoover Dam water, but

under the 1928 Boulder Canyon Project Act, they only had a “senior” right to 4.4 maf. Thus, over 900,000 af of its water supply was reliant upon surplus water being available. California’s primary concern was that, together with further development by the other basin states, the treaty would significantly reduce or eliminate the surplus water available for use within the United States. California did not want to be limited to 4.4 maf per year.

Tipton’s hydrology was based on a wetter climate than we have today. It also was based on the idea of “salvage by use” – that water use in, for example, Arizona, should be apportioned based on the impact of that use on the main stem of the river, not on how much was being consumed at the point of use. This would be then, and remains today, a critical point of unresolved water law.

Tipton’s hydrology was based on studies completed by the Bureau of Reclamation as a part of its comprehensive development study of the basin. The period of record was 1897-1943, with a reconstructed natural flow (the flow that would have been in the stream absent human modifications) at the international boundary with Mexico of 17.75 maf per year (16.3 maf per year at Lee Ferry). Additionally, Tipton told the Committee that because of salvage by use (the reduction of natural losses as the flow of the river is reduced through consumption) a minimum of another 1.25 maf/year would be available for use in the United States. The major portion of his estimated salvage by use water would be generated on Arizona’s Gila River. Tipton believed that Mexico’s consumptive use in 1943 was approximately 1.8 maf/year, more than what the treaty would guarantee. He emphasized that an estimated 8 to 10 maf/year was then currently flowing into Mexico and because the Hoover Dam power operation stabilized the river, Mexico could substantially increase its irrigation use. He was reinforcing the core argument by the supporting states: the treaty would limit Mexico to less water than it was currently using by about 300,000 af per year, and without a treaty, in the future Mexico might acquire more water through international arbitration.

Key to Tipton’s winning argument in support of the treaty was the belief that the agreement would constrain what might otherwise be increasing Mexican demands on the river.

Eric also found a wonderful bit of business from Tipton that echoes in today’s Colorado River Basin arguments over what constitutes “hydrologic shortage”:

[A]s a member of the engineering profession I will admit that engineers have made mistakes many times. One type of mistake that has been made in the West has been recommendations to irrigate areas larger than available water supplies would permit. Some of these errors were made on account of lack of stream flow records. Most overexpansion, however, has resulted from the lack of control of development under our western water laws. In most states no one can be denied the right to appropriate the waters of a stream; and the only control is the administration of the use of the water so appropriated by the system of priorities. So that we have many, many systems that have only flood rights on our streams, which is due to no one’s fault.

We conclude with some advice about what might be done next:

While the Colorado River Basin is currently focused on the negotiations between the Upper and Lower Division States over the future management of the river, the post-2026 operating rules, sooner than later the United States and Mexico will have to negotiate the successor to Minute 323. Both Minute 323 and the 2007 Interim Guidelines expire at the end of 2026. Given the current stalemate between the divisions over the future operations of the river, both hydrologic and political uncertainties loom large.

Rather than waiting for the Basin States to resolve their differences, which may ultimately require lengthy interstate litigation, Mexico and the United States may want to consider a parallel and alternative approach. They should consider abandoning complicated approaches that set annual deliveries based on how much water is being used by the Lower Division States or Lake Mead storage levels and instead negotiate an agreement based on giving Mexico a fixed percentage of the river’s natural flow (probably at Lee Ferry). By doing so, they could set an example for a long-term solution that works for the entire basin.

Section: The May USBR Colorado River 24-Month Study Confirms What We Feared

Source: <https://www.inkstain.net/2025/05/the-may-usbr-colorado-river-24-month-study-confirms-what-we-feared/>

By Eric Kuhn and John Fleck

The Bureau of Reclamation has released its May 24-Month Study. It confirms that 2025 will be another very dry year and the consequences will be significant. Under the minimum probable forecast, active storage in Lake Powell will fall to an elevation of 3530' (5.8 maf), only about 9 feet higher than the February 2023 low of 3521' (5.3maf). Just as alarming, under the "most probable" scenario, 2027 is projected to be another year for a 7.48 maf release from Glen Canyon Dam. This means that the ten-year flows at Lee Ferry are projected to fall well below the 82.0 maf tripwire – the point at which the basin states' disagreement over interpreting the Colorado River Compact's Lee Ferry delivery/non-depletion requirement could trigger interstate litigation.

The May 1st "most probable" forecast for unregulated April to July inflow to Lake Powell was 3.5 maf, down from an April 1 st forecast of 4.3 maf. Since May 1st. However, the runoff forecast has continued to decline, down another ~400kaf as we write this (May 26, 2025). No one should be surprised if we end up with an actual inflow closer to the May 1st "minimum probable" forecast of 2.6 maf.

Even with continued crop fallowing programs, storage in Lake Mead also continues to decline, dropping to an elevation of 1047' at the end of Water Year 2026 under the "most Probable" forecast and to elevation 1041' under the "minimum probable" forecast.

Lower Basin use continues to run well below long term averages, with this year's consumptive use by Arizona, California, and Nevada forecast at 6.3maf, well below the legal paper water allocation of 7.5maf. Yet Mead keeps dropping. The latest analysis of total reservoir storage from our colleague and collaborator Jack Schmidt (here's Jack and colleagues from March, with an update expected later this week) clearly shows that we are once again failing to rebuild reservoir storage. We're draining the system.

Of course, the 2007 Interim Guidelines expire after 2026, so we do not know what the rules will be for Glen Canyon Dam releases in Water Year 2027. Lacking any better information, the Bureau of Reclamation has assumed a continuation of the 2007 Interim Guidelines rules. Another approach would be for the Bureau of Reclamation to assume that absent an agreement among the states, the Secretary of the Interior could return to an annual release of 8.23 maf from Glen Canyon as set by the 1970 Long-range Operating Criteria. And curiously, under the "minimum probable" scenario, assuming a continuation of the 2007 Interim Guidelines, the projected 2027 annual release at Glen Canyon Dam reverts to 8.23 maf. Under a quirk in the 2007 Interim Guidelines, if the December 31, 2026, projected elevation of Lake Powell is below 3525' and the projected elevation of Lake Mead is below 1075,' the release reverts to 8.23 maf. This was referred to as the "sacrifice Lake Powell to save Lake Mead" strategy (seriously!).

Unless the 2025-26 winter is very wet or the Basin States can find consensus, the choices facing the Basin are stark: sacrifice Lake Powell for Lake Mead and perhaps keep ten-year Lee Ferry flows above the tripwire (no guarantee) or reduce annual releases from Glen Canyon Dam to maintain a balanced but small amount of storage in both reservoirs, which risks pushing cumulative 10-year flows past Lee Ferry across the tripwire.

Section: Upon the Retirement of Bob Snow, May 15, 2025

Source: <https://www.inkstain.net/2025/05/upon-the-retirement-of-bob-snow-may-15-2025/>

The following is signed by a stunningly long list of folks:

The Colorado River Basin will feel the loss of Bob Snow with his retirement from the Solicitor's Office at the Department of the Interior. His more than forty-year presence in the most impactful and most difficult conversations in the Basin has been both a comfort to those involved and a catalyst of judicious solutions. It is not an exaggeration to say that Bob is the most respected person of this era in the arena of western water law and policy. When the definitive legal history of the Colorado River is written, Bob Snow will be among the heroes.

Bob's initial work as a stream gager with the U.S. Geological Survey's Water Resources Division provided the technical basis for his deep understanding of the plumbing, capabilities, and limitations of the Colorado River system. After USGS, Bob returned to law school, earning his degree in 1994, and starting at Interior's prestigious Honors program shortly thereafter. During his legal career, Bob has demonstrated an incredible ability to consider and honor the interests of the whole basin and all its various constituent parts, always ensuring that his clients' interests were protected. He knows the history and the legal structure in such detail that he can think through the consequences of various proposals, whether intended or not. He has been adept in finding ways to thread the needle – to get both sides of a controversy what they need and what they want. As a result, he has proven to be amazingly effective at crafting compromises, a trait much admired and valued by his colleagues. Bob's career epitomizes the ideal of a federal public servant working effectively with federal administrations and state governments of both parties, never looking to take credit for himself.

Many of the agreements, programs, and institutions the Colorado River community relies on today bear the mark of Bob Snow. The 2007 Guidelines, the Lower Basin Multi-Species Conservation Program, the Glen Canyon Dam Long Term Experimental and Management Plan and High Flow Experiment Protocol, and various Memoranda of Understanding regarding conservation of water are just some of the many important advances that were forged with his guidance.

Bob's influence in the U.S. relationship with Mexico on the Colorado River cannot be overstated. He became close personal friends with the officials in the U.S. and Mexican section of the International Boundary and Water Commission, not only the commissioners, but also the secretaries, engineers, and other staff, earning the nickname Señor Nieve. He was a close confidant of the late great Commissioner Roberto Salmón Castelo. He helped to craft the seminal minutes between the countries that assisted Mexico after the devastating earthquake in the Mexicali Valley in 2010 and in which Mexico agreed to share in the reductions in water deliveries absorbed by the Lower Basin states in the U.S. He was also instrumental in the implementation of the 2014 pulse flow, sending fresh water to the Colorado River Delta for the first time in decades.

But for all of these and many other outstanding professional achievements, Bob will be remembered most for his friendship and camaraderie. To paraphrase Maya Angelou, people may forget what he said, people may forget what he did, but people will never forget how he made them feel. Bob genuinely listened to everyone with whom he interacted and made them feel important and valued. He forged lasting and meaningful personal relationships with so many in the Colorado River community, and those relationships will long outlast his retirement.

Bob is a lover of well-written books, Hatch green chiles, multiple baseball teams, excellent mole, and all kinds of rock music. He is a fierce and devoted friend, mentor to many of us in the Basin and elsewhere, and incredibly generous with his time. He has anecdotes from rock concerts and Mexican restaurants over the past five decades in all parts of the country that leave his listeners rolling on the floor with laughter. From all his friends in the Colorado River Basin, we wish Bob the best and most fun retirement ever. We hope and expect that he will continue to contribute to the overall sustainability of the Colorado River and know that he will continue to be part of our lives.

From your friends,

Section: Colorado River negotiators won't be appearing at Getches-Wilkinson conference

Source:

<https://www.inkstain.net/2025/05/colorado-river-negotiators-wont-be-appearing-at-getches-wilkinson-conference/>

Alex Hager had a piece today on the decision by the Colorado River basin states principles to not appear at next month's Getches-Wilkinson conference in Boulder. In a process where decisions are being made behind closed doors, outside of public view, Getches-Wilkinson is one of the few places those charged with the decisions show up in public, explain themselves, and answer our questions.

I said stuff:

"The unwillingness to answer the public's questions suggests that negotiations aren't going well," said John Fleck, who teaches water policy at the University of New Mexico. "I think it misses an important obligation in democratic governance of a river that serves 40 million people."

Alex did a nice job of summarizing the weird nature of this process. We tend to take it for granted, but it's worth remembering this weirdness:

Unlike many government processes, Colorado River policymakers work in a space that does not involve a mandate for public access. Their meetings are often held behind closed doors, are not listed publicly and do not yield minutes or records that can be viewed by the public....

Reporters' requests to state water authorities that once yielded interviews with top policymakers are now often met with written statements that tend to be short on detail.

As Joanna Allhands pointed out in Alex's piece, this undercuts the ultimate legitimacy of the results:

Joanna Allhands, an opinion writer at the Arizona Republic who has written about the Colorado River's "bankruptcy of leadership," said more transparency from water policymakers "would be smart as a matter of self preservation."

"Whatever the decision is made," she said, "Whatever alternative gets chosen, if people feel like they've been left out, guess where we're headed? We're going to the Supreme Court."

Section: A modest Colorado River proposal

Source: <https://www.inkstain.net/2025/04/a-modest-colorado-river-proposal/>

A group* of my Colorado River collaborators has put together what we hope can be a useful set of foundational principles as the basin states and federal leadership search for a path toward a negotiated agreement for post-2026 Colorado River management. They're based on a number of key premises:

- The Colorado River Compact will remain the foundation of the river's management, but we have to find a way past the deep disagreement between Upper and Lower basin states on what the Compact actually says.
- Colorado River Basin tribes must be essential partners in crafting the next set of guidelines, including through compensation for foregone water use.
- Shared pain is essential. The path toward a sustainable river system requires everyone to contribute to the solution to the problem of the river we all share.

There's more. I encourage you to read the whole thing. (It's short!)

* In alphabetical order: Anne Castle, John Fleck, Eric Kuhn, Jack Schmidt, Kathryn Sorensen, Katherine Tara.

Section: Wrecking Ball Report: Western Water Assessment

Source: <https://www.inkstain.net/2025/04/wrecking-ball-report-western-water-assessment/>

The Western Water Assessment, a federally funded research and outreach group based at the University of Colorado, sent a note to its stakeholders yesterday informing us that the new administration's plans to eliminate large swaths of federal climate spending include WWA's primary funding source:

We know that it can be hard to keep track of all of the news these days, and wanted to reach out to let you know that proposed changes to NOAA would eliminate the office that funds our work with you, which could mean the end of Western Water Assessment as a NOAA-funded program.

We are hard at work exploring alternative options so that we can continue to partner with and serve you. That includes our work holding community hazard planning workshops, improving rural community wildfire recovery, providing our 'one-stop-shop' Intermountain West Climate Dashboard and Summaries, supporting water planning efforts, and more.

In my Colorado River management world, that work includes the indispensable Colorado River Basin State of the Science report.

Section: On the Colorado River, doing the accounting with care

Source: <https://www.inkstain.net/2025/04/on-the-colorado-river-doing-the-accounting-with-care/>

It's easy to take for granted the accounting innovations in the Colorado River governance regime's 2007 guidelines, which have governed river management and the upstream-downstream relationships between the upper and lower basins. "Intentionally Created Surplus" (ICS) is now part of the lexicon, and the idea behind it shows enough promise that it's at the heart of the current negotiations over the post-'07 guidelines management of the river.

But we need to be careful about the lessons that we learn, and the details of how we implement the successor to ICS. How should the successor to ICS related to action levels for reservoir management? How do we ensure that water in ICS-like accounting pools is really conserved water, part of a sincere effort to reduce basin consumptive use?

Those questions are at the heart of the argument in Floating Pools & Grand Bargains, a new white paper by Kathryn Sorensen from Arizona State University and a group of colleagues, including Eric Kuhn:

While the concept of creating storage in Lakes Powell and Mead is not new, a Floating Pool proposal inherently creates trade-offs between existing and future uses that varies from the traditional shortage sharing priorities found in the Law of the River. We are, however, facing challenges not foreseen by the drafters of the Law. Without some new and imaginative thinking, the alternative may be to turn the future of the river over to the nine justices of the United States Supreme Court. That alternative is the least likely to produce a result which allows for the flexible management of a changing river system and potentially sets up the parties for continuing legal challenges. Cooperation among Colorado River water users now is essential to creating working relationships that can manage changing river conditions. The viability of the economy in the West is at stake.

Highly recommended.

Section: Quoting Jack Schmidt

Source: <https://www.inkstain.net/2025/04/quoting-jack-schmidt/>

"Everybody keeps hoping that the only way we're going to really rebuild storage is if we have another ridiculous, gangbuster year like 2023," said Jack Schmidt, a watershed sciences professor at Utah State University and Director of its Center for Colorado River Studies.

But, he continued, "that's highly unlikely."

11 Comments

-

Duh. ?

-

Duh. ?

-

Apportion 90% percent of the previous year's flow (or mean of previous 3 years' flow) in proportion to the 1922 compact with the Minute giving a share to Mexico. Lake Mead's evaporation comes out of Lower Basin total and Lake Powell's evaporation comes out of Upper Basin total. (Powell is upstream from "Lee Ferry," and so is the Paria.) 5% each is allotted to "refilling" (a joke) or actually slowly adding storage to Powell and Mead. (I actually think it should be 80, 10, 10%). This gives some advantage to Arizona (full disclosure, that's where I live) considering the junior right that Arizona agreed to for the Central Arizona Project. HOWEVER, OF ALL THE STATES, CALIFORNIA IS UNIQUE IN NOT CONTRIBUTING A DROP OF PERENNIAL WATER TO THE COLORADO RIVER and gets 40% of the water. What are the quantitative contributions of the Paria, Little Colorado, Havasu Creek, Gila and smaller tributaries in Arizona, and who gets that water? It worked for 80 years because the Upper Basin did not take its share. That's a two-edged sword. In any other appropriated rights context, they would have lost that water right. Lower Basin States were paid handsomely to leave water in Lake Mead. Upper Basin states got nothing. Seems like a raw deal to me. They gave away water for 80 years. As I understand it, the Upper Basin wants the Lower Basin to absorb all the reductions. Seems to me that they could shrink to what they took, on average, for 80 years, plus some more, without a sweat. BUT, THAT'S WHERE THE WATER COMES FROM. What does hydrology and reality have to do with a poorly thought out and poorly worded agreement 100 years ago? "Lee Ferry"? One mile downstream from the mouth of the Paria River, but the gage is upstream? Please correct me if I am wrong. As I understand it there is no streamflow gage measuring DISCHARGE into Lake Powell, and only one gage remains in the Grand Canyon at Phantom Ranch. LCR, National, and Diamond Creek are gone as well as Havasu. The Lee's Ferry gage is upstream from the Paria, and no gage remains on the Paria. Contributions of these tributaries are usually small, but may be significant in comparison with evaporation and stream loss (which nobody mentions as well as I can tell).

-

I agree with Jack on this point. Our planning for the Colorado River has been baked in hope for many years. We need to plan for the river we have (and may have), not the river we want. If we plan for an 11 or 12 maf river and sometimes get a 16 maf river, then that's a better problem to have.

-

@Mike Carpenter

Your comment: " HOWEVER, OF ALL THE STATES, CALIFORNIA IS UNIQUE IN NOT CONTRIBUTING A DROP OF PERENNIAL WATER TO THE COLORADO RIVER and gets 40% of the water", I would have expected the question to be asked by Delph Carpenter. I've been waiting for someone to ask that question. California actually contributes no sustainable water to the Colorado River. The only occasional contribution happens when there is that rare thunderstorm happening over the Hacienda wash (northwest of Needles) or the Milpitas wash (southwest of Blythe). I've seen only two episodes of any water reaching the river from these natural sources in over the two decades that I noticed. Oh...wait...there are two distinct 24/7 minor streams that flow into the Colorado River from California. Both come from leakage from MWD's reservoirs. As I'm the first to mention this, I'll call them MWD North Fork (which empties into Lake Havasu) and MWD South Fork (which empties about three mile below Parker Dam). The total CFS of these massive streams amounts to less than 10 CFS total. Had the total been more significant I would have called them Dave's Rivers. All kidding asides Mike, your point is well taken. The basic question is why did California end up with the Lion's share of the river? Was it due to that California had more votes in Congress than all of the basin states? Was it due to it had more potential to utilize the resource? If you think about the problem of how to deliver the water to where the population was in the early 1900's, you can see that the distribution relied mainly to areas adjacent to the river. The upper basin states had the disadvantage of limited growing seasons and limited land close to the river to grow crops. California and Arizona had better opportunities due to several valleys adjacent to the river. They also had a better climate to accommodate more crops. In the big picture, the deck was always stacked in California's favor. California was decades ahead of all the other basin states as far as commerce was concerned in the early 1900's. Commerce drives politics. Politics in turn drives how the river was distributed. It's just that simple. Also, California paid for a great deal of the development. MWD and LADWP paid a fair share for Hoover and Parker's construction during the great depression. Historically, Hoover and Parker's construction happened in conjunction with the building of the MWD aqueduct and the building of the All American Canal (IID) system. We all lament about the problems on the Colorado River but ignore the fact that it is commerce that actually is the bottom line.

-

As much as California irritates me in Colorado River issues, water rights are water rights (like deep high capacity wells draining basin aquifers in Arizona as well as the Central Valley in California). An unlined All-America Canal created a recharge mound in Mexico. Mexico even tried to make the case for water rights after the canal was lined. (It also made a mound of sorts on the US side). It will be interesting to see how it plays out. The reason it worked for 80 years was because the Upper Basin didn't take its share. Now they want the Lower Basin to absorb all the reduction. In any other appropriative-rights situation, the Upper Basin would have lost those rights. On the other hand CA and AZ users were handsomely rewarded for leaving water in Lake Mead. The Upper Basin has lots of reservoirs to screw with storage.

-

@Mike Carpenter

Again you make some great points starting with the All American Canal (IID). Yes, I remember when we had proposed that years ago and the Mexicans complaining that how that would ruin the ecosystem around the Alamo Canal. Speaking of ecosystems in Mexico, is everyone here aware that no river water is released below Morelos Dam? Mexico controls that Dam. No water reaches the delta. Yet they complain that any leakage into the porous sand on the American side was sustaining the wildlife in Mexico? I'm pretty sure that they were pumping any and all water adjacent to the border for AG usage. Yes, the AAC got lined. Also Brock Reservoir was built to absorb any over deliveries before the water made it into the IID system. Both of these projects were underwritten by California water agencies to 'save water'. Your next point about ground water is well taken. This falls into what happens within the state. Be it California, Arizona and elsewhere. If the state does not take ownership, there will be trouble in the long run. Arizona is a prime example. You've driven past Vicksburg AZ and have seen this

firsthand. I believe that the state and Senator Mark Kelly are looking into this. And what about using more water over time? It is indeed true that in 1925 the total population in the basin states was around 6 Million. In 2025 that population went up to over 60 Million. AG usage went up too. Are you aware that the farmers in both Southern California and Arizona are probably the most productive farmers anywhere. They have a growing season of nearly a year in duration and total control in when (and how much) water gets distributed to the crop. Alfalfa accounts for 32% for total usage of the Colorado River. They can get up to 12 cuttings per year and it is not a labor intensive crop. Of course, cattle is the main consumer of the crop. When asked about this, the farmer says – “You like Dairy Products...Right?” The bottom line is that one pound of Hamburger the family consumed in last night’s dinner accounted for 1850 gallons of water. Most of which was that Alfalfa. In my earlier response in this thread, I talked about California having the upper hand back when things were being discussed back in 1922. Arizona was always upset on how the distribution would work out. In my profile ‘link’ above, I’ve included a political cartoon that addresses the situation.

-

I didn’t know that the percentage of Colorado River water for alfalfa was 32%. Colorado River Indian Tribes (CRIT) gets almost 700,000 AF. About half is return flow. They (contractors) grow mostly alfalfa and much of that is eventually sold to Saudi Arabia. Senator Kelly and the late Representative Grijalva got a law passed that CRIT could now “lease” water as other tribes already can. I hope Tucson is after as much of that water as they can get. The Saudis leased Arizona state land, bought large swaths, drilled deep high-capacity wells, and raise alfalfa. The state cancelled leases on state land, and the state attorney general is trying to end their pumping on the basis of “nuisance.” I don’t think that will fly because AZ is a “rule of capture” state in unregulated land. I worked on the Lower Colorado accounting surface project (gravity study and other things) ’95-’04, Glen Canyon Environmental Studies ’85-’96 (Jack Schmidt started with Julie Graf in the USGS about ’85), and Upper Colorado Endangered Fish Recovery Program (erosion sensor in spawning beds of razorback suckers) about ’95-about ’04. It’s in my blood.

-

CRIT’s allocation of the Colorado River is greater than the state of Nevada. I was surprised at the amount you quoted as return flow. I’m sure there is a backstory here. Back in the 90’s I was involved in one of these bootstrap projects that Reclamation partnerships with the local irrigation district. In this case it was CRIT and BIA that actually ran the irrigation system. They had no way to monitor their distribution system and measurements were as good as the person making them. We installed a handful of sites that did the measurements using instruments and data loggers and a radio system where they could be monitored in real time. BIA had nobody on staff that knew anything about instrumentation nor radio systems. The only guy was an irrigation engineer with no background in automation. It was a good match for me as I had no experience in Canal Operations. Both of us got an education that summer. My big takeaway was understanding the necessity of the concept of constant monitoring in the collection of data for water accounting. What I learned at CRIT was applied across the board. In 2017 I gave a presentation to BoR’s Regional Office in Boulder City on the Data Collection system we had. We had added up all the measurements we had logged we had made in the previous year. The amount was 44 million logged entries. I said that actual raw instrument readings actually was ten times that, Water accounting standards are actually higher than USGS standards. Be it at MWD’s or CAP’s intakes or some farmers field, we measured constantly.

-

CRIT’s allocation is nebulous, having to do with adequate flow to irrigate crops, etc., rather than a set amount. It precedes the Compact, however. I was sidekick to my friend and USGS colleague Richard Wilson after we retired, consulting for CRIT. The “spillage” control structures were the most bazaar “weirs” I ever saw in my entire life. I kept thinking that any one of several cities would love to have the quantity of water that turned from potable to higher TDS return flow at those structures. The guy who measured them for years did it diligently, but he measured in the downward curving flow rather than the

backwater. Didn't really matter because they were such bizarre structures. We recommended Parshall flumes or standard weirs with telemetered data to automated gates to minimize "spillage." The council changed leadership and we were out. As I understand it, they have since done something like that through an engineering company contracted by BIA. We also addressed some of their root flooding and excessive vertical flow problems. It was apparent to us that the engineering consultants didn't understand multilayered groundwater flow or what kind of piezometers to install to figure it out. They were going to install shallow 2-inch pipes screened over several feet with gravel pack in 4-inch holes. Wells are not piezometers. We went off on a project of mine to monitor an earth fissure in Chino, CA.

-

Mike are you talking about the duckbill weirs that CRIT had? Also CRIT's returns were in two outlets. One was on a ditch just south of the Agnes Wilson Bridge. The other on the south end of the Res just above the Palo Verde Dam. We monitored those sites in the 90's but later farmed them out to the USGS. You're correct about the quality of water in those returns. However that's common with every return drain I've seen.

Section: Wrecking ball report: blowing up 25 years of productive US-Mexico collaboration on the Colorado River

Source: <https://www.inkstain.net/2025/03/wrecking-ball-report-blowing-up-25-years-of-productive-us-mexico-collaboration-on-the-colorado-river/>

Via Annie Snider at Politico:

The Trump administration escalated a water fight with Mexico on Thursday, saying it will cut off Colorado River water deliveries to the city of Tijuana.

In a post on X, the State Department's Bureau of Western Hemisphere Affairs said it would deny Mexico's request to deliver water to Tijuana through San Diego's water system because the country has fallen behind in its water deliveries to the U.S. along a different shared waterway — the Rio Grande.

"Mexico's continued shortfalls in its water deliveries under the 1944 water-sharing treaty are decimating American agriculture — particularly farmers in the Rio Grande valley. As a result, today for the first time, the U.S. will deny Mexico's non-treaty request for a special delivery channel for Colorado River water to be delivered to Tijuana," the post states.

Section: Colorado River Reservoir Storage in Mid-March: Where do we stand?

Source: <https://www.inkstain.net/2025/03/colorado-river-reservoir-storage-in-mid-march-where-do-we-stand/>

1 Center for Colorado River Studies, Utah State University

2 Utton Transboundary Resources Center, University of New Mexico

3 Kyl Center for Water Policy, Arizona State University

4 (sort of) retired

In Short:

Since the onset of the Millenium Drought 25 years ago, water agencies in the Colorado River Basin have been challenged by the overwhelming, yet essential, tasks of balancing total water use with a

reduced supply and recovering some of the reservoir storage lost since the last time the system was relatively full in summer 1999. By monitoring long-term changes in basin-wide reservoir storage, we can readily judge the success of these efforts.

In mid-March 2025, total storage in 46 reservoirs tracked by Reclamation was the third lowest in the 21st century for this time of year. The total amount of storage was the same as it was in late July 2021 when water managers described the situation as “serious” and declared a shortage in Lower Basin water supply. Between late July 2021 and mid-March 2023, water storage further plunged to an unprecedented low, but the exceptional runoff of 2023 provided modest recovery. However, basin storage in mid-March was 2.79 million acre feet, or 9%, less than the summer 2023 peak. In mid-March, 33% of the basin’s storage was in Lake Mead, 29% was in Lake Powell, 29% in 42 federal and non-federal reservoirs upstream from Lake Powell, and 9% in Lake Mohave and Lake Havasu. Water has been accumulating in Flaming Gorge Reservoir since early February and throughout the winter in a few smaller Upper Basin facilities but has been withdrawn from other large federal Upper Basin facilities and from Lake Powell. Lake Mead has been going down since late February. It is likely to be that total basin reservoir storage will decline until the beginning of the 2025 snowmelt runoff season and will only modestly recover, because inflow this year is forecast to be below average and less than in 2024.

System conservation and Assigned Water development by the Lower Division states and Mexico have prevented storage in Lake Mead from being even lower than what it is today. However, recent Reclamation projections indicate that consumptive use in the Lower Basin in 2025 will be larger than in 2024, suggesting that basin reservoir storage at this time next year will be even less than it is today. Projections of Upper Basin consumptive use are not available at this time. The continued decline and lack of recovery of water in reservoir storage conveys the clear message that our efforts to balance use with supply and to recover storage have not succeeded. The Colorado River water crisis endures.

In Detail:

On 15 March, active storage in 46 reservoirs in the Colorado River watershed that are tracked by Reclamation in the Bureau’s Hydrodatabase[1] was 26.9 million af (acre feet), the same amount as on 21 July 2021 (Fig. 1). That amount was the third lowest total basin storage on 15 March of any year of the 21st century[2] and is approximately 45% of the total stored in these reservoirs the last time the basin’s reservoirs were relatively full in late July 1999[3]. On 15 March, 33% of the basin’s storage was in Lake Mead, 29% in Lake Powell, 29% in 42 reservoirs upstream from Lake Powell, and 9% in Lake Mohave and Lake Havasu[4]. These data remind us of the challenge in providing a secure and reliable water supply to the Southwest, southern California, and northwestern Mexico.

Figure 1. Graph showing total reservoir storage in 46 reservoirs reported by Reclamation in its Hydrodatabase (blue line), as well as the contents of Lake Mead and Lake Powell (orange line), 42 reservoirs upstream from Lake Powell (green line), and in Lake Mohave and Lake Havasu (red line). Data are between 1 January 1999 and 15 March 2025. Total basin reservoir storage today is the same as in late July 2021.

It is instructive to remember how today’s small amount of reservoir storage was viewed when it occurred in summer 2021. On 27 July 2021, The New York Times posted the headline, “Two of America’s largest reservoirs reach record lows amid lasting drought.” That story led with these words, “The water level in Lake Powell has dropped to the lowest level since the U.S. government started filling the enormous reservoir on the Colorado River in the 1960s — another sign of the ravages of the Western drought.”[5] In the article, Wayne Pullan, Reclamation’s Upper Colorado Basin Regional Director, said, “This is a serious situation,” and Brad Udall said, “I’m struggling to come up with words to describe what we’re seeing here.” In mid-August 2021, Interior formally announced a water shortage in Lake Mead, triggering cuts on water deliveries, especially to Arizona farmers.

But today, there is less discussion about whether this small amount of reservoir storage represents a crisis. In part that may be because the season of snowmelt is ahead of us rather than behind us, as

was the case in late July 2021. We hope that inflow this coming spring will recover some storage, but, this winter's snowpack is merely average[6], and the basin's soils are very dry. The Bureau of Reclamation's "most probable" forecast of unregulated inflow to Lake Powell in 2025 is only 6.77 million af[7], 70% of the 30-year average and less than in 2024. Additionally, there may be little sense of concern, because we survived these conditions between July 2021 and March 2023. In fact, total basin storage plunged to only 21.3 million af in mid-March 2023. Perhaps, we are distracted by the engineering, legal, and political intricacies of the negotiations concerning post-2026 consumptive use and the seeming dysfunction of those negotiations. Perhaps, we are resigned to low reservoir storage as the new normal. Perhaps, we are the frog in the pot of water whose temperature is gradually rising, and we do not realize the water is about to boil.

Although significant strides have been made to conserve water, further reductions in water use throughout the basin are necessary, should we experience a succession of very dry years such as occurred between 2002 and 2004 and between 2020 and 2022. The post-2026 negotiations primarily have focused on strategies to reduce basin consumptive uses to match the 21st century's declining supply, but today's small amount of storage reminds us that it is critically important to also develop policies to recover reservoir storage to ensure security and reliability of the system.

To date, it has been exceptionally hard to recover storage. Despite the Lower Colorado River Basin System Conservation and Efficiency Program, the Upper Basin System Conservation Pilot Program, the Drought Response Operations Plan, Assigned Water development programs and large expenditures to reduce consumptive use using the Inflation Reduction Act, the Basin's water managers have made no progress in rebuilding storage except that provided by the unusually large inflows of 2023. Between mid-July 2023 and mid-April 2024 (immediately prior to the onset of spring snowmelt inflows), the basin's reservoirs were only drawn down by 2.2 million af, the smallest drawdown of total basin storage of the last 15 years[8]. However, the winter 2023/2024 snowpack yielded below average inflow to Lake Powell, and the basin only gained 2.5 million af of storage (Fig. 2). As of 15 March, the 46 reservoirs of the basin have been drawn down by 3.1 million af since the peak storage of those reservoirs in mid-July 2024. During the remainder of March and part of April, reservoir drawdown will continue to deplete storage originally accumulated in 2023.

Figure 2. Graph showing reservoir storage in different parts of the Colorado River basin between 1 January 2021 and 15 March 2025, summarizing periods of increase and decrease in total storage. Lake Mead, Lake Powell, and the 42 reservoirs upstream from Lake Powell each store approximately 30% of basin's total storage.

Low storage in Lake Mead persists despite 3.94 million af of water savings by the Lower Colorado River Basin System Conservation and Efficiency Program since 2006 (Fig. 3) and despite approximately 3.7 million af of Assigned Water development. Although additional savings are needed in 2025, the prospects of significant savings are not encouraging. Water users in the Lower Basin and Reclamation measure actual use and forecast trends in real time, and we anticipate 6.5 million af of main stem consumptive use by the three Lower Basin states. Commendably, this is less than the states' nominal 7.5 million af/yr allocation under the Supreme Court defined allocation. Arizona is expected to take the largest share of those cuts, with projected main stem use of 2.1 million af, 74% of its nominal allocation. Nevada is projected to use 68% of its 300,000 af allocation, and California is projected to take 96% of its 4.4 million af allocation.[9] However, the Lower Basin's projected 6.5 million af use is more than last year's 6 million af of use. It is unclear the extent to which use in 2025 might fluctuate based on the availability of federal funding to compensate water users for their conservation efforts, given the uncertainty enveloping federal policies under the new administration. We have no comparable set of numbers that allow evaluation of anticipated Upper Basin use and actual savings of wet water.

Figure 3. Graph showing water conserved in Lake Mead resulting from the Lower Colorado River Basin System Conservation and Efficiency Program and conservation efforts in Mexico.

Deficit spending is likely to continue between now and mid-April and will primarily be from Lake Mead and Lake Powell, because the total storage in the 42 reservoirs upstream from Lake Powell is no longer being depleted[10]. The basin-wide spatial pattern of reservoir operations in late winter and early spring 2025 has been storage of water in small upstream reservoirs and continued withdrawal of water from some CRSP facilities including Lake Powell, and recently from Lake Mead. Draw down continues at Granby (the primary storage facility of the trans-basin Colorado-Big Thompson Project), Blue Mesa, Navajo, Fontenelle, and several smaller reservoirs[11]. These Upper Basin depletions have been somewhat offset by small amounts of accumulation at other reservoirs[12]. Flaming Gorge Reservoir, the largest facility upstream from Lake Powell, was at its lowest at the very end of January and increased 41,100 af of storage in February and the first half of March. In contrast, the total contents of Lake Powell and Lake Mead continue to be drawn down. Lake Powell was at its highest on 1 January 2025, has lost 803,000 af of storage since that time, and will probably continue to decline for another month, based on projections by Reclamation[13]. Lake Mead increased in storage after 1 January, peaked in late February, and subsequently lost 74,000 af. Last year, storage in Lake Mead continued to be lost until early August, and the same pattern is likely this year. The total loss of storage in Lake Powell and Lake Mead between 1 January and 15 March was 476,000 af. The rate of loss from the Mead-Powell system for the next few weeks will be determined by the balance between inflows to Lake Powell and releases from Lake Mead.

[1] Data are accessed at https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html.

[2] Total active storage in the same 46 reservoirs on 15 March 2025 was less than on the same date in 2022 (23.5 million af) and in 2023 (21.3 million af).

[3] On 21 July 1999, total active storage in the same 46 reservoirs was 59.5 million af.

[4] Lake Mead stored 9.01 million af, Lake Powell stored 7.85 million af, the 42 reservoirs upstream from Lake Powell stored 7.74 million af, and 2.30 million af were in Lake Mohave and Lake Havasu on 15 March.

[5] On 27 July 2021, active storage in Lake Powell was 7.90 million af, approximately the same as today.

[6] On 20 March 2025, snow water equivalent (SWE) in the Upper Colorado Region was 97% of median with 18 days remaining until the annual peak SWE typically occurs.

[7] Reclamation's 5 March 2025 forecast.

[8] This comparison is for the reservoir drawdown between the mid-summer peak and the following spring's minimum storage prior to the next year's runoff. The smallest draw down of total basin reservoir storage in the recent 15 years was between 13 July 2023 and 17 April 2024 (2.15 million af). The second smallest drawdown was 2.61 million af between 6 July 2014 and 4 May 2015. Drawdown was 2.56 million af between 4 August 2011 and 30 April 2012 and was 2.82 million af between 28 July 2019 and 29 April 2020.

[9] Reclamation Lower Colorado River Basin forecast, 17 March 2025
<https://www.usbr.gov/lc/region/g4000/hourly/forecast.pdf>

[10] Maximum draw down of these Upper Basin reservoirs was 1.49 million af on 26 February 2025.

[11] Net drawdown exceeding 1,000 af between early January and mid-March occurred at Granby (69,600 af), Williams Fork (6,570 af), Dillon (4,520 af), Green Mountain (8,940 af), Ruedi (5,310 af), Taylor Park (1,790 af), Blue Mesa (17,200 af), Fontenelle (54,100 af), Upper Stillwater (1,230 af), and Navajo (28,400 af) Reservoirs.

[12] Reservoirs accumulating more than 1,000 af storage between January and mid-March were Willow Creek (1,300 af), Rifle Gap (2,600 af), Vega (1,400 af), Crawford (1,750 af), Big Sandy (1,670 af), Eden (1,150 af), Meeks Cabin (2,290 af), Red Fleet (1,030 af), Steinaker (2,910 af), Strawberry (7,050 maf), Starvation (15,200 af), Moon Lake (3,180 af), Scofield (4,950 af), and Vallecito (6,060 af) Reservoirs.

[13] Based on the March 2025 24-Month Study Projections
<https://www.usbr.gov/uc/water/crsp/studies/images/PowellElevations.pdf>.

Section: The March 24-Month study and the myth of a “Compact Call”

Source: <https://www.inkstain.net/2025/03/the-march-24-month-study-and-the-myth-of-a-compact-call/>

By Eric Kuhn

The Bureau of Reclamation released its March 24-Month study last Friday and just like last month, the forecast is for big trouble in the Colorado River Basin. Under the “Most Probable” scenario, the ten-year cumulative flow at Lee Ferry will drop below 82.5 million acre-feet (the “tripwire”) by the end of Water Year 2027. If this happens, the odds are high that the Lower Division states will trigger what they referred to in their February 13, 2025, letter to Secretary Burgum as a “compact call.” The nuance, however, is that the Colorado River Compact has no specific provision for a compact call. Under the compact, a call is just another word for interstate litigation.

Although the letter is now over a month old, it just recently received attention from two of the region’s most respected water reporters, Ian James of the Los Angeles Times, and Tony Davis of the Tucson Daily Star. In his piece, ([link: Three states urge Trump administration to fix Colorado River dam – Los Angeles Times](#):) James pointed out that in their letter, the Lower Division states used the term “compact call” 23 times. The term “river call” is commonly used in prior appropriation states that actively administer water rights. For example, the Shoshone Hydroelectric Power Plant, located on the Colorado River a few miles upriver from Glenwood Springs, has a senior water right for 1250 cfs with a priority date of 1902. When the flow at the plant’s diversion dam drops below 1250 cfs, its owner places a “call” on the river. Under Colorado law the Division Engineer, an employee of the Colorado State Engineer, then shuts off sufficient upstream junior uses to bring the flow back to 1250 cfs. A “Shoshone call” is almost an annual occurrence.

The Colorado River Compact places two specific flow obligations on the Upper Division states at Lee Ferry. Article III (d) requires these states to not cause the ten-year cumulative flow to be depleted below seventy-five million acre-feet. Additionally, under Article III (c), if there is not sufficient surplus water available, then each basin is responsible for one-half of the deficiency (the difference the annual treaty delivery and the available surplus water). Assuming there is no surplus water and the 1944 Treaty delivery to Mexico is 1.5 maf per year, the Upper Division states would have to deliver to Lee Ferry, an additional 750,000 af per year.

Thus, using the Shoshone analogy, the Lower Division states claim they have a 1922 Compact water right for up to 82.5 maf every ten years. Note, we say “up to” because in the last few years, pursuant to Minute 323, annual deliveries to Mexico have been slightly less than 1.5 maf. For many reasons, the Upper Division states do not agree that their 1922 Compact obligation is 82.5 maf every ten years, see: “On the Colorado River, there are no Simple Disputes,” ([link: On the Colorado River, there are no Simple Disputes – jfleck at inkstain](#):).

If (or more likely when) the ten-year flow at Lee Ferry were to drop below ~ 82.5 maf, and there is no consensus agreement among the basin states in place, it is clear that the Lower Division will then attempt to place a compact call on the Upper Division states (and perhaps legally challenge the Secretary’s operation of Lake Powell) to increase deliveries at Lee Ferry. Where the Shoshone Plant analogy breaks down is what happens once a call is placed. Colorado law directs the State Engineer/Division Engineer how to administer a Shoshone call, but intentionally, there is no equivalent of the Colorado State Engineer in the Colorado River Compact. The Colorado River Compact negotiators debated and rejected a compact commission with enforcement powers. Arizona’s Winfield

Norviel suggested such a commission, but led by Colorado's Delph Carpenter, it was rejected. Carpenter abhorred the idea of creating what he referred to as a "super agency."

Except for Article V which provides for the Directors of the Reclamation Service and USGS to cooperate, on an ex-officio basis, with the basin State Engineers to collect and publish data on Colorado River flows and uses, the 1922 Compact provides no role for the federal government. The Secretary of the Interior is not even mentioned. Instead, the compact negotiators provided two mechanisms for resolving disputes and enforcing the provisions of the compact. Article VI is a dispute resolution provision which has never been used. The somewhat cumbersome provision provides that when a dispute arises, upon the request of one governor, the resolution process can be triggered. If this happens, each state governor then appoints a commissioner to formally negotiate a resolution with the other states. If the commissioners reach an agreement, it must be ratified by the affected state legislatures, most likely all seven. If a resolution is reached under Article VI, the compact does not require it to be approved by Congress.

The second mechanism is litigation. Article IX states: "Nothing in this compact shall be construed to limit or prevent any State from instituting or maintaining any action or proceeding, legal or equitable, for the protection of any right under this compact or the enforcement of any of its provisions." Thus, if Lee Ferry ten-year flows drop below 82.5 maf, the compact vehicle to implement a "compact call" is for one or more of the Lower Division states to initiate litigation under Article IX and convince the U.S. Supreme Court, or its appointed Special Master, that the Upper Division states are not complying with the compact.

Assuming no agreement among the states to avoid compact litigation, a compact call scenario might occur as follows: The ten-year flow at Lee Ferry is forecast to drop below 82.5 maf tripwire (it might be a little less if corrected for actual deliveries to Mexico). The Lower Division then states demand that the Secretary increase releases from Lake Powell or, alternatively, the UCRC implement a curtailment to bring the flow up to 82.5 maf by the end of the water year. Via the UCRC, the Upper Division states respond that they are in full compliance with the 1922 Compact and insist that the Secretary not increase releases from Lake Powell. Lacking a consensus agreement among the states, the Secretary makes no change to the prescribed annual release forcing the Lower Division states to initiate litigation. Assuming the Supreme Court accepts the case, it would now be up to the court or its Special Master to decide if the Upper Division states are in compliance with the compact. If they are not, a remedy could be the imposition of a compact call by ordering the UCRC to implement a curtailment pursuant to the 1948 Upper Basin Compact. How long might litigation take? It could be decades, or the Lower Division states might succeed with a request for immediate relief. No one knows.

While the 1922 Compact does not give the Secretary of the Interior any special power or authority, under subsequent federal legislation and the 1963 decision in *Arizona v. California* the Secretary has considerable power and authority. For example, under Section 602 of the 1968 Colorado River Basin Project Act, Congress directed the Secretary to promulgate criteria for the coordinated long-range operation of the federal reservoirs. It also set priorities for the annual release of water from Lake Powell. The first priority is "releases to supply one-half the deficiency described in Article III (c) of the Colorado River Compact, if any such deficiency exists and is chargeable to the States of the Upper Division."

The legislation, however, is silent on who or what entity decides if any such deficiency exists and is chargeable to the States of the Upper Division. According to Tony Davis ([link: Arizona water officials, others blast feds for not protecting dam – Our Community Now](#)), a spokesman for the Arizona Department of Water Resources suggested that the Secretary has this responsibility. But even if the Secretary does ultimately decide how much water must be released from Lake Powell to satisfy the obligation of the Upper Division states to Mexico under the 1922 Compact, if the Lower Division states believe the Upper Division states are violating the 1922 Compact, it could result in litigation. In fact, a decision by the Secretary to interpret the compact could be the trigger for litigation. After the Secretary signed the 1970 Long-range Operating Criteria which set a minimum objective release of 8.23 maf per

year from Glen Canyon Dam, the Upper Division states seriously considered litigation. They decided against it because they concluded they could not show any actual injury. The impact of climate change on the flow of the river has now fundamentally changed that dynamic.

The Lower Division State's letter was directed to Secretary Burgum, but it is a message to the entire basin. The March 24-Month study confirms what we already know. The basin has two basic choices: litigation or a basin-wide agreement implementing fundamental change. Let's hope it's the latter.

Section: Taking another crack at Tribal water in Arizona

Source: <https://www.inkstain.net/2025/03/taking-another-crack-at-tribal-water-in-arizona/>

The three Native American tribes in northeast Arizona – Diné, Hopi and San Juan Southern Paiute – are taking another run this year at a water rights settlement that would finally bring water to communities that lack it.

An attempt last year failed in the final days of the last administration after Upper Colorado River Basin states blocked it in the most glaring example of basin governance dysfunction.

Here's Shondin Silversmith's rundown:

The agreement will end the three tribes' claims to the main stem of the Colorado River, the Little Colorado River and relevant groundwater sources in Arizona. The settlement would guarantee the tribes' access to over 56,000 acre-feet of Colorado River water, with specific groundwater rights and protections.

Additionally, the agreement requires the Navajo Nation and Hopi Tribe to leave 17,050 acre-feet per year of Arizona Upper Basin water in Lake Powell for the first 20 years.

It will also allow the Navajo Nation and the Hopi Tribe to lease a portion of their water rights, providing the tribes a chance to create economic opportunities until local demand is met through new infrastructure.

This is a crucial agreement, I hope it can escape the new administration's wrecking ball.

Section: Wrecking Ball Report: At Reclamation, a view from the inside

Source: <https://www.inkstain.net/2025/03/wrecking-ball-report-at-reclamation-a-view-from-the-inside/>

Elevated from the comments, observations from former Reclamation manager Doug Blatchford:

WHEN I joined the Reclamation team in 2005 as the River Operation Manager on the Colorado River, part of my duties was to prepare a business plan to direct future business decisions based on the operations budget and services required (like delivering water to 20 million people). Part of the business plan was succession planning; I estimated it took at least 5 years to hire a replacement for say, the oracle data base manager, and that hire would shadow the retiree to affect a smooth transition. Just letting experienced (20, 30, 40 yr employees...) retire without planning was difficult enough inside the Federal system, let alone just taking a wrecking ball and eliminating 40 percent of the work force.

So what could happen? What kept me up at night was mostly public safety, meaning what could go wrong that would threaten life and limb because of faulty operations, much like managing an air traffic control system, except less dynamic. Imagine :

- an overworked field crew in 120 degree heat installs the a gage shaft encoder that reads the lake going up, instead of going down as it should
- screwing up the distribution of flows at Imperial Dam to Arizona, California, and/or excess flow to Mexico
- not having the modeling staff to provide the 5-yr probabilistic forecast
- reduced skill level in the forecast from CBRFC
- foregoing special requests to raise or lower Havasu water surface elevations, such as the parade of lights
- not being able to respond quickly enough to dynamic operations during a monsoon or hurricane event, and accidentally releasing more water /and or not retaining the water behind the dams, or worse, an atomospheric river that hits a higher water surface elevation at Havasu that pulses the lake over the top of the dam

and on and on and on ...not to mention staff moral, which is a from a dedication based on service, less on monetary incentive

Section: What does it mean for western water management when the federal government becomes an unreliable partner?

Source: <https://www.inkstain.net/2025/02/what-does-it-mean-for-western-water-management-when-the-federal-government-becomes-an-unreliable-partner/>

I got a text message yesterday afternoon about this, which is nuts:

The U.S. Army Corps of Engineers-Albuquerque District announced today that an unintended water release from Cochiti Dam may increase flood risk on the Rio Grande in the river channel, riverbanks, and floodway.

The cause of the unintended water release was a procedural error during routine maintenance.

Accidentally dumping 8,000 cubic feet per second into a river channel that hasn't seen that much water since 1985 is a big deal. The gage data suggests the river level rose four feet basically instantaneously.

I've been thinking a lot lately about the stuff the federal government does in water management in the United States that we used to be able to take for granted, like, for example safely operate the dams.

We all love to complain about the federal government's water management work, but the complaints are based on narrow questions and presume a broad societal consensus that there's a bunch of stuff the federal government can be reliably counted on to do while we argue over details. Reclamation and the Corps are gonna operate the dams, for example. The details we argue about are at important margins, but they're at the margins, based on the presumption that the basic stuff will get done.

Like, for example, spending the money that Congress approved to help us manage shortages in the Colorado River Basin. Which money has now been yanked out from under us by the autocrats who think they know better, as Alex Hager reported yesterday.

HAGER: Colorado River user groups are being told that payments in exchange for their water are on pause for now. It's unclear if those payments scheduled to arrive in coming months will actually show up.

BART FISHER: It's unnerving to think that maybe come August 1, all of our plans will need to suddenly change.

HAGER: Water experts say these projects don't seem to conflict with the Trump administration's executive orders. Anne Castle helped manage water under presidents Biden and Obama.

ANNE CASTLE: These are not woke environmental programs. These are essential to continued ability to divert water.... Having this appropriated funding suddenly taken away undoes years and years of very careful collaboration among the states in the Colorado River Basin and threatens the sustainability of the entire system.

I have no idea what happened at Cochiti Dam yesterday, whether the person who made the "procedural error" was new because the old timer who knows how to run the dam took the early buyout and bailed. But I do know that is exactly the "what if" scenario I was gonna lay out in a blog post that's been percolating in my head about this question of how we in the West go forward in water management when the federal government suddenly becomes an unreliable partner.

I am not saying this because complaining about the stunningly arrogant idiots crashing through the federal government right now is great clickbait. I'm tired of all the angry clickbait, frankly, which is why I hadn't written the blog post until today.

My point here is a serious question, not a rhetorical one: What would it mean for us in Western water management if the federal government becomes an unreliable partner? What must we do to prepare? What does that even look like?

Section: Anne Castle steps down as the federal representative to the Upper Colorado River Commission

Source: <https://www.inkstain.net/2025/01/anne-castle-steps-down-as-the-federal-representative-to-the-upper-colorado-river-commission/>

Worth sharing in full:

January 28, 2025

Re: Resignation as U.S. Commissioner to Upper Colorado River Commission

Dear (addressee redacted)

As requested, I am submitting my resignation as U.S. Commissioner and Chair of the Upper Colorado River Commission, effective January 27, 2025. I was honored to be

appointed by President Biden to this position and serving on the Commission has been a privilege. I am deeply appreciative of the opportunity to contribute to the vital work of managing and protecting the Colorado River. This is an existential time for the river. We are on the brink of putting in place an operating regime that will govern our lives and our economies for decades. We are currently trying to manage an immense river system with tools and structures that were not designed for the conditions we face today. The Law of the River has provided a

stable foundation for over 100 years, and we need to give it credit for the amazing development and thriving populations in the entire southwestern part of the country. But the law has not fully adapted to kind of hydrologic conditions we have seen in the past and expect in the future. Agreements negotiated in good faith decades ago are no

longer providing the intended balance of fairness. The operation of the Colorado River is a complex web. Pulling on one thread of the web vibrates through the entire system, with significant potential for unexpected and

adverse consequences. Sustainable operational solutions must be crafted by those who best understand and appreciate this complexity – the Colorado River Basin States and Tribes, the Bureau of

Reclamation and other affected agencies and offices within the Department of the Interior, and the many dedicated stakeholders and professionals who rely on and care about the River. This type of process is slow, messy, cumbersome, and unscripted, but ultimately results in the best path forward. This is the course being followed now. Edicts imposed from outside the Basin, such as recent proclamations concerning California water, based on an inadequate understanding of the plumbing and motivated by political retaliation, upend carefully crafted compromises, create winners and losers, and unnecessarily spawn the potential to adversely affect the lives of millions of people as well as the ecosystems on which they depend.

During my time as the U.S. Commissioner on the UCRC, I have had the opportunity to witness an evolution of role of the 30 Basin Tribes in Colorado River management. The commitment by the UCRC and its state commissioners to greater inclusion of Tribes in decision-making processes and more robust consideration of Tribal interests and issues has been extraordinary and had tangible results. The Memorandum of Understanding between the UCRC and the Upper Basin Tribes cementing adopted processes of cross communication and committing to shared goals is an historic first in the Basin. I am optimistic that, despite the rescission of the Executive Order on environmental justice, progress toward more meaningful participation by Tribes in Colorado River decision-making will continue.

In this position, and in my former role as the Assistant Secretary for Water and Science at the Department of the Interior, I have had the privilege to work with many dedicated and professional public servants within the Department and particularly in the Bureau of Reclamation. Despite being vilified by the very Administration they serve, these federal employees continue to strive to fulfill the mission of their agencies. The abrupt imposition of disruptive new terms of employment and undermining of civil service entitlements and expectations are explicitly designed to result in a wave of resignations, which are unabashedly welcomed by the architects of such policies. This broad brush, unfocused purge in furtherance of the stated goal of liberating large corporations from regulation they do not care for will result in attrition of expertise, damage to the American public, and specifically, a more disordered and chaotic Colorado River system.

I wish only the best for the Upper Colorado River Commission and its Commissioners. Both the staff and the Commissioners are dedicated and hardworking and fully

committed to their responsibilities. I will be cheering their efforts to reach a negotiated operational compromise that will sustain the River in the decades to come. Sincerely,

Anne J Castle

Section: The January 2025 24-month study is a major caution sign for the Colorado River Basin

Source: <https://www.inkstain.net/2025/01/the-january-2025-24-month-study-is-a-major-caution-sign-for-the-colorado-river-basin/>

By Eric Kuhn, John Fleck, and Jack Schmidt

On January 16th, the Bureau of Reclamation released the January 2025 24-Month Study. Based on the January 1st runoff forecast into Lake Powell, the projected “most probable” annual release from Glen Canyon Dam for Water Year 2026 is now 7.48 maf. This needs to be taken as a significant caution sign because it shows that we are on a clear trajectory to hit what Colorado’s Jim Lochhead first called the 1922 Compact’s first “tripwire” (82.5 maf/10-year) as early as 2027. Given the current stalemate between the Upper and Lower Division States over how the reservoir system should be operated, it means the potential for basin-wide litigation is now in the “Red Zone.”

The January 24-Month study is the first in each water year to be based on a published forecast from the Colorado Basin River Forecast Center’s runoff model – the first to be based on actual snowpack. The

January 1st runoff forecast for unregulated April-July inflow to Lake Powell was 5.15 maf (about 81% of average). This results in a projected 12/31/2025 elevation of Lake Powell of less than 3575' making the annual release for Water Year 2026 7.48 maf. Of course, the forecast will change as the winter progresses. In fact, the January mid-month forecast dropped by 300,000 acre-feet to 4.85 maf. At this point in the winter our confidence in the "most probable" forecast is low and in recent years, the track record has been to overstate future runoff suggesting that we should pay equal attention to the "minimum probable" forecast. (See Wang et al, Evaluating the Accuracy of Reclamation's 24-Month Study Lake Powell Projections.) Also remember that the actual decision on the WY 2026 release is not made until the Spring runoff is over and the August 24-Month study is released.

A 7.48 maf release from Glen Canyon Dam bodes trouble for the basin because it takes us very close to the tripwire. Simply put, the tripwire is the ten-year flow at Lee Ferry at which the Lower Division States can claim the Upper Division States are in violation of the 1922 Compact. Under the compact, the Upper Division States have two specific flow obligations at Lee Ferry: (1) to not cause the ten-year flow to be depleted below 75 maf every ten consecutive years, and (2) to deliver one half of the annual delivery to Mexico under the 1944 Treaty if the "surplus" is not sufficient. If there is no surplus and the delivery to Mexico is 1.5 maf/year, the Upper Division's share is 750,000 af/year, resulting in a total ten-year delivery obligation of 82.5 maf. Since under Minute 323, Mexico shares in mainstem shortages, recent annual deliveries have been less than 1.5 maf. To keep the math simple, let's call the current obligation (with no surplus) 82.0 maf. With a 7.48 maf release in WY 2026, the ten-year flow for 2017-2026 will be about 82.8 maf.

Keep in mind that the obligation of the Upper Division States to Mexico under the 1922 Compact has been a disputed issue since the Treaty was signed in 1944. The Lower Division believes there is no current surplus, thus the obligation is one half of the Treaty delivery. The Upper Division believes that since the Lower Basin is currently overusing its 1922 Compact apportionment, this overuse is surplus, and thus, must be delivered to Mexico. Following this thread, if the overuse is greater than 1.5 maf/year, the Upper Division would have no obligation to Mexico.

With an 82.8 maf ten-year flow at the end of Water Year 2026, the Upper Division States are still slightly above the tripwire with a cushion of about 800,000 af. The problem is what happens in the next one to three years. From 2015-2019 annual the annual Glen Canyon Dam releases were 9.0 maf/year. Note, the annual release from Glen Canyon Dam and the flow at Lee Ferry are not quite the same, the Paria River and leakage around the dam contribute another 100,000 -150,000 af/year to the flow at Lee Ferry (bonus flows). Because of the way the ten-year math works, at the end of WY 2027, the Lee Ferry flow for WY 2017 (~9.2 maf) will drop out and be replaced by the 2027 Lee Ferry flow, thus, to keep the ten-year flow greater than 82.0 maf, the 2027 flow will have to be at least 8.4 maf. AND, there are two more 9.0 maf releases in the pipeline, WYs 2018 and 2019! That means that when those drop out of the sequence, the risk of the basin stumbling across the tripwire into litigation grows.

To stay above 82.0 maf, the total deliveries at Lee Ferry for the three-year period of 2027-2029, the annual release from Glen Canyon Dam will have to average about 8.8 maf/year (factoring in the bonus flows). Since 2012, the average unregulated inflow to Lake Powell is about 8.2 maf/year. After deducting 500,000 af/year of gross evaporation from the reservoir, the "net-of-evaporation" annual inflow is only about 7.7 maf/year. Going back to 2000, the net inflow is about 7.9 maf/year. Thus, to avoid going below the 82.0 maf tripwire, it will take either above average (post-2000) hydrology or continuing to draw down Lake Powell levels. If the hydrology is a bit drier than the post-2000 (or post-2012) levels, maintaining at least 82.0 maf/10-year may require drawing Lake Powell below minimum power. As Lake Powell levels approach minimum power, we approach environmental and power generation tripwires.

The fact that we're on track for another year of below-average inflows to Lake Powell, another 7.48 maf/year annual release from Glen Canyon Dam, and on a trajectory to drop below a 1922 Compact tripwire adds another level of urgency for the basin states to break their current impasse over the how the system will be operated post-2026. The chances of Lee Ferry flows dropping below the 82 maf

tripwire are high. The Colorado River Basin needs to be fully prepared before this occurs. With every 24-Month study, the basin's litigation clock is getting closer and closer to that midnight hour.

Section: Stable on the Colorado River: When “good” is not good enough

Source: <https://www.inkstain.net/2025/01/stable-on-the-colorado-river-when-good-is-not-good-enough/>

By John Fleck and Jack Schmidt

Preliminary year-end Colorado River numbers are stark. Total basin-wide storage for the last two years has stabilized, oscillating between 30 and 27 maf (million acre-feet), where storage sits at the start of 2025[1]. That is lower than any sustained period since the River's reservoirs were built (Fig. 1). Stable is better than declining, but we did not succeed in rebuilding reservoir storage during 2024's excellent snowpack but modest inflow. Although reservoir storage significantly increased after the gangbuster 2023 snowmelt year, we have not protected the storage gained in 2024 when inflow to Lake Powell was ~85% of normal from a 130% of normal snowpack. We can't rely on frequent repeats of 2023; we must do better at increasing storage in modest inflow years like 2024.

Why is this happening?

The phrase “the new normal” can be misleading, suggesting a new, more stable state for the climate. It's not gonna be stable. But by one reasonable measure – total estimated natural flow in the Colorado River at Lees Ferry – Calendar Year 2024 was typical of the first quarter of the 21st century, with a preliminary estimate of 12.1 million acre-feet “natural flow.” Thus, the calendar year average annual natural flow at Lees Ferry between 2000 and 2024 has been 12.4 maf/yr, down from 14.3 maf for the period 1930-1999. An additional 770,000 af/yr in side inflows between Lees Ferry and Lake Mead add to the available water supply[2].

That we made the cuts needed to stabilize reservoir levels with a natural flow at Lees Ferry as low as 12.1maf would have been a substantial achievement in the wetter “before times.” Now, it's table stakes. The most important point is that we absolutely did not rebuild storage in 2024, despite a 130 percent snowpack. We must do better in reducing total basin consumptive use.

Once again in 2024, we saw substantial water use reductions among the states of the Lower Colorado River Basin. Total U.S. Lower Basin main stem use of 6.08 maf is the lowest since 1985 (meaning the lowest since the Central Arizona Project came on line). California's use, based on preliminary numbers published by Reclamation seems to be the lowest since 1950, and use by the Imperial Irrigation District seems to be the absolute lowest in a dataset that goes back to 1941. These are important achievements, to be celebrated.

With regard to the other two major U.S. areas of use – Lower Basin tributaries and the Upper Basin as a whole – we have no idea what 2024 consumptive use was. This is a problem. Lower Basin main stem use is quantified through Reclamation's annual accounting reports and reported on a nearly daily basis during the course of the water year. River flows and reservoir levels across the basin are similarly reported in public, transparent ways. That's how we're able to provide the data you see above. Anyone can download and crunch the numbers. The general public can't readily do that for consumptive use in the Upper Basin or Lower Basin tributaries.

As Elinor Ostrom noted in her classic book *Governing the Commons*, shared understanding of the resource is crucial to successful water management. Increasingly, areas of uncertainty have become contested ground, as the genuine technical uncertainties collide with the motivated reasoning of political actors across the basin.

With respect to the Upper Basin, we note that the rhetoric that Upper Basin water users suffer shortages in dry years has shifted to a broader claim that Upper Basin users always suffer shortages. We quote here from the Upper Basin states' January 2 press release: "There are acute hydrologic shortages in the Upper Basin every year – there simply isn't enough water in any year to satisfy current needs in the Upper Basin every year. The Upper Basin has made uncompensated cuts to their water users every year for the past 24 years." Some of the data to support this assertion was presented at the December 2024 UCRC meeting, and we look forward to a more complete and transparent accounting of these data, because these data are crucial to a robust Colorado River management discussion. The Upper Basin's experience of "acute hydrologic shortages ... every year" is exactly what John Wesley Powell described in 1878 in the first edition of *The Arid Lands Report*. Nothing has changed, and the challenge of agriculture throughout the watershed has been well known for 150 years. We also note that consumptive use data throughout the basin has not been integrated with the important findings of Richter et al (2024) who documented the proportion of water used by different agricultural sectors. They estimated that 55% of all Colorado River water use supplies livestock feed.

We leave a discussion of Lower Basin tributary use for another post but note that in both the cases of the Upper Basin use and Lower Basin tributary use, the numbers are entangled in the current Upper Basin-Lower Basin feud, which makes serious efforts to think about how to manage water at the Basin scale, rather than simply defending parochial interests, much more difficult. It is important that the general public not employed by a state or water agency, and therefore not beholden to local parochial interests, help the basin community as a whole navigate these technical issues.

Conclusion

The stable reservoir levels at the end of 2024, despite another year of deep Lower Basin water use reductions, should be cause for alarm. Deeper cuts are needed. But without a shared understanding of water use elsewhere in the basin, we're flying blind.

[1] Basin-wide reservoir storage reached a peak of 29.7 maf on 13 July 2023 and was subsequently drawn down to 27.5 maf by mid-April 2024. Inflow from 2024 snowmelt rebounded basin-wide storage to 30.0 maf on 6 July 2024, and storage was subsequently drawn down to 27.4 maf by 31 December 2024. Retention of storage in Lake Mead and Lake Powell has been somewhat better during the same period. Combined storage in Lake Mead and Lake Powell peaked in mid-July 2023 at 18.0 maf, declined to 17.1 maf by mid-May 2024, increased to 18.5 maf on 8 July 2024, and was 17.3 maf on 31 December 2024. Thus, storage in the two largest reservoirs at year's end was slightly greater than it was at its spring 2024 minimum just before storage increased when significant snowmelt reached Lake Powell.

[2] This estimate is calculated as the difference between annual flow measured just upstream from Diamond Creek in western Grand Canyon and measured at Lees Ferry.

Section: Water for Navajo is the latest victim of Colorado River Basin governance dysfunction

Source: <https://www.inkstain.net/2025/01/water-for-navajo-is-the-latest-victim-of-colorado-river-basin-governance-dysfunction/>

Winters rights are no match for the current dysfunction of Colorado River Basin governance.

Shannon Mullane at the Colorado Sun has been on this, and last week had some useful details:

Advocates of a deal to secure reliable water for thousands of tribal members in Arizona raced to win Congressional approval until the final hours of the session in December.

They didn't make it.

"We just ran out of time to address all the issues," said Tom Buschatzke, director of the Arizona Department of Water Resources and principal negotiator for the state on Colorado River matters.

Part of the catch is the cost. This is a really expensive settlement, though it's reasonable to point out all the other expensive water projects built over the last century with massive taxpayer subsidy to provide water to non-Indian communities. But, as Mullane (thanks, Shannon!) reported back in December, one of the main hitches is the way the deal is entangled in broader Colorado River Basin uncertainties about who can move what water where:

State officials have asked for clarity on how water will move across state lines. And Colorado River officials in Colorado, New Mexico, Utah and Wyoming have concerns about how the settlement would allow water from their basin to be used farther downstream.

Finger-pointing for the failure to work out an agreement is appropriate, but ultimately doesn't matter. The bottom line here is that the failure of the Colorado River governance community as a whole to work out these details means some of the most water-poor communities in the nation continue to not have water, while the people blocking the agreement do have water.

Let me say that twice, for emphasis: Communities that do have water are blocking access to water for their neighbors who don't have water.

That's not OK.

A functional Colorado River governance community would have been able to solve this. Its failure, rooted in parochial attempts by communities to protect their own water supply, is a concrete demonstration of how hollow talk about Tribal inclusion really is.

Do better.

Section: Can "Floating Pools" be the template for future management of the Colorado River?

Source:

<https://www.inkstain.net/2025/01/can-floating-pools-be-the-template-for-future-management-of-the-colorado-river/>

By Eric Kuhn and Jack Schmidt

The press coverage of the December 2024 Colorado River Water Users Association (CRWUA) meeting mostly focused on the ongoing stalemate between representatives of the Upper and Lower Division States over their competing proposals for how the Colorado River Systems' big reservoirs will be operated after the 2007 Interim Guidelines terminate in 2026. The headlines included words such as "turbulent", "bitter", "bluster", and "spar". Indeed, there was tension in the air, and the potential for interstate litigation was a topic of much discussion both on the formal agenda and in the hallways where, traditionally, progress is often made between competing interests.

While the press focus on the tension and divisiveness was unavoidable, I believe that there were good reasons for some guarded optimism.

For the ongoing effort to renegotiate the post-2026 operating guidelines, a consortium of seven environmental NGOs has also made a detailed proposal. Their proposal is referred to as the "Cooperative Conservation" proposal. One of the four action alternatives that Reclamation will analyze, Alternative #3, is patterned after the NGO submittal. At CRWUA, John Berggren of Western Resource Advocates, who along with Jennifer Pitt and others prepared the proposal, made a presentation on the proposal. Like the other submitted proposals, the cooperative conservation alternative proposes sophisticated operational rules for Lakes Mead and Powell based on combined system storage and actual hydrology. Where the Cooperative Conservation proposal breaks new ground is the concept of a

Conservation Reserve Pool, and this idea could lead the basin toward a practical on-the-ground solution. Indeed, the Gila River Indian Community introduced at CRWUA a similar concept in the form of a Federal Protection Pool made up of stored water in both Lake Powell and Lake Mead. These proposals, taken separately, together, or in some combined and moderated form, might serve as a catalyst for compromise.

As proposed, both the Conservation Reserve Pool and the Federal Protection Pool would be filled with water conserved by reductions in consumptive use and perhaps augmentation from programs in both basins and this water could be stored anywhere in the system. This water would be “operationally neutral” and thus invisible to the underlying system management operating rules. From an accounting perspective, this Pool would “float” above other water in the reservoirs. Floating Pools operate separately from and above the prior appropriation system of water allocation on the Lower River and are invisible to the rules that dictate annual releases from Glen Canyon Dam. Thus, these proposals impart important operational flexibility. In many ways, Floating Pools split the baby—they incentivize innovative conservation measures that allow participants to find value they would not have been able to realize under the prior appropriation system—yet they insulate the prior appropriation system and thus are more protective of higher-priority water users than operationally non-neutral ICS. It’s a stretch to say there is something here for everyone, but there may be enough to kick-start otherwise stalled conversations.

In their proposal, the Lower Division States have offered to take up to 1.5 maf/year of mainstem shortages. Where the two basins remain deadlocked is what happens in those years when shortages exceed the amount the Lower Division States are willing to accept. The Lower Division States have proposed that the two basins share the additional required shortages up to a maximum shortage of 3.9 maf/year. The Upper Division States have said, “No, because we already suffer large hydrologic shortages in dry years, and we have not used our full compact entitlement; the Lower Division should cover all of the shortages.” In their presentation, however, the Upper Division Commissioners (UCRC members) left the door open for continuing discussions between the two divisions. In his remarks, New Mexico Commissioner Estevan Lopez stated that under what he referred to as “parallel activities”, the Upper Division States might be willing to discuss conserving “100,000, maybe 200,000 acre-feet per year.”

Water in Floating Pools could be used for a variety of purposes including environmental management, fostering binational programs, and supplementing scheduled water deliveries. During his CRWUA presentation, John Berggren mentioned an obvious use for this pool. Water stored in the Pool by conserved consumptive use programs in the Upper Division States could be used as an Upper Division contribution during years when mainstem shortages to the Lower Division States exceed a negotiated amount. Of course, the Lower Basin is unlikely to accept Upper Basin creation of Floating Pools made up of water for which there is no current consumptive use. This water is already “system water” and is now being used by existing Lower Basin water agency. Thus, it would be necessary to develop a program to account for and certify savings in the Upper Basin. Further, the thorny problem of shepherding (legally protecting the conserved water so that it ends up in system storage) needs to be overcome. For a perspective on this issue, see Heather Sacket. Undeveloped Tribal water is a controversial sticking-point in this regard, with strong feelings and strong arguments on all sides.

If the Upper Division States were to conserve 200,000 acre-feet per year for five years and deposit that saved water in a conservation reserve “Floating Pool”, something like 900,000 acre-feet could be available for shortage sharing (after accounting for reservoir evaporation). (We use 900,000 af as an example only, how much water the Upper Division States would have to contribute and maintain in a Floating Pool would have to be negotiated between the two divisions.) In their presentation, the Lower Division principals pointed out that had their proposal been in place beginning in 2007, there has yet to be a year when shortage sharing would have been required. Note, this conclusion is very sensitive to “initial conditions.” In 2007, total storage in Lake Mead and Lake Powell was about 8 maf more than it is today. If the 21st century hydrology continues, shortages greater than 1.5 maf/year are likely to occur.

What would the Upper Division States get in return? During the term of the new post-2026 operating guidelines (which we all assume will also be “interim”), the Upper Division would benefit by the Lower Division agreeing to remove the threat of litigation over a “compact call.” For a perspective on the potential impacts of a “call” in Colorado see *The Risks and Potential Impacts of a Colorado River Compact Curtailment on Colorado River In-Basin and Transmountain Water Rights Within Colorado*.

Carefully crafted with appropriate guardrails, Floating Pool concepts can be a catalyst for compromise between the two divisions that give both parties something they need.

How do Floating Pool alternatives fit with the Schmidt, Kuhn, Fleck management approach? Based on our conversations with the authors of the cooperative conservation proposal, we believe the two approaches agree — that our management proposal fits on top of and complements their proposal quite well. In my presentation at CRWUA, I emphasized that, like future hydrology, there is great uncertainty in the future needs of the river’s ecosystem and society’s values. It’s almost a certainty that in the future, prescribed annual releases from Glen Canyon Dam will cause an unacceptable and unanticipated outcome to some river or reservoir resource. When that happens, our flexible management approach and accounting system keeps the basins “whole.”

Is using the concept of Floating Pools as a catalyst to break the stalemate between the two basins without warts? – of course not. There are important considerations regarding the use of undeveloped water—Tribal or otherwise, and the devil is in the details when it comes to developing appropriate guardrails for annual and total accumulation in such a Pool, the number and type of participants, annual debits, and other important qualifications. Even conserving 100,000 acre-feet per year in the Upper Division States, with acceptable verification, could be a stretch, especially if there is less federal money in the future, as there almost certainly will be. Finally, it might put off addressing fundamental problems with the law of the river until the new post-2026 operating rules again expire. When they do, the 1922 Compact and 1944 Treaty with Mexico will still be in place, and these agreements collectively allocate 17.5 maf/year of consumptive use on a river that is only producing 13-13.5 maf/year of water at the international boundary (and runoff continues to decline). What the Floating Pool concept might accomplish is to significantly reduce the temptation and threat of unpredictable interstate litigation, keep the basin’s stakeholders talking to each other, and give us time to move toward more foundational change in how the river is managed.

Section: Inkstain 2024 – y’all really like Eric Kuhn’s writing

Source: <https://www.inkstain.net/2024/12/inkstain-2024-yall-really-like-eric-kuhns-writing/>

If traffic to this blog is an indicator, y’all are really interested in thoughtful independent analysis of the standard talking points in the Colorado River negotiations. Also, the work of my longtime collaborator Eric Kuhn (which is unsurprising, I’ve long been really interested in Eric’s work).

Three of the four most-read Inkstain posts this year were Eric’s (one of the three a double-byline piece with me):

- No Simple Disputes, Eric’s explanation of the problems posed by the failure to define “consumptive use,” and how that failure weakens the Upper Basin’s argument about Lower Basin tributary use. (Eric’s version is wonky, I came back a day later to de-wonkify it for Inkstain readers, though Eric’s technical version got way more reads.)
- The Compact Tripwire, a post Eric and I wrote together on the risk of total Upper Basin deliveries past Lee Ferry dropping below 82.5 million acre feet over a ten-year period, and the risk of Compact litigation that sets up.
- What happens if there is no agreement on post-2026 Colorado River Management is Eric’s dive into the rules that would apply if we don’t have any agreement or federal intervention – Eric’s version of the

“no action alternative.”

(The fourth was a weird riff on the Russian anarchist thinker from the late 1800s-early 1900s Peter Kropotkin, a pair of wire cutters, and a pedestrian alley behind the gas station by my house. It had a bit of a viral moment, though my efforts to think through the connection between Kropotkin and Colorado River governance failed.)

Section: Lousy start to the Colorado/Rio Grande 2024-25 snowpack season

Source: <https://www.inkstain.net/2024/12/lousy-start-to-the-colorado-rio-grande-2024-25-snowpack-season/>

I was talking to Eric Kuhn Thursday (write a book together – bonded for life) who pointed out that the Colorado Basin River Forecast Center has started running its models for 2025 runoff. They don't look good.

It's way too early to think of this as a “forecast.” But they provide a feel for where we're at now: Do we have a good head start? Are we already behind? The error bars are still huge, with a lot of upside potential, but we are already behind – 1.4 million acre feet below median for Lake Powell inflow.

The current climate forecast headlights, which can at least dimly illuminate the next month for us, don't look good. The US Drought Monitor folks publish an experimental forecast tool called the Evaporative Demand Drought Index (EDDI).

EDDI uses the federal Climate Forecast System model, an operational model to help gauge conditions over the coming months. CFS is then coupled with tools to estimate evaporative demand – not simply how much snow we're going to get, but how rain and snow interact with temperature and atmospheric moisture, all of which play roles in the system that sends water from the snowpack in the Rockies to headgates and kitchen taps across the West.

EDDI says that over the next month, we should expect the CBRFC's runoff forecast to go down, not up. We're falling behind.

Why this matters

The obvious reason this matters is its direct relationship with this year's water management. Will Powell and Mead go up or down? What does that mean for near term water supply?

But we're also all playing multiple games of four-dimensional chess trying to anticipate how the near term runoff scenarios influence long term negotiations over Post-26 river management. One of my little projects right now is to step back from my normative angst (where “normative angst” == John's super pissed off about the negotiators' abject failure) to think about the deeper negotiation theory stuff going on.

Section: White Christmas somewhere but maybe not here

Source: <https://www.inkstain.net/2024/12/white-christmas-somewhere-but-maybe-not-here/>

It's that time of year when I'm looking for a storm a week to build a snowpack. More is better. Zero is not better.

It's that time of year when I'm looking for a storm a week to build a snowpack. More is better. Zero is not better.

New Normal™:

Changes in store for atmospheric rivers : Warming climate will affect the storms differently in California, Pacific Northwest [National Center for Atmospheric Research/University Corporation for Atmospheric Research 12/19/24] <https://www.eurekalert.org/news-releases/1068888>

Section: Do we linger less?

Source: <https://www.inkstain.net/2024/12/do-we-linger-less/>

Research comparing pedestrian behavior in Bryant Park and outside the Met, 1979-1980 versus 2008-2010:

The biggest change in behavior was that lingering fell dramatically. The amount of time spent just hanging out dropped by about half across the measured locations.

I personally am a fan of “lingering” and “just hanging out.”

Via Tyler Cowen

Section: A sign of hope on the Colorado River

Source: <https://www.inkstain.net/2024/12/a-sign-of-hope-on-the-colorado-river/>

One of the hopeful notes coming out of the recent Colorado River discussions is the way the operation of Glen Canyon Dam in a more flexible way, to accommodate a broader range of values, is back on the table. The USBR alternatives released ahead of this week’s Colorado River Water Users Association, while requiring some tea leaf divination because of their brevity, seem to leave the door open for this discussion.

Jack Schmidt and I have a new white paper offering some assistance, based on our understanding of the legal and regulatory structure around Grand Canyon National Park and Glen Canyon National Recreation Area. The idea behind what we’re arguing isn’t to wag a regulatory finger and say, “The law requires us to do X.” Rather, we’re saying, “The law enables us to do X,” where for “X” we argue for the consideration of a wider range of social, cultural, and environmental values as we make decisions about how to divide the water up between Lake Mead and Lake Powell.

The resources in each part of this spectacular region are unique. Cataract Canyon was known to river runners as the “graveyard of the Colorado” because of its challenging rapids. Glen Canyon held sublime beauty amid the tranquility of its countless side canyons until they were inundated, and those side canyons are now reemerging. Grand Canyon is one of Earth’s greatest geological statements and is sacred to many native American tribes. In the eyes of the Bureau of Reclamation in the 1960s, and to millions of visitors today, Lake Powell is the “Jewel of the Colorado.” Lake Mead, visited by four million people each year, is the largest reservoir in the United States and allowed transformation of much of the American Southwest.

In 2025 we have an opportunity to establish policies for the Colorado River that honor these diverse and evolving environmental and societal values. The legal and administrative foundations that guide river and reservoir management in Cataract Canyon/Lake Powell/Glen Canyon/Grand Canyon/Lake Mead appear to encourage, and perhaps even require,

consideration of the many environmental and societal resources throughout this diverse region.

Section: Deficit Spending on the Colorado River

Source: <https://www.inkstain.net/2024/12/deficit-spending-on-the-colorado-river/>

By Jack Schmidt | December 3, 2024

Drawdown of the Colorado River's reservoirs now slightly exceeds the amount of gain that occurred during the 2024 snowmelt season. For the next four months until snowmelt begins again, the basin's reservoirs will be drawing from the excess accumulated in 2023, demonstrating the immense challenge in balancing water consumption with supply.

In Detail...

On 30 November 2024, total basin reservoir storage was 27.5 million af (acre feet)¹, approximately two years' supply at today's rate of consumptive use and loss (Fig. 1). That amount is 43% of the maximum system contents of July 1983² and is the same amount as at the beginning of July 2021 when the basin's water managers were beginning to get worried. Conditions are not quite as bleak as in summer 2021, because that year's snowmelt season had already passed. Now, we can hope that the 2025 snowmelt season might be a good one. Nevertheless, reservoir storage is the bank account from which we draw to maintain the economy of the American Southwest and parts of northwestern Mexico. It would be preferable for there to be more water in that account.

Figure 1. Graph showing total storage in 46 reservoirs (blue line) in the Colorado River basin since 1 January 1999. Also shown are the total contents of Lake Mead and Lake Powell (orange line), total contents of Lake Mohave and Lake Havasu (red line), and 42 reservoirs upstream from Lake Powell (green line) which includes reservoirs managed by the federal and state governments, municipalities, and water districts.

Approximately 63% of current total reservoir storage is in Lake Mead and Lake Powell. Presently, there is approximately 400,000 af more water in Lake Powell than in Lake Mead, but the contents of Lake Powell are slowly being depleted. The contents of Lake Mead held fairly constant during the past month. The contents of Lake Powell decreased by approximately 4300 af/day during November, but the contents of Lake Mead decreased by only 800 af/day (Fig. 2). Upstream from Lake Powell, Colorado River Storage Project (CRSP) initial unit reservoirs, as well as Fontenelle Reservoir, decreased by only 600 af/day in November, and other Upper Basin reservoirs lost even less (400 af/day).

Figure 2. Graph showing total basin reservoir storage (blue line), and storage in different parts of the Colorado River watershed between 1 January 2021 and 30 November 2024.

Basin reservoir storage must be increased to improve the security of our water supply. We need to increase the balance in our "bank account," and the only way to do that is to spend less than the amount of our actual water "income." Most of our income arrives during the snowmelt season of late spring and early summer. Mid- and late-summer, fall, winter, and early spring is the period when we spend the snowmelt-season income, although summer rains and groundwater inflow offset some of our uses.

Occasionally, we have an unusually snowy winter, and the basin's reservoirs significantly refill. 2023 was one of those years. Reclamation estimates that the natural flow of the Upper Basin³ was 17.4 million af in 2023, the third largest of the 21st century (after 2011 and 2019), and total basin storage increased by 8.38 million af, only exceeded by the increase in storage in 2011⁴. 2024 was a moderately snowy winter.

2024 was a different story, however. The NRCS estimated that the peak snow water content in 2024 was 14% greater than the 30-year average, but dry soils and other effects of a warming climate limited natural flows to between 11.9 and 12.1 million af, which is less than the average for the 21st century⁵. In 2024, the basin's reservoirs increased in storage by 2.45 million af. The drawdown of the basin's reservoirs as of 30 November was 2.46 million af, slightly more than the gain from snowmelt (Fig. 3). The contents of Lake Mead and Lake Powell increased by 1.39 million af in 2024, and the drawdown in those two reservoirs has been 1.07 million af this year. During the next four months, the basin will begin drawing from storage that accumulated in 2023.

Figure 3. Graph showing reservoir storage between 1 January 2023 and 30 November 2024, highlighting the amount of reservoir recovery during the past two snowmelt seasons and the amount of intervening reservoir drawdown. The drawdown of the basin's reservoirs since July 2024 slightly exceeds the recovery that occurred due to snowmelt in 2024.

Drawdown of the basin's reservoirs has been much greater in 2024 than in 2023. The amount of drawdown between early summer and today is slightly more than the median drawdown for the past 15 years⁶ and is 43% greater than the drawdown at this time last year (Table 1). The drawdown of Lake Mead and Lake Powell in 2024 is slightly less than the median for the past 15 years⁷ but is 98% greater than it was at this time last year.

The total reservoir drawdown between early summer and 30 November is now 14% greater than in all of last year, and drawdown in Mead and Powell also exceeds the total drawdown in those reservoirs last year (Table 2). We did well last year, but not so well this year.

There are many details ignored in this overview. Reservoir drawdown in the Upper Basin is not only determined by consumptive use, but also by reservoir operating rules that require winter drawdown and by requirements to provide environmental flows. Water use in southern California is significantly affected by water supply available from northern California, the Owens River, and locally. Nevertheless, every drop of water released from upstream is used, lost, or stored in a downstream reservoir, and total basin storage is the only available supply to make up the shortfall between annual precipitation and annual use.

Conservation in the Lower Basin and in Mexico is reducing drawdown in Lake Mead, and the storage contents of Lake Mead are likely to increase during the next few months as water is delivered from upstream. Drawdown of the total contents of Lake Mead and Lake Powell is still 0.32 million af less than what accumulated there from the 2024 inflow season and a 2024 deficit might not occur if Lower Basin water use is drastically reduced or if Upper Basin reservoirs are emptied.

Efforts to date to reduce water consumption in the basin have been significant, and required a significant investment by the federal government. Despite those efforts, we have four months ahead of us before snowmelt in 2025 begins, and we are likely to begin deficit spending unless radical changes in use are immediately implemented. The challenge faced by the federal government, Mexico, the seven basin states, every tribe, and every water user is immense and is not solely restricted to negotiating the post-2026 agreements. We remain in a water crisis today, and the time to greatly reduce water consumption is right now in the present moment.

- [1] Basin reservoir storage is for 46 reservoirs reported by the Bureau of Reclamation in its Hydrodata base https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html.

- [2] There was 63.6 million af of storage in the basin on 15 July 1983. Some of this storage exceeded the generally accepted capacity of some reservoirs, notably Lake Powell.

- [3] at Lees Ferry

- [4] Basin reservoir storage increased by 8.78 million af in 2011.

- [5] Natural flows for calendar year and water year 2024 were 12.1 and 11.9 million af, respectively, based on Reclamation's 12 September 2024 estimate. The average natural flow at Lees Ferry between 2000 and 2024 was 12.4 million af/yr, based on Reclamation's estimates.

- [6] The median drawdown between the summer peak and 30 November during the past 15 years was 2.26 million af for the 46 reservoirs of the watershed.

- [7] The median drawdown between the summer peak and 30 November during the past 15 years was 1.16 million af for the total contents of Lake Mead and Lake Powell.

- [8] Includes drawdown of Lake Mohave and Lake Havasu.

Section: Does our current approach to Colorado River accounting hide a looming problem?

Source:

<https://www.inkstain.net/2024/12/does-our-current-approach-to-colorado-river-accounting-hide-a-looming-problem/>

My colleagues with the Colorado River Research Group have a new policy brief out today taking another whack at the question of “assigned water” – water kinda sorta conserved, but left in storage so water agencies can pull it out again at some future date. Think “Intentionally Created Surplus” (ICS). At this point, nearly 40 percent of the water in Lake Mead is tagged as some agency’s private storage account, rather than being available for general system use.

By so effectively propping up reservoir elevations, Assigned Water delays or completely prevents the triggering of some mandatory shortage-based curtailments spelled out in the current rules.... [T]his approach has the unintended consequence of hiding the current paucity of shared (i.e., System) water available in Lake Mead, something that will ultimately become obvious when those waters are someday recalled.

This is the issue Arizona State’s Kathryn Sorensen (one of my CRRG colleagues) has been raising, and that Kathryn (with help from Sarah Porter and I) wrote about in October. The new CRRG paper argues that, as we move toward expanding the assigned water programs available to basin water users, we need to be mindful of the risks.

Section: A minor blog redesign, which is a metaphor

Source: <https://www.inkstain.net/2024/12/a-minor-blog-redesign-which-is-a-metaphor/>

As near as I can tell from a lazy visit to the Wayback Machine, it was sometime around 2014 that I added the old Metropolitan Water District Colorado River Aqueduct map to my blog header. It’s a lovely map that has served me well these many years, but it doesn’t match my brand any more.

I was playing today with this 1823 map by Jose Narvaes (Narvaez?), but it didn’t pop the way I need a blog header image to pop. So I went instead with the 1938 USGS topo map of Albuquerque, which is gorgeous. Look at the loving detail of the Rio Grande as it makes the bend past “Old Albuquerque,” the Atrisco and Armijo ditches meandering down the South Valley, the swamps flanking the river.

I’m in the middle of some final choices about the art to include with the revisions to the manuscript of the new book Bob Berrens and I are writing – Ribbons of Green: The Rio Grande and the Making of a Modern American City, and I’ve been staring at both the Narvaes/Narvaez and the 1938 topo, trying to decide whether to include either. Narvaes/Narvaez I probably won’t. It’s lovely, but doesn’t move our narrative. The 1938 topo is on the bubble. It’s so damn pretty!

Finishing the Manuscript

I’ve been holed up for the last six weeks in a third floor garret (metaphor alert) doing the final revisions on the book, which is a full brain activity. I haven’t been paying much attention to the world around me, and I came down to the corner bodega (metaphor) to get some bread and cheese and ask the newsboy what’s been happening. He pointed me to Tony Davis’s story from the Tucson Star last week.* What the actual fuck, y’all? Is this the best you can do? I am so profoundly disappointed in Colorado River Basin leadership right now. Will I lose blog readership if I say “fuck” again?

The network is failing. Yeah, I’m talking to you. You are failing us. There are 40 million of us, and we have to figure out how to share this river.

Do better.

Another semester done

Luckily I'm not going to CRWUA this year. It's early, conflicts with the last week of the UNM Water Resources Program class I'm co-teaching with Bob Berrens. It has once again been a joy to get to know a new cohort of the water managers of the future – a dozen joys, the years Bob and I have spent teaching the class together.

Spending the last three months teaching the class while I'm simultaneously deep in book revisions has been fun because of the interplay. Central to both is the idea that we have a multiplicity of values around water. Good water management does not mean pursuing my values. It means recognizing that there are many communities, and a multiplicity of values, and the institutions we have created to manage our way through that diversity, and whose evolution we are entrusted to steward, matter. I guess this has become my "meta-value." I have my own personal values about the where and when and how of the use of water. But what is more important to me is that we have institutions that reflect the breadth of communities and values and interests other than mine.

* Sorry, I know it's behind a paywall, but a) I appreciate the Star's efforts to figure out a way to pay Tony to continue to do the important work of journalism, and b) the people I'm talking to here all have the pdf.

Section: Clouds and Mountains

Source: <https://www.inkstain.net/2024/11/clouds-and-mountains/>

Clouds and mountains were dancing this morning. I think they're in love.

2 Comments

-

"Never forget: we walk on hell, gazing at flowers." (Kobayashi Issa 1763~1828) "Unexplained heat-wave 'hotspots' are popping up across the globe : So extreme, they cannot be explained by global warming models" Columbia Climate School, November 26, 2024

<https://www.eurekalert.org/news-releases/1066249> "...Areas of Texas and New Mexico appear on the map, though they are not at the most extreme end."

"Global emergence of regional heatwave hotspots outpaces climate model simulations." Proceedings of the National Academy of Sciences of the United States of America (PNAS) November 26, 2024
<https://www.pnas.org/doi/10.1073/pnas.2411258121>

-

See also: "Warming has accelerated due to the Earth absorbing more sunlight : If it's a trend, then future warming will be at the high end of estimates." <https://arstechnica.com/science/2024/12/new-study-blames-recent-spike-in-global-warming-on-lack-of-low-clouds/>

Section: The alley behind Aldo Leopold's house

Source: <https://www.inkstain.net/2024/11/the-alley-behind-aldo-leopolds-house/>

There's an alley off Albuquerque's Central Avenue, old Route 66, between the Southwest Capital Bank and St. John's Thrift Store. You can't go down the alley on Google Street View. Google Street View mostly doesn't go down alleys. Alleys mostly don't have names. You have to go there for yourself.

Down past the “Drug Free Zone: Purchase or Sale of Drugs in this Area is a Felony” sign, past the house with the backyard pond with the lovely sound of tinkling water, past rogue bamboo, past the shed with the big sign that reads “Bicyclist” above the door, you’ll get to Aldo Leopold’s house.

In her book *How to do Nothing*, a meditation on the attention economy (thanks R!), the writer and artist Jenny Odell talks about “the ability not just to withdraw attention, but to invest it somewhere else, to enlarge and proliferate it, to improve its acuity.”

I find alleys to be excellent tools in this pursuit.

Arroyo Mascaras

On the south bank of the Arroyo Mascaras in Santa Fe, New Mexico, is a foot path that you get to by crossing a pedestrian bridge from the old De Vargas Mall.

L and I had rented a casita a few blocks to the south last winter, taking a break. I packed my foldable bicycle, and we both brought piles of books. We had no plans other than the suggestions offered by the things we had carried.

It snowed.

I found my way to the Arroyo Mascaras path by accident, entering what historian and photographer John Stilgoe calls a “secret corridor” – one strand of the web of paths, formal and informal, connecting parking lots and slipping through back alleys of a city.

The secret corridor proves an amazing fast shortcut to any explorer determined to brush aside low-hanging branches and to risk broken glass and rusted wire. And it offers the probing and poking explorer another view of the chrome-and-glitter commercial strip, even of the regional mall, for it makes clear the stealth with which change comes.

From the new path that opened before me, I could look back on the concrete wall bordering the arroyo, elegantly tagged by Santa Fe’s graffiti artists.

In 2013, the municipal government of Bogota, Colombia approved “responsible and artistic graffiti” in public spaces, and promoted state-sponsored murals. This prompted debates about the meaning of the art form, so rooted in a culture of oppositional place-making. It was the anthropologist Mary Douglas who defined “dirt” as “matter out of place.” This is foundational in understanding graffiti. Its meaning arises from the tension between those who think it doesn’t belong there, and those who think it does.

Stilgoe is a fan of secret corridors because you see stuff like this, and think stuff like this.

Alleys are great places to think about matter out of place.

The scrawling on the walls became more chaotic the deeper we moved into the tunnel. The unspoken rules that once governed canvas boundaries no longer apply; borders have collapsed, and pieces are free to exist in any patch of grey concrete their creators can find, or even on top of another artist’s work. It raises several questions: How can spatial freedom co-exist with respect for another person’s space? Is seniority alone a worthy basis for earning somebody’s respect?

That’s from Aaron Cantuu’s 2018 piece in the Santa Fe Reporter. Cantuu found his way to the Arroyo Mascaras too.

Aldo Leopold’s Albuquerque

When Aldo Leopold lived on 14th street in Albuquerque in the teens and 1920s, there was no alley there. Aldo’s house was the last one on what was then the edge of town, an alkali swamp extending from the Leopold back door to the Rio Grande three quarters of a mile away.

This is where Aldo was living as he developed his pioneering ideas about wilderness. This is also where Aldo was living when he helped create the institutions we needed to build levees to protect his home from flooding, and to drain the swamp. That swamp – today we would call it a wetland, or

perhaps a wilderness? – is gone.

The old Leopold house has four steps up in front, built at a slight elevation to protect it from flooding. The next street over, the houses built later don't have the steps up. No need. Drainage and flood control made it easier to build a city here.

Aldo Leopold's Morning Commute

I like to imagine Aldo Leopold walking to work at the old Commercial Club building, past the Stern Apartments and Temple Albert, past the auto showroom and repair shop on Fifth Street, past the Post Office. In 1919, the Commercial Club building had a beauty parlor in one corner, and at one point was home to a library. If Aldo rode his bike to work (consider this speculative fiction), there was a bike shop across the street, a half block in the direction of the Alvarado Hotel.

The Alvarado Hotel is no more, replaced in the 21st century with an architectural facsimile that houses the city bus hub, which is how I got down there in the first place. I love to ride the bus for the same reason I love to walk: It is a perfect way to meet the world on its own terms.

Section: Tools for better environmental adaptation as we manage the Colorado River

Source:

<https://www.inkstain.net/2024/11/tools-for-better-environmental-adaptation-as-we-manage-the-colorado-river/>

I put up a slide for my University of New Mexico water resources graduate students during class yesterday afternoon with two pictures – the emerging canyons at the upper end of Lake Powell, and a smallmouth bass.

When Lake Powell gets low, we get a) the remarkable emergence of Cataract Canyon, and b) warm water invasive smallmouth bass sneaking through Glen Canyon Dam's outlets, headed downstream to dine on the endangered humpback chub. My University of New Mexico colleagues and collaborators Benjamin Jones and Bob Berrens famously dubbed these "green-vs-green" tradeoffs:

The Anthropocene epoch is characterized by extensive interactions among natural systems and managed human systems and institutional arrangements. Complex coupled human and natural systems, such as river systems with managed flows intended to produce a mix of community, ecosystem, energy, water and recreational services, will necessarily implicate multiple dimensions of societal value....

Managing for one – keeping Lake Powell high to keep smallmouth bass out of the Grand Canyon – inevitably conflicts with the other – keeping Lake Powell low to protect the emerging environmental values of Cataract Canyon.

In a new white paper out today, my colleagues Jack Schmidt, Eric Kuhn, and I argue for the creation of a process to better incorporate and manage the multiplicity of values along the Cataract Canyon/Lake Powell/Glen

Canyon/Grand Canyon/Lake Mead stretch of the Colorado River as we develop new post-2026 river operating guidelines. We recognize that keeping water flowing to taps and headgates across the Colorado River Basin is the primary motivation behind the new operating guidelines being developed by the Bureau of Reclamation. We argue that, as the community is writing those rules, we have an opportunity to incorporate a broader set of community values.

In particular, we argue that more creative water accounting methods would allow water to be either held upstream in Lake Powell for later delivery, or send downstream early to Lake Mead, in order to better take into account what Benjamin and Bob called the "multiple dimensions of societal value."

The white paper elaborates on our formal proposal submitted in March to Reclamation as part of the agency's Post-2026 decision process.

Section: Arizona's 50,000 acre-feet of Upper Basin water has always been destined for tribal use

Source: <https://www.inkstain.net/2024/11/arizonas-50000-acre-feet-of-upper-basin-water-has-always-been-destined-for-tribal-use/>

Eric Kuhn, Rin Tara, John Fleck

The pending Northeastern Arizona Indian Water Rights Settlement Agreement settles Navajo Nation, Hopi Tribe, and San Juan Southern Paiute Tribe claims to the Upper Colorado River Basin in Arizona. To do so, Arizona's 50,000 AF entitlement of Upper Colorado River Basin water will be allocated.

Although Arizona's testimony during the ratification of the 1948 Upper Colorado River Basin Compact indicated that Arizona's cut would be used for tribes, Arizona fashioned the deal to benefit the Central Arizona Project. Charles A. Carson, Arizona's Upper Basin Compact Commissioner, originally requested 136,200 AF/yr for Arizona in the negotiation but ultimately accepted 50,000 AF/yr in the interest of sweetening the deal for the rest of the states to sign on to stream depletion theory as the means for measuring system use. Stream depletion theory, under Arizona's interpretation of the 1922 Colorado River Compact allowed Arizona to consume two million acre-feet per year on the Gila River system, while only being charged for one million acre-feet of compact apportionment.

This theory, in combination with the Upper Basin relationships strengthened by Carson's choice to accept only 50,000 AF/yr, is what Carson envisioned would be used to convince Congress Arizona had a sufficient legal water supply for the Central Arizona Project. The CAP project was approved in 1968 and completed in the 1990s, though tribal water in Arizona's northeast corner was not quantified. Even after CAP was built, a portion of the power generated at Navajo Generating Station, which consumed a significant portion of that 50,000 AF/yr apportion, powered the pumps that transported CAP water from Lake Havasu to central Arizona.

Eight decades after Arizona acknowledged that the 50,000 AF of Upper Colorado River Basin water was destined for tribes, Congress is on the cusp of approving the settlement that would resolve some water rights for Navajo Nation, Hopi Tribe, and San Juan Southern Paiute Tribe. This settlement is critical and long overdue, especially considering Arizona's acknowledgement of tribal entitlement in the 1940s.

The "backstory" behind Arizona's 50,000 acre-feet of Upper Basin water.

At the recent Water Education Foundation Colorado River meeting in Santa Fe, we heard an update on the status of Congressional approval of the water rights settlement among Arizona the Navajo, San Juan Southern Paiute and Hopi nations. Among other things, the settlement divides up the use of the 50,000 acre-feet of water apportioned to Arizona by the 1948 Upper Colorado River Basin compact. How Arizona ended up with 50,000 acre-feet of Upper Basin water is a fascinating story. At first blush, it may seem somewhat arbitrary, but the reality is that it was based on a well-conceived and executed strategy by Arizona's negotiators. At the time, the deal was cut, Arizona's negotiator made clear that the only likely users of the water would be Native American communities in northeast Arizona. The deal was not designed for their benefit, but rather for the ultimate benefit of Arizona's quest to build the Central Arizona Project.

While Arizona's motives may have focused entirely on cutting an interstate deal to enable construction of the CAP, the state's leadership were frank in acknowledging that the only people who might put Arizona's Upper Basin allotment to use were Native Americans.

“There is not much possibility of using water on that land except ... on the Navajo Reservation,” Arizona’s Charles A. Carson told members of Congress during the 1949 Upper Basin Compact hearings.

The 1922 Colorado River Compact divides the basin into two sub-basins: the Upper Basin and the Lower Basin. The dividing point is Lee Ferry, located in Northern Arizona, a mile downstream of the confluence of the Colorado and Paria Rivers. Lands that drain into the Colorado River above Lee Ferry are in the Upper Basin, including about 7,000 square miles of lands in northeastern Arizona. Today all but a small portion of these lands are located on the Navajo reservation. Likewise, both Utah and New Mexico have lands that drain into the river below Lee Ferry. The Upper Gila River in New Mexico and Kanab Creek and the Virgin River in Utah are Lower Basin streams.

Although Arizona has lands in the Upper Basin, it is not a State of the Upper Division, a critically important distinction under the 1922 Compact. As a state with Upper Basin lands, Arizona is entitled to use some portion of the beneficial consumptive use apportioned to the Upper Basin under Article III(a) of the 1922 Compact, but since it is not a State of the Upper Division, it does not share in the joint obligations of the Upper Division States to provide certain flows at Lee Ferry under Articles III(c) and III(d). This is a nuance the negotiators of the 1948 Upper Basin Compact understood from the get-go (UCRBCC Official Record, 1st meeting, pages 25-26).

Arizona’s Upper Basin Compact Commissioner was Charles A. Carson. He was the state’s special counsel for Colorado River matters. Carson, an accomplished lawyer and skilled negotiator, began representing Arizona in the mid-1930s. By the 1940s, Arizona’s top water priority was obtaining Congressional approval of the Central Arizona Project (CAP). Carson negotiated the 1944 contract between Arizona and the United States for 2.8 million acre-feet of Hoover Dam water. He orchestrated his state legislature’s ratification of the Colorado River Compact a few weeks later. In 1945 he chaired the legal sub-committee of the Six-State Committee that successfully lobbied for Senate ratification of the 1944 Water Treaty with Mexico. During this time, he became a close associate and friend of Colorado’s Clifford Stone and Royce Tipton. Stone was Colorado’s Upper Basin Compact Commissioner, its first Executive Director of the Colorado Water Conservation Board, and the long-time chair of the Committee of Fourteen that advised the U.S. State Department on the treaty negotiation with Mexico. Tipton was a consulting engineer that worked for Colorado on four major interstate compacts. He was an engineering consultant to the State Department during the negotiations of the 1944 Treaty. With Stone’s blessing, Carson hired Tipton to help with Arizona’s efforts to advance the Congressional approval of the CAP.

Carson’s appointment as Arizona’s Upper Basin Compact Commissioner was likely welcomed by the negotiating teams from the other states. He was designated as Chair of the Commission’s Legal Committee, which would ultimately make numerous recommendations to the Commission on the language and structure of the Upper Basin Compact. Among the many important recommendations the legal committee made were the decisions to include the water requirements of the Upper Basin’s tribes within the apportionments made to each individual state (rejecting an option by New Mexico to consider the tribal needs as a “sixth state”) and the language of Article IV which prescribes how the UCRC will determine the timing and amount and distribute among each Upper Division State a curtailment (aka – “compact call”), if necessary to be in compliance with 1922 Compact.

Carson first spelled out what Arizona wanted from an Upper Basin Compact during the second meeting of the Upper Basin Compact Commission in September 1946, almost two full years before the other four (?) states put their cards on the table during the marathon seventh meeting in July 1948. Carson suggested Arizona be apportioned “all of the waters (on its Upper Basin lands) precipitated thereto, and in addition thereto, 1000 acre-feet from the Paria River” (Official Record, 2nd meeting, page 4). When the other states were finally ready to negotiate the allocations (Colorado had insisted that the Commission not address this core issue until the Engineering Committee had completed its report), Carson reiterated his request—Arizona wanted the right to use all the water that fell on its lands as precipitation plus an additional thousand acre-feet from the Paria River. Now that the Engineering

Committee had completed its report, this number was now quantified—136,200 acre-feet (Official Record, 7th Meeting, page 69). According to the Engineering Committee, these 136,200 acre-feet represented 0.87% of the natural (virgin) flow at Lee Ferry (Official Record, 7th Meeting, page 22). This number may seem very high based on our recent experience, but it was the number the Commission had in front of it and in the 1940s the estimated natural flow of the river at Lee Ferry was about 16 million acre-feet per year.

The problem facing the Commission was that collectively the states had requested a total of 117% of the available water. Since Arizona had requested a fixed amount, the problem was with the four Upper Division States, but that did not prevent the other states from suggesting that Arizona consider taking less. Wyoming's legal advisor Bill Wehrli asked Carson if Arizona would accept an apportionment of 49,200 acre-feet. Carson responded, "I am willing to do that in order to try to help make a compact." Interestingly, Wehrli responded, "we would be willing to be a little more generous and give you one percent" (7th meeting, page 109). The 49,200 acre-feet referenced by Wehrli was taken from the 1947 comprehensive basin report prepared by the Bureau of Reclamation. The report included very little detailed backup information. Although no tribal members were consulted or invited to the negotiations, the Office of Indian Affairs (now the BIA) provided some input to the Commission on tribal needs. It suggested that present and future depletions from tribal use on Arizona's Upper Basin lands would total about 25,000 acre-feet per year but cautioned that this estimate was preliminary (Official Record, 5th Meeting, pages 49-51).

When the dust settled, Carson accepted a fixed 50,000 acre-feet per year, only 37% of Arizona's contribution to the flow of the river at Lee Ferry. The only other state that accepted an apportionment smaller than its contribution was Colorado. It produces 70% of the river's flow at Lee Ferry but accepted a 51.75% apportionment (~72% of its contribution). In contrast, New Mexico which contributes only 1.6% of the river's flow, got an apportionment of 11.25%. What made Arizona happy was the package deal that accompanied the agreement on the state apportionments. The three other Upper Division States accepted a proposal by Colorado and Arizona that apportionments be measured by the stream depletion theory. Under the stream depletion theory, the Upper Basin's compact apportionment is measured as the net impact of man-made depletions on the natural flow of the Colorado River at Lee Ferry. The agreement on the stream depletion theory was made a part of Article VI of the Upper Basin Compact. Article VI is applicable to the Upper Basin only, but Upper Basin officials, including Stone and Tipton, would later testify before Congressional committees that it was their opinion that the Lower Basin's 1922 Compact apportionment was supposed to be measured as the net impact of the Lower Basin's man-made depletions on the natural flow of the Colorado River at the international boundary with Mexico.

Why was adoption of the stream depletion theory an important victory for Arizona? Simply put, under Arizona's interpretation of the 1922 Compact at the time, using the stream depletion theory, Arizona's could consume two million acre-feet per year on the Gila River system, but only be charged for one million acre-feet of compact apportionment. In its natural state, the Gila River loses an average of one million acre-feet per year as it flows from the Phoenix area to its confluence with the Colorado River at Yuma. Under the stream depletion theory, the net impact of consuming two million acre-feet per year of Gila system on the natural flow of the Colorado River was only a million acre-feet. If Arizona was going to be limited under the 1922 Compact to the use of about 3.8 million acre-feet (2.8 million under its Boulder Canyon Project Act allocation plus all one million acre-feet of III(b) water (less a small amount set aside for Utah and New Mexico), using the stream depletion theory freed up a million acre-feet that the CAP could pump from Lake Havasu to Central Arizona.

California, of course, had a different theory on how 1922 Compact apportionments were supposed to be measured. It advocated for the "diversions minus return flows" theory. Under this theory, all two million acre-feet of Arizona's Gila River use would be charged to Arizona as compact apportionment. Under this method, the water available for the CAP would be a million acre-feet less, likely making the project economically unfeasible. For more details on the different theories and why Colorado believed the stream depletion theory benefited the Upper Basin, see Science Be Dammed chapter 12.

During the negotiations of the Upper Basin Compact, Wyoming had initially opposed using the stream depletion theory. Speaking for its delegation, Wehrli questioned the basic legal assumption that the 1922 Compact apportioned depletions, noting that California's legal argument had merit. He concluded that stream depletion theory benefited the Lower Basin more than the Upper Basin and he stated, "Wyoming is desirous of staying completely out of the controversy between Arizona and California, Lower Basin States" (Official Record, 7th Meeting, pages 58-60). To reach a final compact agreement, Wyoming ultimately accepted the stream depletion theory, but unlike Colorado, it never championed it in Congressional testimony or court filings. Note, with perfect hindsight, Wehrli was mostly right, but also note that the question of how to measure apportionments under the 1922 Compact has never been resolved.

What Carson accomplished by accepting a small apportionment was to strengthen the close working relationship between his state and the four States of the Upper Division. It's apparent that Carson believed that this relationship would help Arizona in its battle with California over the Congressional authorization of the CAP. In his compact report to the Governor, Carson writes that his engineers and the Indian Service believed that Arizona would never use more than about 30,000 acre-feet per year on its Upper Basin lands and therefore Arizona received 50,000 acre-feet as a measure of safety. He makes no mention of any input or consultation with the Navajo Nation, nor does he refer to the 49,200 acre-feet estimate made by the Bureau of Reclamation. Concerning Article VI and the stream depletion theory he writes, "[t]his of course is in complete accord with Arizona's construction of the Colorado River Compact, and it is believed will be helpful to Arizona in opposing California's arguments on the Gila River." Carson concluded his report with "I believe it to be fair, just, and equitable to all of the States, and particularly valuable to Arizona in that it supports Arizona's position in opposition to the arguments made by certain California interests" (Carson's report is included in the record of the 1949 Congressional Hearings on the Upper Basin Compact, pages 128-139).

After a minor kerfuffle with California's Congressional delegation which was settled by report language making it clear that by approving the Upper Basin Compact, Congress was not committing the United States to any interpretation of the 1922 Compact, it was approved by Congress and became effective on April 6, 1949.

Was the Carson strategy to minimize its claims for Upper Basin water and enlist the Upper Division States as close allies in Arizona's quest to build the CAP successful? The short answer is yes, the result was that the CAP was authorized in 1968 and has been fully operational for about three decades. The path that Arizona used to get there, however, differed from the one that he envisioned. Carson who died in 1951, believed that Arizona would use the stream depletion theory to convince Congress, or if that failed, the Supreme Court, that Arizona had a sufficient legal water supply under the 1922 Compact to build and operate the CAP. Indeed, in 1952 when Arizona filed suit against California, one of its claims for relief was that it asked the Supreme Court to find that the stream depletion theory was the proper method of measuring apportionments under the 1922 Compact.

In one of the first major turning points in the case, in 1954 California filed a joinder motion to bring the Upper Basin States into the case arguing that the compact issues that Arizona wanted the court to interpret such as how apportionments are measured, how the surplus is measured for Mexican Treaty purposes, and how mainstream reservoir evaporation is handled were basin wide issues that impacted all seven states. From today's perspective, California's logic seems obvious, but that was not the case in the 1950s. In a coordinated response, Arizona and the Upper Divisions States convinced Special Master George Haight that the case involved Lower Basin matters only. While the Upper Division States had independent reasons to stay out of the case, they were concerned that their participation in the case would delay Congressional approval of the Colorado River Storage Project, clearly had they decided that they needed to be in the case, the Special Master would most likely have let them in. Note, New Mexico and Utah were parties to the case as to their Lower Basin interests only.

Ultimately, Haight's successor, Simon Rifkind determined that the 1922 Compact did not need to be interpreted to decide the case. Thus, the disputed compact issues raised by Arizona in 1952 remain

unresolved. In 1963 the Supreme Court agreed, ruling that the 1928 Boulder Canyon Project Act allocated 2.8 million acre-feet per year of mainstem water to Arizona, but subject to water availability under the 1922 Compact. This gave Arizona sufficient water to gain Congressional approval of the CAP. Had Arizona not followed Carson's advice and chosen a more adversarial approach to dealing with the Upper Basin States, it is unclear if the CAP would exist today.

Arizona's 50,000 acre-feet apportionment was also used to support the CAP because a portion of the power generated by the Navajo Generating Station was used to power the pumps used by the CAP to move water from Lake Havasu to central Arizona. At its peak the coal-fired plant consumed over 28,000 acre-feet per year. The plant was shut down in 2019. Today, the average use of Arizona's Upper Basin apportionment is about 11,000 acre-feet per year (U.S. Bureau of Reclamation, Upper Colorado River Basin Consumptive Uses and Losses Spreadsheet, 20240624 version. Note the consumptive use numbers on this spreadsheet are not necessarily the same as Arizona's use under Article VI of the Upper Basin Compact).

Now nearly eight decades after the water needs of the Native Americans living in Arizona's Upper Basin lands were subordinated to its interests in the authorization and construction of the CAP, Arizona, the Navajo Nation, the Hopi Tribe, the San Juan Southern Paiutes, the United States and others have reached a water rights settlement. The settlement divides Arizona's 50,000 acre-feet of Upper Basin water into three ways: 44,700 acre-feet per year for the Navajo Nation, 2,300 acre-feet per year for the Hopi Tribe, and 3,000 acre-feet per year for the City of Page. Note, the settlement goes beyond just allocating Arizona's Upper Basin water, it also addresses the Lower Basin Colorado River, groundwater, and the Little Colorado River (a Lower Basin stream).

We thus have finally reached the point where the water Carson told Congress was intended for use by Native Americans might actually be theirs to use.

At the Santa Fe meeting, there was general support and enthusiasm for Congressional approval (and funding) of the settlement. Tom Buschatzke, Director of AWDR missed the meeting because he was in Washington advocating support of legislation authorizing the settlement. The Salt River Project, IID, and CAP all support the settlement.

The settlement of tribal rights has always raised difficult issues for both basins. The rights of Navajo, Hopi, and Southern San Juan Paiute people to use the Colorado River for both consumptive and cultural purposes predate statehood of all four Upper Division States. Their rights are "pre-compact" rights and thus, under Article VIII, are not impaired by the 1922 Compact. The use of water by tribal sovereigns has never fit well into the state-centric water use rules established by the Euro-American settlers. When the Navajo Nation, a sovereign entity, signed its 1868 Treaty with the United States, it was one community. Today, it is crisscrossed with boundary lines. It has land and water in three Upper Division States, in both the Upper and Lower Colorado River Basins, and in the Rio Grande as well.

Whatever the detailed issues are, it is now time to get past them and move the Basin toward unanimous support for approval and implementation of this important settlement.

Section: Jack's Graph is Better Than Mine

Source: <https://www.inkstain.net/2024/11/jacks-graph-is-better-than-mine/>

This version of the Colorado River graph, courtesy of Jack Schmidt, more clearly illustrates the narrative of the talk I'm giving later this week:

- Early 20th century pluvial, when we built the institutions
- Mid-century baseline, when we built all the dams and farms and cities
- Millennium drought, when we emptied the reservoirs

Section: The Colorado River Problem, In Brief

Source: <https://www.inkstain.net/2024/11/the-colorado-river-problem-in-brief/>

The Colorado River Problem, In Brief

Putting together a slide deck for a talk next week, borrowing Brad Udall's trick of a horizontal line for visualizing the mean during different time periods.

1950 through 1970 look like it might even be lower. Just proves the vicissitudes of flow. Just pick your proof of any position you want to justify.

Funny that you show this 'trick' at this time. I've seen the same method used in the immigration statistics for different political administrations to show cause and effect of policy.

In the case of the river's flows, this method could be considered as cherry picking to prove one's point. There's bias in this. Just my two cents here...

To be more informative you should do another version with 20 year means in the 30-40's and 50's-60's so you see the droughts then. Also do 1900-1920ish and 80's-90's which were very wet decades that pulled up the 20th century mean.

Your graph is oversimplifying a complex system

Okay John...I get your point. I'd like to add some points as well to the argument if that's okay with you?

Data Collection: My general statement here is that there are no absolutes in figuring out exactly how much water flowed during any given year. Why do I say this? The key word is 'absolute'. We can only make close assumptions of what flowed based on the instruments and the number of readings over time. In the case of the USGS station at Lee's Ferry, I'd get the history of the instruments used over the timeframe and the number of times per day that the measurements were made. Prior to 1920 there were very few strip chart recorders used. Some sites were read twice a day and the results were written down. I recall that this was done at Yuma at the Union Pacific Railway bridge and was done by the railway agent. The accounting on this amounts to less than 800 measurements per year. That data figured into the big picture. Like I said, it would be interesting to see the history of how Lee's Ferry was operated over time. I would also look at the discharge at Glen Canyon Dam for comparison after the 1990 timeframe. That's the type of data that you need to see a more accurate accounting.

Something else about Data Collection. The standing rule is that it is being done with the best equipment obtainable at the time and done using the best techniques known at the time. Case in point, the data of yesteryear is not as accurate as what is being collected by today's technology. There is more precision, better accuracy in the product but it still isn't perfect or absolute.

One other factor for the drop in the average flows from the river might be attributed to better methods and the instrumentation over the last few decades. I got introduced into Data Collection on the Colorado River in 1992 as a side project (not my main tasking). I can tell you firsthand about the strides that we made in the product due to the integrity of the equipment we used. The same is true about knowing about the amount of water that entering the system. The point I want to get across is I honestly think the numbers went down because we became better at recording a better picture of the situation.

Another point that you brushed on was 'we could have managed the reservoirs' better. I wholeheartedly agree. The trade off here is that when the bottom line goal is to regulate the flows for optimal capacity in the reservoirs, keeping in mind that nobody can predict the weather no more than ten days in advance. Remember 1983? We almost lost the Glen Canyon Dam due to the levels in Lake Powell. Now put yourself in the shoes the Water Operations Manager for Powell or Mead. You tend to be conservative rather than taking a chance. You have to consider that in the overall equation.

I have to agree with some aspects of mismanagement in regard to releasing more water from the system. I remember one year of non stellar flows in the river yet there was a 'surplus' declared. I attribute that to politics and wanting to please the stakeholders.

My last point about mismanagement is about distribution. Once that precious water is released to the States, Tribes or Mexico in distribution, the usage is determined by the stakeholders. I often look at the disconnect between the States and their local municipalities on land development and water usage. It's an ugly situation as the demand is greater than a reliable supply.

Anyway, the decisions made was far above my pay grade. If anything is to blame, it's politics and money.

Section: Reservoir Drawdown in 2024: Are We on Track to Recover Storage?

Source: <https://www.inkstain.net/2024/10/reservoir-drawdown-in-2024-are-we-on-track-to-recover-storage/>

By Jack Schmidt

Unfortunately, water use between now and next April is on track to exceed the inflows of the snowmelt season, resulting in a net loss of reservoir storage. The persistent decrease in runoff is severely challenging the quest to rebuild reservoir storage.

Summary

Reservoir storage in the Colorado River basin is now approximately equal to two year's average annual consumptive use. In the three months since reservoir storage peaked in July 2024, drawdown of those reservoirs lost more than 80% of the increase accomplished by the 2024 snowmelt inflow season, which had increased basin reservoir storage by only 2.5 million acre feet despite the Upper Basin snowpack having peaked at a snow water equivalent that was 13.5% greater than the long-term average¹. If this rate-of-use continues for the next six months, there will be a net loss in basin reservoir storage. Water supply reliability and security for Colorado River water users can only be accomplished if we replenish the amount of water stored in reservoirs and not further deplete the declining supply.

Details

On 15 October 2024, total contents of the reservoirs of the Colorado River Basin upstream from the Gila River were 27.8 million af (acre feet). This amount of reservoir storage would support two years of consumptive use of the Colorado River², assuming that basin consumptive uses remain approximately 13 million af/yr, the average between 2021 and 2023. Reservoir storage today is comparable to conditions in mid-June 2021 (Fig. 1) when there was increasing concern among the basin's water managers about the security and reliability of water supplies provided by the Colorado River. Today, we should be just as concerned as we were in 2021.

Figure 1. Graph showing total basin reservoir storage (blue line), and storage in different parts of the Colorado River watershed between 1 January 2021 and 15 October 2024. CRSP reservoirs are those authorized by the Colorado River Storage Project Act.

The only way to increase the security and reliability of the water supply is to increase reservoir storage, and we are not doing a very good job of achieving that goal. There is no doubt that the large reservoir inflows of 2023 benefitted the basin water supply, allowing us to take a step back from the edge of the cliff of crisis. Basin reservoirs in mid-March 2023 were the lowest they had been (21.3 million af) since late May 1965, when the Colorado River Storage Project's reservoirs were just beginning to fill and other reservoirs had yet to be built. Snowmelt runoff in 2023 recovered 8.4 million af of reservoir storage, nearly a 40% increase from the March 2023 low point (Fig. 2).

Figure 2. Graph showing reservoir storage between 1 January 2023 and 15 October 2024, highlighting the amount of reservoir recovery during each snowmelt season and the amount of reservoir drawdown during intervening periods.

However, little additional progress in reservoir recovery was made in 2024. We were encouraged that reservoir drawdown during the nine months immediately following the 2023 inflow season was remarkably small, only 2.15 million af and only 26% of the preceding gain in storage. However, snowmelt inflow only resulted in 2.5 million af of gain in reservoir storage in 2024 (Fig. 2).

In contrast to last year, basin uses and losses are much greater this year. In the first three months following the 2024 inflow season that ended in mid-July, reservoir drawdown has been 2.14 million af, more than 80% of the gain of the preceding inflow season (Table 1). Slightly more than half of the drawdown during the last three months has been from the 42 reservoirs upstream from Lake Powell. Those releases supported the needs of mid- and late summer irrigated agriculture, were exported out of the basin, or flowed into Lake Powell. It is likely that the drawdown from these reservoirs will decrease during winter. Slightly more than 30% of the drawdown has been from the combined contents of Lake Mead and Lake Powell. Recent agreements to decrease diversions in the Lower Basin hopefully will reduce drawdown from Mead-Powell combined storage during the next six months. The continued drawdown from Mead-Powell storage will be a robust test of the effectiveness of recent drought management measures.

Table 1. Reservoir drawdown during the first three months following the 2024 snowmelt compared to the total drawdown during the nine months following the 2023 snowmelt season.

Basin water use between now and April 2025 is on track to exceed the inflows of the 2024 snowmelt season, resulting in a net loss of reservoir storage since the bounty of 2023. The persistent decrease in runoff in the 21st century is severely challenging the quest to rebuild reservoir storage. We desperately need to accomplish that goal to avoid another water supply crisis such as occurred between 2020 and 2022.

The only way to replenish the amount of water stored in reservoirs is to decrease reservoir drawdown to match or exceed each year's gains that occur during the inflow season. For the next six months, that is our goal.

- Based on data from the Natural Resources Conservation Service snow water equivalent for the Upper Colorado Region for 2024 and for 1991-2020. https://nwcc-apps.sc.egov.usda.gov/awdb/basin-plots/POR/WTEQ/assocHUC2/14_Upper_Colorado_Region.html
- Also including losses from reservoir evaporation.
- Between 6 July and 15 October 2024
- Between 13 July 2023 and 17 April 2024.
- Includes drawdown of Lake Mohave and Lake Havasu.

Section: Federal Judge Cites Upper Colorado River Basin's Compact Call Risk

Source: <https://www.inkstain.net/2024/10/federal-judge-cites-upper-colorado-river-basins-compact-call-risk/>

A federal judge this week criticized the federal government for failing to consider the risk of a Colorado River Compact call in its environmental review of the planning for Denver Water's expansion of Gross Reservoir in Boulder County.

Wrangling over the risk of a compact call – which the judge said could force water use reductions in the Upper Basin if the Upper Basin states fail to deliver enough water past Lee Ferry to the Lower Basin –

has been a key point in current negotiations between the two basins over future Colorado River operations.

The ruling, in a lawsuit against Gross Reservoir expansion by Save the Colorado River and others, allows construction to proceed, but criticizes the project's planners for not considering the fact that the risk of a compact call means there might not be enough water to fill it. (Here's Michael Booth's Colorado Sun story on the decision.)

In the decision, federal judge Christine Arguello noted that the Army Corps of Engineers environmental review of the project "rests on the assumption that there will be no compact call.... However, considering the American West's last few decades of severe aridity, such an assumption warrants considerable scrutiny."

Here's the full language from Arguello's ruling. I've bolded the key bits:

The American West's viability rests in large part on the Colorado River, which is an undeniably overutilized water source. The Colorado River's over-allocation is due to a fatal flaw baked into the 1922 Colorado River Compact—the bedrock-level agreement that forms the basis of the various overlapping management systems dictating the River's death by a thousand cuts. The Colorado River Compact rests on a politically unpalatable truth—the Compact promised the basin states water that simply does not exist. According to water experts, the Compact's apportionment scheme draws from deliberately misleading hydrological modeling of the Colorado River's flow rates. See generally Eric Kuhn & John Fleck, *Science Be Dammed: How Ignoring Inconvenient Science Drained the Colorado River 1* (Univ. Ariz. Press 2019). That fatal error now sits enshrined within the Compact, and all of the basin states' various agreements with each other (and with the Department of the Interior) are forced to overcorrect for the Compact's flawed hydrology because revisiting the agreement would require a virtually unthinkable amount of time, energy, money, and political will. However, should Colorado River diversions exceed the river's available water supply—which appears inevitable, at this rate—the 1922 Compact provides a "compact call" mechanism whereby the upper basin states are bound to curtail their water users' diversions to ensure enough water reaches the lower basin states and Mexico to satisfy the artificially inflated water delivery obligations set forth in the Compact. This Court emphasizes this context for good reason: the cracked foundation of the Colorado River's management system all but demands skepticism over any proposal that will affect the hydrology of the Colorado River basin. In the instant case, the Corps noted in its FEIS that its reasoning rests on the assumption that there will be no compact call. AR 159092; but see AR 0014667. However, considering the American West's last few decades of severe aridity, such an assumption warrants considerable scrutiny. See, e.g., Anne Castle & John Fleck, *The Risk of Curtailment under the Colorado River Compact* 5, 34–35, 40 (2019) (emphasis added) ("Even if the risk of curtailment of Colorado River rights were assumed to be low, the consequences are not. Cities, and farmers and ranchers on the West Slope, would lose economic activity, jobs, income, and community benefits."); *Cherokee Water Dist. v. City of Colo. Springs*, 519 P.2d 339, 340–41 (1974) ("Water is essential to the existence of a community." (emphasis added)). In light of this context, it is perplexing to this Court that the Corps dismissed the possibility of a compact call in its analysis of a proposed water management project.

Further reading of the judge's sources:

Section: Assigned Water on the Colorado River: Arizona Department of Water Resources response

Source: <https://www.inkstain.net/2024/10/assigned-water-on-the-colorado-river-arizona-department-of-water-resources-response/>

The Arizona Department of Water Resources has published a thoughtful and also delightfully testy response to Enduring Solutions on the Colorado River, the white paper Kathryn Sorensen et al. (I'm

one of the et alias) published in August.

First a reminder of our core premise:

As we work to reduce water use on the post-2026 Colorado River, two paths lie open before us.

One is to incentivize conservation by giving water users the chance to bank saved water for later use. Known most commonly as Intentionally Created Surplus (ICS), and more broadly in a series of increasingly creative implementations as “Assigned Water,” this creates short term savings. But in the long run, the approach entitles the users to take the water back out of the bank.

The other involves permanent reductions – “System Water.” Water use is reduced for the benefit of the Colorado River as a whole.

Investment in Assigned Water, attractive to water managers because of the allure that they can get their water back, has crowded out investment in the more durable System Water reductions that will be needed to bring the Colorado River into balance.

ADWR agrees with a lot of what we said (yay!):

ADWR agrees wholeheartedly with the general premise that System Water is preferable to any category of Assigned Water. Increased volumes of System Water will improve outcomes for water users across the entire Basin, as well as the environment. ADWR also agrees with the need to divorce decisions regarding system operations from any Assigned Water stored in the system.

ADWR takes issue with a lot of what we said (also yay!), and also was kinda mean – “Baseless Accusations and Little Substance” (also yay! because we need to be having these frank discussions – I worry being too polite isn’t helping right now).

I’m delighted this discussion is underway in a public forum. Go read the whole thing.

Section: Rubber Soul, the path to elevation 1,040, and the game of chicken on the Colorado River

Source: <https://www.inkstain.net/2024/09/rubber-soul-the-path-to-elevation-1040-and-the-game-of-chicken-on-the-colorado-river/>

Two years ago, when the level of Lake Mead was hovering near elevation 1,040, my artist wife Lissa Heineman and I drove out over UNM’s fall break to see it for ourselves.

Out beyond the old Boulder Harbor, we walked a half mile across mud flats to get to the water. I could look out across the water to see the elbow of the old Southern Nevada Water Authority intake, above the water line. I was gut-punched by the visceral reality.

On the walk back to the car, Lissa carefully picked up some pieces of cracked mud. Her art has always been wrapped up in the conceptual properties of her materials. So she carefully packed up the cracked mud in a box and took it home. It’s been sitting in her studio ever since, and last month she tried firing some of it atop some small ceramic plates in her kiln.

It worked, and she gave me the results to give to my Lower Basin/Lake Mead friends. The texture of the mud, with ripples across the sandy and muddy reservoir bottom, captures a moment in history I hope we never repeat.

So last week, with the Colorado River brain trust in Santa Fe for the Water Education Foundation’s always-fascinating Colorado River symposium, I drove up to see folks and stuck a couple of Lissa’s pieces in my backpack.

I shared them with a message: That was scary. Let’s not go back there again. Please don’t fuck this up.

I've got a lot going on – revisions to the new book, teaching my fall semester graduate-level water resources class, nervously eyeing the levels in Lake Powell and Lake Mead, and the gridlock in Colorado River negotiations. So when my brain suggested listening to Rubber Soul Friday night, I was resistant. But we've been together for a long time, and I trust my brain's judgment. So Rubber Soul it was.

What a great album.

This post is lengthy and rambling, so for those who are annoyed by my discursive side trips and just here for the Colorado River stuff, I've added anchors to the key material:

- Rubber Soul
- Path to 1,040
- Game theory and the game of chicken now being played on the Colorado River
- Buy me a coffee

Rubber Soul and my fascination with innovation

When Eric Kuhn and I set out to write *Science Be Dammed*, the project arose in part out of a mutual fascination with E.C. LaRue, the early 20th century hydrologist who first tried to map out the supply of water, and possible uses of it, across the entire Colorado River Basin. The thing that first drew me to LaRue, long before I knew Eric, was the fundamental innovation of what LaRue and the others working at the time on similar projects were doing. No one had ever tried to envision managing a continental-sized river at the full basin scale.

In an entirely different context and framework, it's a theme Bob Berrens and I take up in our new book *Ribbons of Green*, about the making of a city.

The first time I remember thinking hard about this was when Lissa, my sister Lisa, and I saw the *Hermitage* exhibition at the Los Angeles County Museum of Art in 1986. It was a magnificent sweep of early modern painting that had been collected by rich Russians before the revolution. I remember rounding a corner and being gobsmacked by a big Picasso canvas, *Three Women*, one of the first few cubist paintings that he and George Braque had been making in Paris in 1907-08. Lissa, who understood the history, took me back through the rest of the exhibit to see the roots – the impressionists breaking one way, Matisse another, and Cezanne sweeping them all away with the beginnings of the deconstruction of the picture plane that led to Braque and Picasso.

My own father had been deeply influenced by the reverberations of that work, and I had always seen it in Dad's work, but it wasn't until Lissa held my hand and walked me through the history that I began thinking about pathways. How does this happen? Once I saw, read, and learned about it once, I became hungry for examples. My intellectual life is now littered with them. I have long since soured on Picasso himself (what an asshole!), but the genre of intellectual journey continues to fascinate.

The most interesting books I've read in recent years all document this – Patti Smith's *Just Kids*, about the birth of punk and her invention of Patti Smith; Amartya Sen's memoir *Home in the World*; Henry Threadgill's *Easily Slip Into Another World* (I still can't grasp the music, but his story of innovation is a joy); Stanley Crouch's biography of early Charlie Parker, *Kansas City Lightning*. In each case (three memoirs, one not), the innovation is rooted in a deep understanding of the past and foundations, and then the ability to see, out of that, something entirely new. And all four books are ripping good reads.

I love playing this game with the Beatles, because thanks to streaming services it is possible to dive in and listen to them learning on the fly, to watch the way the bar band Beatles learned how audiences responded to the old things and began envisioning something new.

This is metaphor.

The path to elevation 1,040

As we near the Sept. 30 end of the water year, Lake Mead is at elevation 1,064 feet above sea level, twenty feet above where it was when Lissa picked up the cracked mud two years ago.

In 2021-22, it took one year to drop from the 1,060s to the 1,040s. Could this happen again?

The short answer is probably not in a single year, because of a couple of things that have changed since then. But in two years? Yup. Lissa and I could have a chance to collect more 1,040s cracked mud.

The first thing that has changed since 2022 is the release from Lake Powell. In 2022 the Basin was in the midst of its hair-on-fire crisis management because of fears of Powell dropping dangerously low, so the Powell release that year was just 7 million acre feet. This year, it's 7.48 million acre feet. So more water coming into Mead.

Things are also better on the outflow side. In 2022, the three Lower Basin States used 6.66 million acre feet. This year, the latest forecast number is 6.09 million acre feet.

Between the higher inflows and lower use, the latest midpoint forecast has Mead ending next year at 1,059 feet above sea level, with Reclamation's most pessimistic model runs (the "minimum probable") at ~1,054. But the min probable clearly shows risk out at the edge of what our headlights can illuminate right now, of dropping back into the 1,040s again by the summer of 2026.

The game of chicken on the Colorado River

The "game of chicken" is a game theory classic. It involves a conflict which, in the classic storytelling version, involves two drivers headed toward one another on a collision course. We'll call them "U" and "L". Each has the option to swerve or stay on course. The best outcome for each driver is for the other to swerve and lose face (water), while the driver who stays the course demonstrates dominance (keeps its water). But if neither swerves, we end up with a catastrophic collision. In the game theory matrix, it looks like this, with the payoffs for each:

| Driver U Swerves | Driver U Stays | |

| Driver L Swerves | (0,0) | (-1,1) |

| Driver L Stays | (1,-1) | (-10,-10) |

I'm obviously talking about the Upper Basin and the Lower Basin here, which are at impasse over the Lower Basin's proposal to cut deeply up to a point (1.5-ish million acre feet total) and, if any deeper cuts are needed, to share them among the two basins.

The Upper Basin's counter is basically "no." If deeper cuts are needed, the Lower Basin should make them.

The payoff matrix, though, is a lot more complicated than my toy example above. First, both sides can gamble on good hydrology, which could avert the crash. So even if the impasse remains, the collision is not a sure thing. (In this regard, it'll be interesting to see how the players' strategies shift if we have a really bad winter.)

The second is the nature of the collision itself. No one knows quite what it will look like.

In the classic chicken game, both drivers know about the crash that happens if neither swerves. But part of the risk calculation we all have to live with right now is the uncertainty about what happens if the Upper and Lower Basin states can't come to an agreement. We also have a situation where the nature of the game is changing over time.

Walking down a Santa Fe sidewalk Wednesday evening after dinner, one of my Colorado River friends observed that both sides seem to think that, if the collision comes, they have a winning legal argument.

If you think that, your understanding of what happens in the bottom right quadrant of the matrix, the crash scenario, is very different.

The Upper Basin seems to have convinced itself, at least based on public pronouncements, that it has a winning legal argument in terms of its obligation, or lack thereof, to send water downstream past Lee Ferry. This seems dangerous to me given my understanding of the history and the law, but it doesn't matter what I think. The Upper Basin seems happy to keep hammering down the road.

Once deliveries past Lee Ferry drop below one of the "tripwire" triggers (82.5maf / ten years or 75/10), the Lower Basin states have nothing to lose by suing. And when that happens, my community's water supply is at risk if my basin's lawyers aren't right.

Crash!

A few years back Kaveh Madhani and Bora Ristic wrote a paper working out the details of a game theory example that seems to fit what we're currently seeing in the Lower Basin's proposal to "own" the 1.5 million acre feet of structure deficit, and to try to negotiate some sort of sharing arrangement if the cuts need to go deeper. This would appear to be what Ristic and Madhani describe as "strategic loss."

Without foresight into game changes over time, players are blind to the fact that they are in a game of chicken. We model agents with foresight by interconnecting games across time and show how this creates opportunities for "strategic loss" early on, allowing players with foresight to reduce total costs. High future costs can thus be avoided with foresight if the rising costs of inaction are made apparent.

Which seems to model what the Lower Basin has done.

I would prefer to live in the upper left quadrant of the chicken game matrix, where both sides compromise. My values: mindful shared reductions across the basin can leave us all with healthy, thriving communities. I wrote a whole book about this path. But one of my smart friends pointed out something that is a reasonable hypothesis: The model I laid out in that book, written a decade ago, was sufficient on a river that shrinks some, but seems to be failing on a river that has shrunk a lot.

Risti? and Madhani seem to be suggesting a game theoretic path that could get us back on track.

Buy me a coffee

Thanks so much to Inkstain's supporters. You can join them here.

Section: The 2024-2025 Season of Water Consumption: Can We Retain Our Gains?

Source: <https://www.inkstain.net/2024/09/the-2024-2025-season-of-water-consumption-can-we-retain-our-gains/>

By Jack Schmidt

The decrease in reservoir storage following the 2024 inflow season has been thankfully modest, but not as favorable as it was at this time last year. Perseverance reducing consumptive uses and losses is needed for reliability and security in the water supply and to regain reservoir storage.

Between mid-April and early July 2024, reservoir storage in the Colorado River basin increased by 2.45 million acre feet (af). Now we are in the nine-month period of progressive decline as reservoir storage supports consumptive uses and losses throughout the basin until the 2025 spring snowmelt season begins. As of 1 September 2024 basin reservoir storage was 28.9 million af, and the combined storage in Lake Mead and Lake Powell was 18.0 million af. Those amounts are similar to conditions from spring 2021 when media outlets began reporting on the emergence of a water crisis. That crisis continues.

It is useful to monitor changes in basin reservoir storage because it is the "bank account" from which we can make withdrawals during dry years. Basin water managers have little control over each year's watershed runoff, but they have a continuing ability to reduce water consumption.

Basin water managers have a long way to go to replenish reservoir storage to amounts that ensure a secure and reliable water supply. Today's water in the basin's reservoirs is slightly more than a two-year supply, based on the average rate of water consumption and losses[1] in the basin. It remains in a precarious state should a string of very dry years occur, as was the case between 2002 and 2004 and between 2020 and 2022.

Although the ultimate cause of the ongoing crisis in water supply is a declining watershed runoff associated with a warming climate, the proximate cause is the inability to reduce consumptive uses to match the declining supply[2]. John Fleck summarized recent progress in reducing Lower Basin water use[3]— that is the kind of progress needed throughout the basin.

Where We Stand Today

Figure 1 is a reminder that present reservoir storage remains low in relation to conditions throughout the 21st century. Today, 62% of total basin storage is in Lake Mead and Lake Powell, 30% of storage is in reservoirs upstream from Lake Powell, and 8% of storage is in Lake Mohave and Lake Havasu. Storage in reservoirs upstream from Lake Powell increases during each year's snowmelt season, and subsequently decreases to sustain consumptive uses. Storage in Lake Mohave and Lake Havasu change little. The big changes in the basin are mostly due to changes in storage in Lake Mead and Lake Powell.

Figure 1. Graph showing reservoir storage in the Colorado River basin between 1 January 1999 and 31 August 2024.

The water supply in Lake Mead and Lake Powell, as well as Lake Mohave and Lake Havasu, supports water use in the Lower Basin and in Mexico. Lake Powell is downstream from virtually all Upper Basin water use. Essentially, Lake Powell and Lake Mead are one reservoir, separated into two parts by the Grand Canyon. Nevertheless, Lake Mead and Lake Powell are operated differently, as is evident in Figure 2. In spring and early summer, snowmelt runoff is captured in Lake Powell, and storage increases there even though storage at the same time decreases in Lake Mead in some years. Once the snowmelt season ends, water is transferred to Lake Mead, and Lake Powell storage slowly declines. Figure 2 demonstrates that changes in water storage in Lake Mead occur over longer cycles than do the annual cycles of storage change that occur in Lake Powell. Because of the different operating rules of the two reservoirs, basin water storage conditions are better reflected by the combined storage contents of the two reservoirs rather than conditions in either Lake Mead or Lake Powell.

Figure 2. Graph showing reservoir storage in different parts of the Colorado River basin since 1 January 2021. Note that water storage in Lake Powell increased greatly during the 2023 inflow season, declined thereafter until the beginning of the 2024 inflow season, increased again in spring 2024, and is now declining.

Despite the modest inflow season of 2024 when unregulated inflow to Lake Powell was only 83% of average, reservoir storage increased by 300,000 af, because losses from the basin's reservoirs between mid-July 2023 and early April 2024 were less than the gains in storage that occurred in spring 2024[4]. The total decrease in storage between mid-July 2023 and early April 2024 was the smallest in the past decade and was primarily due to reduced consumptive uses in the Lower Basin.

One way to keep track of the loss in reservoir storage due to consumptive uses and losses is to monitor changes in storage that occur after the early summer peak occurs, as is depicted in Figure 3. For example, the dark blue line in Figure 3 was computed by subtracting the total basin reservoir storage on each day from the peak value of 30.0 million af that occurred on 6 July 2024. On 31 August 2024, total basin storage of 28.8 million af was 1.12 million af less than the early July peak. This amount of loss is midway in the range of reservoir loss that has occurred during the past decade. Reservoir storage declined little following inflow in 2017 (2017-2018); 2019 (2019-2020); and 2023 (2023-2024). Storage declined by large amounts following inflow in 2018 (2018-2019); 2020 (2020-2021); and 2021 (2021-2022). These data demonstrate that the current rate of decrease in reservoir storage has been

“average” for the last decade but is much greater than the remarkably small rate of loss last year.

Figure 3. Graph showing the decrease in total basin reservoir storage in 2024 (2024-2025) following the early summer peak, compared with the decrease in some other years of the past decade. Loss in reservoir storage was greatest following the 2020 inflow season (2020-2021) and least following the 2023 inflow season (2023-2024). This year's loss is midway between those extremes.

The rate of decrease in the combined contents of Lake Mead and Lake Powell since early July 2024 has been comparable to the loss in other years of small decline, as is evident in Figure 4. It is especially encouraging that storage in Mead and Powell greatly slowed since mid-August.

Figure 4. Graph showing the decrease in the combined contents of Lake Mead and Lake Powell following peak storage of 18.5 million af that occurred on 8 July 2024, compared with the decrease in some other years of the past decade. Loss in reservoir storage was greatest following the 2020 inflow season (2020-2021) and least following the 2023 inflow season (2023-2024). This year's loss is similar to years when the loss in combined storage was relatively small.

- [1] Basin consumptive uses and losses averaged 13.0 million af/yr for 2021-2023, based on the latest published reports of the Bureau of Reclamation.

- [2] Schmidt, J. C., Yackulic, C. B., and Kuhn, E. 2023. The Colorado River water crisis: its origin and the future. WIREs Water 2023;e1672.

- [3] Fleck, J. 2024. Imperial Irrigation District's water use on track for a record low, as is U.S. Lower Basin use. Inkstain, 9 September 2024, <https://www.inkstain.net/>.

- [4] The gain in reservoir storage during the 2024 inflow season was 2.45 million af, and preceding decreases in storage between mid-July 2023 and mid-April 2024 were only 2.15 million af. Thus, inflows in 2024 added 300,000 af to total basin reservoir storage (Schmidt, 2024. The 2024 runoff season comes to an end- how did we do? Center for Colorado River Studies, 17 July 2024, <https://qcnr.usu.edu/coloradoriver/>).

Section: Imperial Irrigation District's water use on track for a record low, as is US Lower Basin use

Source: <https://www.inkstain.net/2024/09/imperial-irrigation-districts-water-use-on-track-for-a-record-low-as-is-us-lower-basin-use/>

Taming the Lower Basin Structural Deficit

The federally funded water use reductions approved last month by the Imperial Irrigation District and the federal government have made their way into the Bureau of Reclamation's annual forecast model (updated Sept. 6 as I'm writing this), and the numbers are remarkable.

Imperial's projected 2.2 million acre foot take on the Colorado River in 2024 is on track to be the lowest on record, with data going back to 1941.

California's total projected main stem withdrawals are again under 4 million acre feet, the lowest they've been since the 1950s. Arizona's main stem withdrawals remain under 2 million of their nominal 2.8 maf allocation for the second year in a row, basically the lowest they've been since the Central Arizona Project was built. Nevada is once again hovering around 200,000 acre feet of its 300,000 acre foot allocation.

Taken together, water use by the three lower basin states is currently on track to be the lowest since detailed record keeping began in 1964.

A note on the data

The Bureau of Reclamation has complete reported data back to 1964, when the modern accounting system was established as a result of the Supreme Court's Arizona v. California decree. I have stitched that data together with a separate dataset that pushes California records back into the 1940s, assembled some years ago by the Metropolitan Water District of Southern California and kindly shared with me. For my current version of the dataset, I extend a huge thanks to Sami Guetz, who spent time QA'ing it as part of her masters project at UC San Diego.

Section: Emerging Values and Institutional Reform on the Colorado River

Source: <https://www.inkstain.net/2024/09/emerging-values-and-institutional-reform-on-the-colorado-river/>

Lorelei Cloud and John Berggren had a really important piece on Colorado River governance in the Colorado Sun last month that has not received sufficient attention.

The challenge, they argue, is the lack of the institutional framework we need to address evolving societal values around the river's management in a changing world.

Cloud is Vice-Chairman of the Southern Ute Tribe and has become a major voice in the effort to rethink the role of indigenous people in management of the Colorado River. Berggren, now at Western Resource Advocates, is the author of one of the most insightful analyses of Colorado River governance we've had in recent years. (I hope that link works for folks, this might also.)

They catalog the remarkable efforts within the last decade or more to create new frameworks for Tribal involvement in Colorado River governance, notably the Ten Tribes Partnership and the Water and Tribes Initiative. Here's Cloud:

When I was first elected to the Southern Ute Tribal Council in 2015, I was asked to participate in the Ten Tribes Partnership, or TTP, which is a coalition of the 10 Tribes along the Colorado River focused on securing and using tribal water. After one year, I was asked to chair TTP.

I drew on my personal and spiritual connection to water and started learning about the complex legal and technical issues related to managing water in the American West. I was stunned to learn that Tribes have historically delegated to have little to no role in managing Western water, and that tribal needs and interests are often marginalized.

The challenge, as Berggren documented in his thesis, is a set of water management institutions – by “institutions” here I mean the formal rules we wrote to manage water – which are antecedent to the government agencies and political power centers that emerged to carry them out – created to allocate water for municipal and agricultural use.

Because those rules were allocative in nature, the government agencies and political power centers that emerged to carry them out focused almost entirely on carving up the water supply and getting it efficiently to farms and cities. Which worked great, until it didn't. As the twin challenges of climate change and evolving values emerged, those institutional structures have proven maladaptive.

But it's a path dependence from which it is hard to dislodge ourselves as new, changing values emerge. These new values (“New” here seems weird, the indigenous communities represent the oldest values! Maybe “newly recognized”?) don't have a seat at the table.

Like Tribes, environmental interests have mostly taken a backseat to the use of the Colorado River for municipal and agricultural purposes. Most adjustments to address cultural and ecological values have been treated as subservient to the allocative laws that largely service municipal and agricultural interests.

Returning to the primary purpose of the 1922 Compact, we believe that providing for the equitable use of water includes substantive and procedural elements. There's a huge difference between how the Colorado River is managed for multiple values (substance) and how people who care about such issues determine what ought to happen (process).

I don't know if their proposed solution, is the right one:

an ongoing, whole-basin roundtable that would embrace the entire transboundary watershed, address the major water issues facing the basin, and, importantly, provide an equitable process to engage all four sets of sovereigns (United States, Mexico, seven basin states and 30 Tribal nations), water users and stakeholders.

But if not this tool, then what should we do instead?

Section: The search for enduring solutions on the Colorado River

Source: <https://www.inkstain.net/2024/08/the-search-for-enduring-solutions-on-the-colorado-river/>

Kathryn Sorensen & Sarah Porter, Kyl Center for Water Policy, Morrison Institute for Public Policy, Arizona State University; John Fleck, Utton Transboundary Resources Center, University of New Mexico School of Law

Colorado River Basin governance is increasingly struggling with a deep question in water management: When we reduce our use of water, who gets the savings?

If I install more efficient irrigation equipment, should I get credit for the saved water to expand my acreage, or save the water in an upstream reservoir as a hedge against next year's drought? If I tear out lawns, can I use the saved water to help build the next subdivision, or save the water in an upstream reservoir as a hedge against that next year of drought?

Or should the savings contribute, not to my own resilience and well-being, but to the resilience and the well-being of the system as a whole by simply reducing overall water use?

In a deeply insightful 2013 book, British scholar Bruce Lankford bestowed the unfortunately wonky name of "the paracommons" to this question, and it dogs water policy management around the world.

This issue has been lurking in Colorado River management for a long time. Should we create legal structures that allow users to bank the savings for their own use later? Or should the reductions benefit the health of the system as a whole? There are advantages and disadvantages to both approaches, and we need to design new rules for managing the Colorado River with our eyes open on this question.

Assigned Water

In a new paper, we explore the implications of the two paths for the management of a post-2026 Colorado River.

One is to incentivize conservation by giving water users the chance to bank saved water for later use. Known most commonly as Intentionally Created Surplus (ICS), and more broadly in a series of increasingly creative implementations as "Assigned Water," this creates short term savings.

The other involves permanent reductions – "System Water." Water use is reduced for the benefit of the Colorado River as a whole.

In more than a decade of experimentation with these policy tools, we have seen the results. Investment in Assigned Water, attractive to water managers because of the allure of getting their water back, has crowded out investment in the more durable System Water reductions that will be needed to bring the Colorado River into balance.

As we develop new operating rules for the river, we need to be mindful of the differences involved.

Assigned Water does not solve the problem of overallocation because when it is deployed we are borrowing against our own bank. Enduring solutions on the river can only be found by addressing overallocation.

- Assigned Water creates critically important operational flexibility; it allows its owner to either forgo water deliveries in one year—or pay someone else to—and take delivery of that water during another potentially desperate time.
- Assigned Water is generally insulated from shortage, forfeiture and abandonment.
- Protection from shortage and forfeiture has value; Assigned Water creates individual resilience for its owner. Because of this, the availability of Assigned Water appears to crowd out investment in collective resilience in the form of System Water.
- In conversations about post-2026 operations negotiators are contemplating extending, enlarging and/or enhancing Assigned Water and/or creating an operationally neutral form called Top Water. In any form, Assigned Water lives outside of the existing priority system. In this regard, the conversation involves the reallocation of water in Lakes Powell and Mead.

Critics of the West's priority system of water delivery can rejoice—nearly 40% of the water in Mead in 2023 was Assigned Water, meaning that Assigned Water is replacing priority to a significant degree. But is the priority system like capitalism in that it has its warts but the alternatives are far worse? As the expansion of the rights of municipal water providers, irrigation districts, foreign nations and tribes to own even more and different kinds of Assigned Water is contemplated for a post-2026 world, consideration should also be given to how these changes may also inure to the benefit of environmental non-governmental organizations, hedge funds and water speculators. Those who share John Wesley Powell's fears will understand the implications because the expansion of Assigned Water in Lakes Powell and Mead may bring about the ultimate divorce of priority-based water rights from arid lands in the Colorado River Basin.

There are important elements of transparency and fairness at play. The large, powerful players on the River received Assigned Water through negotiations not available to others—meaning, there was no open bidding process or invitation to smaller entities to acquire this valuable water. Apparently, there still isn't. Thought ought to be given to those other stakeholders—smaller cities, farmers, tribes and others—who have made investments and built economies based on the priority system. Imagine a restaurant that operates on a first-come-first-serve basis and a hungry patron who waits patiently in line for the doors to open only to be told that the rules changed while he was waiting and all of the reservations have been claimed through a process from which he was excluded.

It is helpful to continue to deploy a tool as flexible and alluring as Assigned Water, particularly in the form of operationally neutral Top Storage, so there's no need to throw the baby out with the bath water. A reasonable path forward may be to allow the creation of Top Storage with appropriate guardrails while including a 50% cut for System Water. Post 2026, Assigned Water will be so valuable that entities likely will be willing to take a big haircut to get it, and such a required contribution solves the problem of developing enduring funding for System Water to a significant degree. Maybe ultimately environmental non-governmental organizations, hedge funds and water speculators get a piece, but if so, it will be at the price of protecting and respecting the priority system upon which so many depend.

Section: Tipping Point: Colorado River Reckoning

Source: <https://www.inkstain.net/2024/07/tipping-point-colorado-river-reckoning/>

Out toward the top left corner of this picture – maybe a little bit left, out of the frame – is the point where the Southern Nevada Water Authority gets its water out of Lake Mead. There's nothing to see – the

intake is at the bottom of the reservoir.

Completed in 2015, with a new pump station first turned on in early 2022, the intake system represents a ~\$1.5 billion investment in shoring up the reliability of the Las Vegas, NV, water supply as Lake Mead and the Colorado River decline.

I hitched a ride yesterday on SNWA's water quality sampling boat, and they took me over to the face of "Saddle Island" (not an island any more with the reservoir this low!) to see the areas where the old intakes are. Back in 2022, one of the old intakes emerged above water. Had Las Vegas not invested in the new system, we would have faced huge risk to the water supply of a community of 2 million-plus people, and terrible choices, as Mead dropped.

I tried to peer down into the water to see the pipe, but Mead's up high enough now that it's no longer visible. But it's an important place, I wanted to try to see it, which is why I hitched the boat ride. Mead here is doing double duty: water storage for Phoenix, Los Angeles, and the farm districts of the Lower Colorado, but also forebay for the pumping system for Las Vegas's water supply. Multiple purposes.

Las Vegas spent the \$1.5b not for new water, but to provide reliability for the water they've got. But in doing that, removing (reducing?) the risk to a city of 2 million people, Las Vegas also removed the risk that the basin as a whole would have faced if it had to choose between Las Vegas's supply and the needs of downstream users as the reservoir's levels dropped toward the intake pipes.

Newshour link

I'm here for today's (July 24, 2024) PBS Newshour "Tipping Point" show. 5 p.m. Mountain Time (7 eastern, 4 Pacific), livestream link [here](#).

Section: Hoover Dam and the social nature of infrastructure

Source: <https://www.inkstain.net/2024/07/hover-dam-and-the-social-nature-of-infrastructure/>

A water nerd friend and I made a pilgrimage yesterday evening to Hoover Dam, spanning the Colorado River on the Arizona-Nevada border.

We'd had dinner at one of the restaurants on the docks at Hemenway Harbor, and driven up to the old abandoned boat ramp at Boulder Harbor, two Lake Mead landmarks for me, places I often visit. It was late, and hot as hell, and we were tired, but it didn't take much persuasion for my friend to say "yes" to a last stop at Hoover Dam.

I'm here in Southern Nevada to be part of a PBS Newshour livestream Wednesday: "Tipping Point: Colorado River Reckoning." I came out a couple of days early to see some friends and do the sort of "place-based" stuff that's at the heart of my Colorado River work. I've been coming to this place – Lake Mead and Hoover Dam – for the last 15 years, watching the reservoir rise and fall, using this place as a conceptual anchor as I write my books.

My best guess based on hazy childhood memory is that I first visited Hoover Dam 60 years ago. We were on our annual summer car-camping trip, and I remember five-year-old John being both awestruck and uneasy – a bit frightened? – at the scale of the thing. Many years later, on one of many return trips, I stumbled on the thing that I remember most vividly: off the side of an embankment, massive concrete plugs that had been removed from the turbine tubes, dumped off the old rail line, lying today in a ravine.

It was a vague thing, fuzzy and just out of reach, but seeing them again on a bike ride along the Hoover Dam rail trail, the memory jumped up out of the ravine at me. As my friend remarked when I told them the story last night, I got bitten early.

Deb Chachra and the social nature of infrastructure

In an interview published last week in Public Books, the engineer Deb Chachra said a kaching thing about Hoover Dam that captures why I keep coming here, why I anchor so much of my thinking in the dam itself:

To borrow an idea from Joan Didion, who used it to describe driving in Los Angeles, these older forms of infrastructure, especially charismatic megastructures like bridges and train stations, can be secular cathedrals. We recognize that these are things that we made for ourselves and for each other. People have been visiting the Hoover Dam in droves since it was brand-new, so I don't think it's nostalgia per se—I think it's more about celebrating this idea of the collective, these collective or communal projects. If we're nostalgic, I think it's less about the physical structures and more about the idea of working collectively.

Five years ago, on my umpty-first tour of Hoover Dam, a friend steeped in the “law of the river” led me to the elevator tower on the dam's Nevada side to look at the decorative frieze. It's a stone carving (or maybe cast concrete), visualizing the statutory purposes embodied in the Boulder Canyon Project Act, the act of Congress that authorized construction of the dam.

Be it enacted by the Senate and House of Representatives of the United States of America in Congress assembled, That for the purpose of controlling the floods, improving navigation and regulating the flow of the Colorado River, providing for storage and for the delivery of the stored waters thereof for reclamation of public lands and other beneficial uses exclusively within the United States, and for the generation of electrical energy as a means of making the project herein authorized a self-supporting and financially solvent undertaking, the Secretary of the Interior, subject to the terms of the Colorado River compact hereinafter mentioned, is hereby authorized to construct, operate, and maintain a dam and incidental works in the main stream of the Colorado River at Black Canyon or Boulder Canyon.... (emphasis added)

This is what Chachra is talking about. We often view Hoover Dam as a remarkable achievement of physical engineering, and it is. But it is, more importantly, an act of collective will. We came together in the 1920s, in community, to envision a desired future – of cities and farms in a place with too little water for either without infrastructure. We wrote a set of rules – the Colorado River Compact, the Boulder Canyon Project Act – that instantiated that collective will in the construction of a physical object – Hoover Dam.

And then we did the same thing, again and again, across the West. This enabled the places that tens of millions of us now call home.

Yesterday evening it was 110F (43C) after sunset, the parking lots were closed, and the tourists still came. My friend, who may be more of a Hoover Dam/Lake Mead/Colorado River nerd than I am (this is saying a great deal) was talking about the dam's aesthetics – the care paid to the art deco flourishes – like the Boulder Canyon Project Act frieze. It's a testament to the understanding of the people who built it that it was more than physical infrastructure – that it was a monument to collective action, the point Chachra so eloquently made.

This is why I'm confident, despite the current conflicts over how to share the shortages we face as a result of climate change, that we'll figure things out. Collective action is at the heart of what we've been doing, it's often hard, but we do it.

A note on Chachra's book

Chachra's book, *How Infrastructure Works*, is a terrific look at the social nature of infrastructure. Highly recommended.

Section: The 2024 Runoff Season Comes to an End – How Did We Do?

Source: <https://www.inkstain.net/2024/07/the-2024-runoff-season-comes-to-an-end-how-did-we-do/>

By Jack Schmidt, Utah State University Center for Colorado River Studies

How did we do in the continuing effort to recover reservoir storage? How much reservoir storage accumulated from this year's snowpack, and how does that accumulation compare to other years?

In Summary:

Total basin-wide reservoir storage is an appropriate metric to describe the status of the regional water supply and its year-to-year changes. Reclamation provides data on the storage contents of 46 reservoirs in the basin that are primarily managed by the Bureau of Reclamation but also by municipal water agencies and water conservancy districts. Whether destined for within-basin use or for trans-basin diversion, the total amount of water in these reservoirs is the carryover storage available to sustain use during dry times.

Accumulation of storage in those reservoirs occurred between mid-April and early July 2024, and basin-wide storage increased by 2.5 million acre feet. This amount is only 30% of the increase in storage that occurred during the same period in 2023. Nevertheless, basin-wide reservoir storage increased by 300,000 af when the summer peak of 2024 is compared with the summer peak of 2023, because consumptive uses and losses in the intervening time between the two runoff years was only 2.2 million acre feet. Despite the modest runoff of 2024, water managers were able to increase reservoir storage, because they had done such a good job of limiting consumptive uses following the 2023 runoff season.

Today, 62% of the total basin-wide storage is in Lake Mead and Lake Powell. The combined contents of these two largest reservoirs in the United States peaked on 8 July at 18.5 million af. Most of 2024's snowmelt runoff was stored in Lake Powell, and storage in Lake Mead declined during spring and early summer 2024. Now that the runoff season has ended, some of the contents of Lake Powell will be gradually transferred to Lake Mead.

The Details:

On 6 July, storage in the Colorado River basin peaked for the year at 30.0 million acre feet (af), approximately 50% of capacity of the reservoir system¹ (Fig. 1). The combined contents of Lake Mead and Lake Powell peaked on 8 July at 18.5 million af, approximately 37% of the capacity of those two reservoirs. The last time total basin reservoir storage was as much as this was in early January 2021, and the combined storage of Lake Mead and Lake Powell had last been at its present volume in very late April 2021. Thus, reservoir storage has not yet recovered to the average conditions between 2005 and 2020.

Figure 1. Graph showing reservoir storage in the Colorado River watershed between 1 January 1999 and 15 July 2024.

The season of water accumulation, when inflow to reservoirs exceeds outflow, that began in mid-April has now ended. How did we do in the continuing effort to recover reservoir storage? How much reservoir storage accumulated from this year's snowpack, and how does that accumulation compare to other years?

The snowpack of the Upper Basin peaked on 3 April at 16.8 in of snow water equivalent (SWE)². For comparison, the median peak SWE for the past 30 years, as computed by the Natural Resource Conservation Service, was 16.0 in., so 2024 was a pretty good year. Nevertheless, 2024 was not nearly as good a year as 2023 when the peak SWE was 23.9 in. Preliminary estimates of natural flow at Lees Ferry for Water Year 2024 are that this year's natural flow³ will be 12.1 million af, although that estimate may change slightly by the end of the water year.

The relation between peak estimated SWE in the Upper Basin and natural flow at Lees Ferry has a reasonably good correlation for data from 2000 and the years of the 21st century (Fig. 2). There is year-to-year variation in this relation caused by springtime weather that affects the rate of melting and

the amount of sublimation. Variation is also caused by the intensity of the summer North American monsoon that augments the natural flow but is unrelated to snowmelt. Estimated natural runoff in 2024 was well predicted by the general relation.

Figure 2. Graph showing the relation between peak annual snow water equivalent for the Upper Colorado River basin and natural flow at Lees Ferry, estimated by Reclamation. The solid line is an exponential relation fit to these data.⁴

The amount of water added to reservoir storage is very well predicted by the natural flow at Lees Ferry (Fig. 3), and this relation shows that much more of the natural runoff is captured by reservoirs in wet years than in dry years. In 2024, approximately 20% of the estimated natural flow was stored, consistent with comparable years (2010, 2014, 2015, and 2016) (Fig. 3). A higher proportion of the natural runoff was stored in the wet years of 2005, 2011, 2019, and 2023, when more than 40% of the natural runoff was captured in reservoirs. More than 30% of the natural runoff was stored in 2008, 2009, and 2017.

Figure 3. Graph showing increase in basin-wide reservoir storage as a function of natural flow at Lees Ferry. The solid line is linear relation fit to these data⁵.

The combined contents of Lake Mead and Lake Powell peaked on 8 July, but storage in each reservoir followed very different trajectories (Fig. 4). Lake Powell, which is upstream from Lake Mead, captured the inflowing snowmelt runoff and increased in storage by 2.2 million af between mid-April and early July while Lake Mead lost approximately 900,000 af. Now that the inflow season has ended, storage will gradually decline in Lake Powell and increase in Lake Mead.

Figure 4. Graph showing the distribution of reservoir storage in different parts of the Colorado River basin between 1 January 2021 and 15 July 2024.

The storage gains resulting from the 2024 runoff compensated for the consumption of reservoir storage that had occurred after the 2023 runoff season (Fig. 5). This year's peak of 30.0 million af is slightly more than the peak storage in summer 2023 that was 29.7 million af. This small increase in storage occurred despite a modest 2024 runoff season, because the basin's water managers had done a good job in conserving the gains of last year (see blog post of 21 May 2024). Only 2.2 million af was consumed or lost following the 2023 runoff season, and the gain of 2.5 million af in 2024 exceeded the preceding loss. Thus, basin-wide storage is ever so slightly better than last year, because we used so little water last year. We now begin a 9-month period when the job in front of us is to continue to reduce consumptive uses and losses until the onset of the 2025 snowmelt season. Let's not lose focus. We're all in this together.

Figure 5. Graph showing changes in reservoir storage between 1 January 2023 and 15 July 2024.

2. Natural Resources Conservation Service. Snow water equivalent data accessed at: https://nwcc-apps.sc.egov.usda.gov/awdb/basin-plots/POR/WTEQ/assocHUC2/14_Upper_Colorado_Region.html.

3. Bureau of Reclamation. Natural flow data accessed at: <https://www.usbr.gov/lc/region/g4000/NaturalFlow/provisional.html>. Lees Ferry data were released on 22 April and are provisional and based on the April 24-month study. This estimate will be revised in August.

4. This relation is $y = 4,048,600 * e^{(0.06632 x)}$, where y is the annual natural flow at Lees Ferry for the water year, in acre feet, and x is the peak snow water equivalent of the year, in inches. The R² of this relation is 0.73.

5. This relation is $y = -6,484,000 + 0.75833 X$, where X is the annual natural flow at Lees Ferry for the water year, in acre feet, and y is the increase in basin-wide reservoir storage that occurred during the snowmelt season typically between mid-April and mid-July. The R² of this relation is 0.92. Data used to calculate this relation do not include 2002 and 2012 when basin-wide storage decreased during the snowmelt season.

Section: Colorado River 2023 Water Use: An Optimistic Narrative

Source: <https://www.inkstain.net/2024/07/colorado-river-2023-water-use-an-optimistic-narrative/>

Preparing for A Thing I'm doing next week, I updated the Crazy Fleck Spreadsheet this morning of data from Reclamation's annual Lower Basin decree accounting reports.

Amid all the angst and rhetoric, it is easy to miss the salient fact made clear by this graph: Lower Basin water users have reduced their take on the Colorado River substantially since the early 2000s.

- Nevada's use was the lowest since 1992.

- Arizona's use was the lowest since 1991.

- Records that far back in time are tricky*, but California's take on the river in 2023 was appears to have been the lowest since the late 1940s.

To be clear, the use in the late 1990s and early 2000s was unsustainably large. Praise is due for shrinking Lower Basin use, but the praise should be tempered by the fact that that they didn't do it until the reservoirs had dropped to scary low levels.

But – crucially – everyone's economy is doing fine. We've absorbed dramatic water use reductions without harming the basic structure and function of the Lower Colorado River Basin communities that depend on the river.

Upper Basin Data

Working with Upper Basin data is trickier*. There's a new dataset from Reclamation that uses remote sensing to estimate consumptive use from 1971 to 2023. There's been a good deal of back-and-forth among Colorado River data nerds because of some confusing aspects of the data, which we hope to sort out soon to enable a more useful analysis. But the top line numbers tell a different but also ultimately an optimistic story.

The curve appears (see data nerd confusion caveat above) to show an upward trend since the 1970s with a huge amount of interannual variability. So we haven't hit the conservation brakes yet, at least at the basin scale. But it also is clear that the Upper Basin is using far less than the 7.5 million acre feet tagged for us in the 1922 Colorado River Compact.

We Can Do This

It is easy to get tangled these days in the anger and finger pointing about who should cut, and by how much, about who has already cut and how and why, about questions that are both technical but more importantly deeply emotional about equity and fairness. We need to remember and learn from our successes.

* A Note on Data

The Lower Basin data from 1964 to the present is contained in the decree accounting reports, prepared by the Bureau of Reclamation since the Supreme Court ruling in the Arizona v. California case back in the 1960s. Prior to that, I have stitched in a dataset created by the numbers wizards at the Metropolitan Water District of Southern California.

The Upper Basin data comes from the new Upper Basin Consumptive Uses and Losses reports published by the Bureau of Reclamation. The were originally published in May, then a revised set was published in June, I'm cautious in my analysis and citation because there are still some things I don't understand about them.

Section: A reminder to be careful how you think about “wasted” water

Source: <https://www.inkstain.net/2024/07/a-reminder-to-be-careful-how-you-think-about-wasted-water/>

A team out of Wyoming, including my Colorado River Research Group colleague Kristiana Hansen, has a new paper that reminds us that we need to be careful about how we thinking about conserving water that is being “wasted.”

Their case study is an area on the New Fork in Wyoming, a tributary of the Green, which is a tributary of the Colorado, where producers use flood irrigation on timothy grass to grow livestock forage.

Flood irrigation is often seen as “wasteful.” One approach is to install “more efficient” irrigation technology. But – and this is one of my repetitive talking points with students in the graduate water policy course I teach every fall – you need to flag the word “waste” when you see it in a water policy discussion and think carefully about how you’re using it.

That water is going somewhere, and doing something. You have to include this in your analysis. Maybe it’s really being “wasted”. But you may find that the place the water is going, and the thing that it’s doing, is valuable!

That’s what the Wyoming team found. Flood irrigation recharges the shallow aquifer – reducing the spring peak in the area’s streams, and slowly releasing that water back into those same streams in late summer. Which is crucial, in this case, for economically valuable fisheries – recreational brown trout fishing, to cite their analysis.

This is at the heart Bruce Lankford’s oddly named work on the “paracommons,” which has provided an enormously helpful analytical framework for my thinking about this stuff.

Cleaning up our urban sewage for reuse is super popular right now, and can in some cases be an enormously powerful water policy response to scarcity. But we’ve got to be mindful about where that “wasted” water is going and what positive benefits it is providing. Lots of inland urban cities in the southwestern United States treat their wastewater and return it to rivers, where it feeds ecosystems and downstream users.

We always have to consider the tradeoffs.

Section: Important update on the Bureau of Reclamation’s Boulder City lawn

Source: <https://www.inkstain.net/2024/07/important-update-on-the-bureau-of-reclamations-boulder-city-lawn/>

In my book *Water is for Fighting Over*, I delighted in this cheap shot at the Bureau of Reclamation’s Boulder City office –

... a grandiose white building atop a hill ... surrounded by an expanse of lawn that is embarrassing in a desert city that averages less than six inches of rain a year.

I am happy nine years after I wrote that, via Daniel Rothberg’s excellent water news, to correct the record. In response to Southern Nevada’s “non-functional turf” regulations, Reclamation will be tearing out the lawn. In classic federal fashion, the project is accompanied by a 400-page NEPA analysis:

According to the report, about 4 million gallons of water were used to maintain the landscapes at the two building. Redesigning them as desert-friendly xeriscape could result in water savings of more than half, or about 66%. The bureau started this work, at a cost of more than \$4 million, and it recently opened up the project to the public.

I'll be back in Boulder City week after next, I'll check on how things are going.

Section: Hustling to get Imperial Irrigation District water reduction tools in place

Source: <https://www.inkstain.net/2024/07/hustling-to-get-imperial-irrigation-district-water-reduction-tools-in-place/>

Janet Wilson had a super helpful piece this week in the Desert Sun about steps being taken (in a hurry) to get the institutional widgets in place to meet Lower Basin commitments to reduce water use under a deal hashed out in spring 2023 to head of Colorado River NEPA litigation.

If all goes as planned, growers and owners of farm fields could be paid \$300 per acre-foot for not irrigating alfalfa and other perennial feed crops for between 45 and 60 days. The plants would be stressed but would survive, and substantial water supply would instead be left in drought-depleted Lake Mead, which provides water for millions of people and millions of acres of farmland in California, Arizona and Nevada.

I wrote (with youthful enthusiasm) in my book *Water is For Fighting Over* about the potential of "deficit irrigation" as a water use reduction tool in Imperial and places like it. One of the reasons we have converged on alfalfa as a crop in the arid southwestern United States is how robust it is when the water runs short. From the book:

Farmers have known for years that when their water supply runs short, they can get away with skipping an irrigation cycle and their plants will survive. They'll just have fewer bales to feed to their cattle or send off to their dairy industry customers.

Do that intentionally, for money, and you have an adaptation tool that avoids fallowing entire fields or "buy and dry". This also works with Bermuda grass and Klein grass, two other forage crops grown in Imperial. Taken together, the three crops accounted for 233,000 of Imperial's 333,000 acres under active irrigation in June, according to IID's latest irrigation acreage report. (Total Imperial "farmable" acreage is 436,000 acres, the rest is either being fallowed or between crops right now.)

Deficit irrigation is one of three water conservation tools on the table for Imperial, as discussed in a draft Environmental Assessment released last week. Also on the table are on-farm efficiency improvements and straight up fallowing.

All involve federal money to compensate farmers (and their irrigation district).

Section: Texas v. New Mexico ruling creates interesting questions for Arizona v. Colorado on the Colorado River

Source: <https://www.inkstain.net/2024/06/texas-v-new-mexico-ruling-creates-interesting-questions-for-arizona-v-colorado-on-the-colorado-river/>

The U.S. Supreme Court's decision Friday in a lawsuit between Texas and New Mexico over Rio Grande water reframes discussions in Colorado River Basin, where the threat of a similar Supreme Court action looms.

Texas and New Mexico schemed a water-sharing agreement to settle the thing, but the Department of Interior intervened to say "Nope" to the deal, and the Supreme Court said "Yup" to Interior's "Nope," essentially saying the federal interest in water management was sufficient to block the states' attempt to settle their own affairs.

The Rio Grande Compact is fundamentally a contract among the three states that share the Rio Grande – Texas, New Mexico, and Colorado. (Colorado is technically a party to all of this, but the lawsuit's a Texas versus New Mexico dispute.)

But while it's a state-to-state thing, the Supreme Court's rulings in this case make legally clear that the United States government has a "distinctively federal" interest in water management on the Rio Grande in southern New Mexico and Texas, parallel to the states'.

This interest arises from the federal government's responsibilities under the legislation creating the Rio Grande Project, and the contracts flowing from that project.

To ensure that Texas receives its share of water, the Compact relies on the United States Bureau of Reclamation to operate the Rio Grande Project, an irrigation system in southern New Mexico.

So we have here a) states arguing over the meaning and interpretation of key terms of an interstate compact, and b) the federal government with a legal standing to step into that argument resulting from a "distinctively federal" interest in the outcome of that argument.

This seems analogous to the situation on the Colorado River.

First, as in the initial phases of Texas v. New Mexico, we have states in disagreement – in the Colorado River case over the meaning of Article III(d) of the Colorado River Compact, which discusses the flow of water past Lee Ferry. Is it a delivery obligation, or a non-depletion obligation?

Second, we have at least the potential for a "distinctively federal interest" flowing from the 1928 Boulder Canyon Project Act, which authorized construction of Hoover Dam and the resulting federal contracts for the delivery of water therefrom. To ensure that the Lower Basin states receive their share of the water, the Colorado River Compact relies on the United States Bureau of Reclamation to operate Hoover Dam and the irrigation and river management works downstream.

The parallel to the Rio Grande situation is striking.

If for whatever reason Colorado River management ends up at the Supreme Court when deliveries past Lee Ferry drop below a legally argued "delivery obligation," it would be consistent for Interior to assert a federal interest in ensuring that flows from the Upper Basin are sufficient to meet the requirements of the federal government's "distinctively federal interest."

It also would be consistent for the U.S. Supreme Court to be OK with that.

That would be bad for an Upper Basin being squeezed by climate change.

I am Not a Lawyer

There's a legal tangle in the decision's footnotes that I don't understand, but that's probably important. In the Rio Grande case, Texas filed suit, and the U.S. government intervened. Might the federal government have standing to initiate a lawsuit if the deliveries at Lee Ferry cross the tripwire? From said footnote tangle:

As with our 2018 decision, today's opinion says nothing about whether the United States could have initiated a Compact suit on its own.

I am a writer

While I am not a lawyer, I am enough of a water writer to have been charmed by the cheesy use of water imagery in the ruling. My favorite bit:

Conventional wisdom posits that, because time changes all things, no one can step into the same river twice.

My original joke here was going to be about my disappointment that Justice Jackson didn't cite to Heraclitus. But then upon further research (You didn't think I'm just wingin' it here, did you? You're payin' good money for this blog, I do my research!) I find that there is question about whether Heraclitus

really said it.

Revised joke:

I'm just disappointed that Justice Jackson didn't cite to Wittgenstein.

Section: Gila River Indian Community proposal for post-2026 Colorado River Management

Source:

<https://www.inkstain.net/2024/06/gila-river-indian-community-proposal-for-post-2026-colorado-river-management/>

Given the apparently unproductive state-to-state negotiations over post-2026 management of the Colorado River, it's worth examining, in our search for a path forward, some of the other proposals submitted to the Department of the Interior. (If you need some bedtime reading....)

One of the most interesting comes from the Gila River Indian Community. (Their March 29 letter to Reclamation lays it out.) Spanning the Gila River along the southern edge of Phoenix, the Gila River Indian Community has long done a masterful job of leveraging water in defense of its sovereignty – or maybe sovereignty in defense of its water?

This 2021 piece by Sharon Udasin does a great job of explaining the GRIC's success:

A riverbed that has been parched since the end of the 19th century — a portion of the historic lifeblood of the Gila River Indian Community — is now coursing again with water, luring things like cattails and birds back to its shores.

"You add water and stuff just immediately starts coming back naturally. Birds have returned and it's just such a different experience," says Jason Hauter, an attorney and a Community member. "It's amazing how much has returned."

The water, the cattails, and the birds, are part of a complex legal tangle that led to the 2004 Arizona Water Settlements Act, which ensured Central Arizona Project water – Colorado River imports, pumped up into the Gila River Valley from the river's main stem – to replace water stolen from the Gila River Indian Community by settlers a century before.

Access to that water has made the Community a power player in central Arizona water politics. But that water is now at risk as a result of efforts to cope with declining flows on the Colorado River. In particular, the Community views the Lower Basin States' proposal for post-2026 river management as (my words, not theirs) an assault on their sovereignty. Here's how they put it in their March letter to Reclamation:

Our initial analysis of the LB Alternative indicates that under their static reduction proposals, at a minimum, the Community's Colorado River supplies that we are currently using would likely be reduced by 130,000 acre-feet, 42% of our entire Colorado River supply, even after Arizona's firming obligations are met. These cuts

would become more draconian under both the LB Alternative and the UB Alternative when additional reductions beyond 1.5 million acre-feet are needed, which would be often based on our review of the various models we have seen.

Allocation of CAP water is crazy complicated, and I'm still trying to get my head around these details. But the Gila River Indian Community is essentially arguing that in the current proposal, which calls for cuts deep enough to eliminate the "structural deficit," Arizona is essentially bargaining away the Community's water.

So the Community has concerns with the Lower Basin proposal. But it has even greater concerns about the Upper Basin proposal, which argues that if even deeper cuts are needed, they should all fall on the Lower Basin.

The UB Alternative does not appear to us to present a reasonable or balanced approach at all, shifting all of the burden for addressing the current drought crisis onto the Lower Basin States.

The footnote to that paragraph is wonderful:

Reclamation has indicated that it will model any reasonable set of assumptions presented for consideration. The Community, for its part, does not view the UB Alternative as a reasonable set of assumptions, so the Community would understand and support Reclamation if it determined not to model the UB Alternative.

The Gila River Indian Community offers an alternative suggestion for managing Lower Basin cuts that's super interesting. Rather than what the Lower Basin Proposal offers – essentially a negotiated who-cuts-what set of numbers based on talks among the three states – the Community suggests cuts across the Lower Basin proportionally, based on the calculation of evaporation and system losses (which we're now supposed to shorten, apparently, as ESL):

We believe the United States has the authority to apply ESL to consumptive use in the Lower Basin in a proportional manner. It is our understanding that United States believes it has the authority to apply ESL to consumptive use in the Lower Basin on a proportional basis and, as such, we believe this should be modeled.

The Community's letter includes a strong emphasis on the federal government's trust responsibility to the basin's 30 Tribal Sovereigns. The letter makes clear that the federal government has a legal obligation "to find alternative water supplies for tribes that will be negatively affected by the Post-2026 Operations." As Gila River Indian Community Gov. Stephen Roe Lewis put it at this month's Getches-Wilkinson conference in Boulder, "First peoples of this land should be the last to be cut."

Section: Water Flowing, Again, in the Colorado River Delta

Source: <https://www.inkstain.net/2024/06/water-flowing-again-in-the-colorado-river-delta/>

Forgotten in all of the noise around the Colorado River right now is this moment of hope – water again flowing in the Colorado River Delta.

Under the 2017 agreement between the United States and Mexico known as Minute 323, we have 210,000 acre feet of water set aside for environmental flows through 2026 – one third provided by the United States, one third by Mexico, and one third by environmental NGOs – in the long-dry river channel through the Colorado River Delta.

Audubon's Jennifer Pitt's mention of the flow came during the last panel of last week's Getches-Wilkinson Center annual Colorado River conference at the University of Colorado Law School. Managing the pulse flow to maximize environmental benefit requires, ironically, the same sort of engineering that on a much larger scale dried the delta river channel in the first place – routing water through an irrigation system to deliver it at the point of maximum environmental benefit, feeding a strip of riparian vegetation. That's how we do environmental flows now.

It made me smile, remembering the joy of watching the pulse flow a decade ago, an event that was a pivot point in my life. It was a reminder that, amid sturm und drang of the current Colorado River, good stuff is possible.

Out of the shadows

Co-sponsored by the Water and Tribes Initiative, the conference again moved the role of the Colorado River Basin's 30 sovereign Tribal Nations into the foreground, in particular celebrating the new water settlement among the Navajo, Hopi, Southern San Juan Paiute tribes, the state of Arizona, and the federal government. It's a sweeping agreement that could, if it can cross the next hurdles before it, ensure water supplies for what one a member of my brain trust once described as the place of greatest water poverty in the nation.

"We refused to be in the shadows any longer," Hopi Chairman Tim Nuvangyaoma said during a Friday morning session.

Lorelei Cloud, vice chair of the Southern Ute Tribal Council and a member of the Colorado Water Conservation Board, talked about the importance of normalizing tribal voices at the decision making table, which should have been obvious a century ago, but is increasingly a no-brainer today.

As the Colorado River community debates where and how cuts should be made to bring water use into line with a climate change-shrunk supply, "First peoples of this land should be the last to be cut," Gila River Indian Community Gov. Stephen Roe Lewis said.

Still in the shadows

The shadowy ongoing discussions among the seven U.S. basin states got a brief airing in a panel of six of the seven states' principals – the governors' representatives in the ongoing negotiations. They appear as stuck as they've been since December's CRWUA fireworks and the competing proposals of March.

I am sympathetic to the difficult position these people are in – political demands from the home crowd to fight for their water colliding with the reality that the water the homers want is simply not there in the quantity they would like. The result was a litany of "praise us for the conservation we've already done" without much clarity beyond everyone's public negotiating positions: the Lower Basin's "We own the structural deficit and if deeper cuts are needed they need to be shared," and the Upper Basin's "We already suffer cuts, it's on the Lower Basin to cut deeper if needed."

On its face that's a conflict being readied for the Supreme Court, and I've begun to think seriously about what such a path might look like in practice. Everyone says they want to avoid this, yet seem powerless to prevent it. A friend noted the seeming powerlessness being voiced by the state officials, helpless to keep the bus they're driving out of the ditch.

They're like Howard the Duck, "Trapped in a world they never made."

Back out of the shadows: A C-SPAN for Colorado River Basin water management talks?

My favorite question of the day came from an audience member asking whether there should be some sort of a C-SPAN-like public forum so we could all watch the discussions now conducted behind closed doors. There was a time not that long ago that I would have seen that as a terrible idea. The people involved need a safe space to explore the sort of compromises that would get them crucified back home if they did it in public, I used to think.

But given the current logjam in the states' discussions, which seems to leave us at increasing risk of potentially disastrous litigation, I'm not so sure that the safe space is serving us particularly well. While the Basin States' discussion remains opaque and unproductive, in a way that increasingly doesn't seem to be serving me as a "stakeholder" whose community's water depends on the river, a bunch of parallel processes happening in far more public ways – see for example the discussion of tribal issues above, and the work on restoring environmental flows in the delta – seem increasingly to be where the useful action is.

John Berggren, from Western Resource Advocates, made this point in a talk about Colorado River process, quoting here from a chapter he and I and some other folks wrote for the book Cornerstone:

Inclusivity does not threaten the special status of existing decision-makers; inclusivity puts those decision-makers in a setting where they have more resources, information, and perceived legitimacy to bargain. The Colorado River Compact, after all, is a bargain—a negotiated agreement born from the realization that it was better to make a deal than to do nothing, to litigate, or to delegate important regional decisions to a distant national government.

Along the C-SPAN lines, Berggren noted the work of the Arizona Reconsultation Committee, which has been super C-SPAN-like in the way it has created a framework for a big messy public discussion in Arizona about the important questions.

The flaw in the current process is made clear by basin states' impasse.

Berggren warmed my heart with this quote from Reuel Olson, whose 1926 doctoral thesis was the first detailed academic look at the Colorado River Compact:

It does not help much to say that what is needed is an equitable division of water between the states. What is needed is a standard which may be applied in order to determine what is an equitable division.

(Fun aside: Comparing notes after his talk, both John and I seem to have bought our copies of Olson's book from the same Salt Lake City used bookstore.)

A century after Olson said that, we seem to have the same impasse. Tough negotiations by the various states trying to protect their own interests leaves out all kinds of equities, all kinds of values – including mine.

To paraphrase California's J.B. Hamby, all water users in the basin can reduce their use. I'm sure Hamby wasn't paraphrasing me when he said that, but he could have been. Here's how I put it in the concluding chapter of my book *Water is For Fighting Over*:

Recent years, especially during the drought in the twenty-first century, saw a rash of ... articles: "Scarce Water and the Death of California Farms," "The Dust Bowl Returns," "A 'Megadrought' Will Grip US in the Coming Decades", "Colorado River Water Supply to Fall Short of Demand."

These dire stories fuel fears that the only way to save yourself is to fight your neighbors for every last drop. But if you are able to sidestep the crisis narrative and recognize that your community can thrive with less water, then the fight with your neighbors seems less necessary and the risks of water wars and a crash diminish. It's time to stop fighting over water....

Section: Kuhn-Tara-Fleck on what comes next – the foundations of the Law of the Colorado River, shaky heading into the post-2026 world

Source: <https://www.inkstain.net/2024/06/kuhn-tara-fleck-on-what-comes-next-the-foundations-of-the-law-of-the-colorado-river-shaky-heading-into-the-post-2026-world/>

ALAMOSA, COLORADO – Meandering toward Boulder for this week's Getches-Wilkinson Center Colorado River conference, I stopped this evening in Alamosa, Colorado, in the San Luis Valley. I love the drive up the back way, through the San Luis Valley and into the heart of the Rockies, and I split it up into a couple of days this year to get some bike riding in.

Long western drives have always been a part of my process, quality thinking time, and the San Luis Valley is a great writing prompt. It's broad, high, pan flat, and a really good place to grow alfalfa and potatoes. (There's a flatbed of alfalfa in the Walmart parking lot next to my motel, headed for a dairy somewhere – future burgers and pizza cheese.)

When the railroad and the Mormons arrived in the 1800s, they started growing a lot of stuff to export, reducing the flow in the Rio Grande which, through a series of knock-on effects, led us in central New Mexico to import Colorado River water via the San Juan-Chama Project, which is why I'm headed to Boulder. For want of a nail....

Foundations of the Law of the River: Shaky

It's the San Juan-Chama linkage – critical to Albuquerque's water supply – that got me started working on Colorado River issues nearly 20 years ago, which led to a couple of books (Water is For Fighting Over, Science be Dammed) a growing list of academic publications, and this crazy blog, which I'm happy to report Emily Guerin called “influential”! The second book was a collaboration with Eric Kuhn, and during the years working at it we more than once met up at the Holiday Inn Express in Alamosa, midway between his home in Glenwood Springs and mine in Albuquerque, holed up in the breakfast area working through chapters. Is it possible to have fond memories of a Holiday Inn Express breakfast area? I do.

The collaboration continues, joined by my Utton Center colleague Rin Tara, with a couple of new papers digging into the history of the development of the Upper Colorado River Compact and its implications for 21st century river management. A preprint of the first of the two papers, a deep dive into the negotiation history, went up over the weekend and I already blogged about it.

A preprint of the second paper, Unfinished Business: 21st Century Questions Posed by Ambiguities in the Upper Colorado River Compact and the Law of the River, went up this morning. It's our attempt to work through the modern implications of that history for 21st century river management:

The Law of the River, particularly those components relating to Upper Colorado River water allocation, usage, and measurement, is rife with ambiguities and uncertainties. In the 21st Century, facing a future with less water, these ambiguities pose serious problems. The ongoing process for the development of the Post-2026 Colorado River Operating Guidelines presents an ideal opportunity to address these issues. This article gives a brief history of the Law of the River, with an eye towards the origins of the Stream Depletion Theory, beneficial consumptive use, and Tribal water allocations. The authors then discuss potential resolutions to these three problems. The article concludes by urging water leaders in the Upper Colorado River Basin to consider these solutions in their development of Post-2026 guidelines

All three of us will be in Boulder for Getches Wilkinson, say hi, we'd love to talk about this stuff!

Section: Horse Trading in the negotiation of the 1948 Upper Colorado River Basin Compact (and its implications today)

Source: <https://www.inkstain.net/2024/06/horse-trading-in-the-negotiation-of-the-1948-upper-colorado-river-basin-compact-and-its-implications-today/>

There was deep tension in Cheyenne, Wyoming, in September 1947 as the negotiators for the Upper Colorado River Basin states gathered to try to hammer out a deal to divvy up their share of the Colorado's water.

The 1922 Compact had split the river in two at Lee Ferry, and left for later the question of how to divide up the upper and lower divisions' shares of the river among the respective regions' states. By September 1947, the Upper Division states had been meeting off and on for more than a year, but progress was slow, stymied by a lack of the detailed water supply and use data some negotiators had hoped for.

Wyoming wanted to move quickly. As Eric Kuhn, my Utton Center colleague Rin Tara, and I write in a new paper:

Alluding to comments made by Utah's Governor Maw earlier in the day, (federal commissioner Harry) Bashore suggested "the Commission attempt to negotiate this compact on the basis of percentage without attempting a lot of fine-haired calculations as to what the water is because it will change from year to year." Adding that he was putting it to them plainly and "was not pulling any punches," Bashore noted he would like "you folks on the Commission to do the same thing."

Wyoming's L. C. Bishop immediately responded that there was no need to wait for the report of the Engineering Committee. It was now time "to go ahead and negotiate the compact with the information we have." He further noted that there are more lands than there is water, so it was time for the horse trading. Colorado's Stone objected, noting "maybe in the end there will be an element of horse-trading in it, but Colorado feels we shouldn't trade horses in the dark," but added "there is an end and a limit to engineering studies and adding that there was much that could be done while the engineering work was being completed."

The paper builds on Eric's exhaustive review of the Upper Basin Compact negotiations, something which strangely we couldn't find that any Colorado River Basin scholar had done in at the level of detail we felt was needed to understand the roots of today's conflicts over questions still shrouded in ambiguity:

- How do we quantify consumptive use in the Colorado River Basin?
- How much water must the Upper Basin send down past Lee Ferry to users in the Lower Basin?

A Horse Named the Stream Depletion Theory

Stone's argument against horse trading held for a time, and it was not until the marathon seventh negotiating meeting the following July that the horse trading finally began. The results have profound implications for modern Colorado River management.

To get to a final agreement, at the urging of Wyoming's Wehrli, Bashore suggested the Commission break for a series of states only and state-to-state caucuses. It was also now time for Bashore to conduct some shuttle diplomacy. Stone, who up until this point had resisted any "horse trading," realized he now needed to make some trades. He needed a horse named "The Stream Depletion Theory." It appears that to get that horse, he gave Wyoming a horse named "A 14% Apportionment."

The meaning of that language – "stream depletion theory" – is arcane, but the implications today are profound. Colorado pushed hard to argue that Upper Basin use should be measured not be diversions less return flows at each point along the river, but rather by the total impact of each state's actions on the river's flow at Lee Ferry. To win Wyoming's backing, Stone offered Wyoming a juicy entitlement.

The result, eventually enshrined in the Upper Basin Compact, has profound effects on today's arguments over Colorado River water.

Section: End of Season Wrap-Up – Holding on to What We've Got

Source: <https://www.inkstain.net/2024/05/end-of-season-wrap-up-holding-on-to-what-weve-got/>

A guest post by Jack Schmidt, director of the Center for Colorado River Studies at Utah State University

By Jack Schmidt | May 22, 2024

Opportunities to rebuild basin-wide reservoir storage have been rare in the 21st century.

On April 3 2024, the snow accumulation season in the Colorado River watershed ended and the snow water equivalent of the snowpack of the Upper Basin peaked. Two weeks later on April 17, the watershed's reservoirs¹ dipped to their lowest level of the year. Now runoff is underway, and the watershed's reservoirs are beginning to refill. This is a good time to assess how well water managers

did during the past nine months to retain the bounty of 2023's excellent runoff season, an essential part of rebuilding reservoir storage and regaining basin-wide water supply security.

The good news is that water managers did quite well, and reservoirs lost only 26% of the total amount accumulated during the 2023 runoff season. This was the smallest loss of any year in the last decade. Most of the decrease in storage that followed last year's snowmelt inflow occurred in Upper Basin reservoirs, and Lake Mead and Lake Powell lost only 5% of the storage that accumulated in those two reservoirs. It is imperative that water managers continue to work to reduce consumptive uses, reduce losses, and retain the bounty of the few unusually wet years of the 21st century, as they did following the 2023 snowmelt.

Opportunities to rebuild basin-wide reservoir storage have been rare in the 21st century, and there have been many years in which there is significant risk of basin-wide reservoir storage depletion. Hydrologic and reservoir storage data between 2014 and 2023 indicate that annual snowmelt-derived gains in reservoir storage exceeded losses when natural flow at Lees Ferry exceeded 13.7 million acre feet per year (af/yr). Annual flows less than this amount occurred in 16 years of the 21st century. Opportunities to significantly rebuild basin-wide reservoir storage existed when natural flow exceeded 15.8 million af/yr, which only occurred six times in the 21st century. Development of a sustainable policy for managing Colorado River reservoir storage must focus on reducing consumptive uses and losses in both wet and dry years.

To recap, the natural flow of the Colorado River at Lees Ferry in 2023 was the third highest of the 21st century and was exceeded only in 2011 and 2019 (Table 1). Unregulated inflow to Lake Powell in Water Year (WY) 2023 was ~13.4 million af².

In response to this large runoff, the basin's reservoirs recovered a significant amount of storage. The watershed's reservoirs reached their maximum in mid-July (13 July 2023) when total storage was 29.7 million af. The increase in basin storage between mid-April and mid-July was 8.38 million af and was the largest single-year increase in storage in the last decade, and approximately 1 million af more than the increase in storage that had resulted from the inflows of 2019 (Table 2).

However, the runoff in 2023 did not eliminate critically low reservoir storage conditions. The increased reservoir storage that peaked in mid-July 2023 recovered storage to the amount it had been in mid-February 2021 in the early stages of the 2020-2022 water crisis (Fig. 1). Based on average annual water consumption³ 2023's runoff would need to be repeated five more times to refill the reservoir system. Good runoff years rarely occur consecutively. The projected unregulated inflow to Lake Powell in 2024 is estimated to be only 81% of average.

Figure 1. Graph showing reservoir storage in the Colorado River basin between 1 January 1999 and 1 May 2024. Note that at the peak of storage in mid-July 2023, the total stored water supply was the same as it had been in mid-February 2021.

When we entered mid-summer 2023, I expressed concern about water managers' ability to conserve the benefit of 2023's runoff season, because we had not done so in previous years of good runoff. In those years, the benefit of reservoir storage recovery was not retained for more than two years (see blog post from October 2023). The benefit of 2011, the largest runoff of the 21st century, had been completely consumed in 19 months, and the benefit of large runoff in 2019 had been consumed in 24 months. I suggested that public understanding about the status of reservoir storage and the need to conserve the bounty of good years would be improved if water managers regularly reported how much of the previous year's inflow benefit was retained. Such a metric could highlight success in rebuilding water storage or could be used to sound a warning of the need for additional conservation.

Throughout winter and early spring 2023 and 2024, I reported on the status of reservoir storage and showed that water managers were successfully conserving reservoir storage. Between mid-July (13 July 2023) and mid-April (17 April 2024), total basin-wide reservoir storage lost only 2.2 million af (Fig. 2) which was 26% of the total "gains" of the 2023 snowmelt season. Most of this decrease in storage occurred upstream from Lake Powell, where reservoirs lost 1.4 million af. In contrast, storage in the

Lake Powell-Lake Mead reservoir system decreased by only 0.83 million af.

Figure 2. Graph showing reservoir storage in the Colorado River basin between 1 January 2023 and 1 May 2024.

The percentage of the accumulated snowmelt in 2023 that was consumed or otherwise lost from reservoirs in the subsequent months was less than in any other year of the past decade and was less than following the 2019 runoff season and significantly less than the years between 2014 and 2017 when runoff was moderately good (Table 2). I compared the rate and magnitude of decrease of reservoir storage in 2023-2024 with similar data for the previous nine years. The results are presented in a complicated Figure 3. Each line on this graph is the loss in storage in each year, plotted as the cumulative decrease in storage from the peak that had occurred in early summer. Lines that plot higher on this graph reflect smaller decreases in basin storage. The decrease in storage was notably large after the 2020 snowmelt season; total basin storage was nearly 7 million af less in spring 2021 than it had been in summer 2020. There were also large reductions after the snowmelt inflows of 2018 and 2021. In contrast, the reduction in storage after the 2023 runoff season (the thick blue line) was smaller than in the other years; this pattern is reflected by the thick blue line that plots higher on Figure 3 than in most other years.

Figure 3. Graph showing the decrease in reservoir storage during late summer, fall, winter, and early spring following each year's snowmelt season.

Although the combined storage contents of Lake Mead and Lake Powell reflect the balance (or imbalance) between basin water supply and consumptive use, the trajectories of individual reservoirs also result from reservoir operational rules specific to each facility. Lake Powell reached its peak storage of the year in early July (8 July 2023; 9.67 million af) and subsequently lost 2 million af by mid-April, because water was transferred downstream (Fig. 4). Storage began to accumulate again in Lake Powell in mid-April (18 April 2024). In contrast, storage in Lake Mead steadily increased between August 2022 and early March (4 March 2024), gaining 2.7 million af of storage. Lake Mead has been losing storage since early March.

Figure 4. Graph showing the distribution of reservoir storage in different parts of the Colorado River basin between 1 January 2021 and 1 May 2024.

The trajectory of storage in Upper Basin reservoirs differed between those facilities authorized or linked to the Colorado River Storage Project (CRSP)⁴ in contrast to other facilities (Fig. 4). Peak storage upstream from Lake Powell peaked in early (facilities unrelated to the CRSP peaked on 5 July 2023 at 3.69 million af) to mid-July (CRSP related facilities peaked on 15 July 2023 at 5.79 million af). Storage in facilities unrelated to the CRSP was quickly reduced to approximately 3 million af by mid-September, and storage was maintained at that quantity until the beginning of the 2024 snowmelt season. In contrast, storage in CRSP related facilities progressively lost storage of approximately 0.8 million af until mid-February 2024 when storage stabilized at approximately 5 million af. The longer period of declining storage in CRSP-related facilities was caused by policies related to transferring water to Lake Powell.

Insights about the Future

The data and analyses presented above provide insight about the likely trajectory of future Colorado Basin reservoir storage if no changes are made in policies concerning consumptive use and reservoir operations. During the past decade, the increase in basin-wide reservoir storage is well predicted by a power function based on the natural flow at Lees Ferry⁵ (Fig. 5).

Figure 5. Graph showing the relationship between annual natural flow at Lees Ferry and increase to basin-wide total storage during the snowmelt inflow season between 2014 and 2023.

The proportion of snowmelt-derived gain in storage subsequently lost during the following nine months is well predicted as an inverse power function⁶ of the increase in storage. The greater the increase in storage, the smaller the proportion of that increase subsequently lost. In years when there is little

increase in storage, basin-wide consumptive uses and losses far exceeded the annual increase in storage (Fig. 6). Such was the case in 2018 and between 2020 and 2022.

Figure 6. Graph showing the proportion of the annual accumulated reservoir storage consumed or lost during the following nine months prior to the beginning of the next runoff season.

These correlations indicate that annual consumption and losses in excess of annual storage gains occurred when gains were less than approximately 3.2 million af. Between 2014 and 2023, storage gains were less than this amount when natural flows were less than approximately 13.7 million af, which occurred in 16 years of the Millennium Drought. Significant retention of reservoir storage, defined as retention of at least 50% of the annual accumulation, occurred when storage increased by at least 5.7 million af. Such an increase of storage only occurred when natural runoff exceeded 15.8 million af (Fig. 5), which only occurred six times between 2000 and 2023.

Take-Home Messages

The essential purpose of negotiating new reservoir operational guidelines for the Colorado River basin is to maintain sufficient reservoir storage to provide a reliable and secure water supply. At the beginning of the 2024 snowmelt season, basin-wide reservoir storage is comparable to what it was in late spring 2021, demonstrating that the Millennium Drought water crisis persists. The opportunity for significant retention of the benefits of significant increases in reservoir storage exist when natural flow exceeds approximately 15.8 million af, a situation that has rarely occurred since 2000. When natural flow is less than approximately 13.7 million af, there is significant risk of depletion of basin-wide storage. Development of a sustainable policy for managing Colorado River reservoir storage must focus on reducing consumptive uses and losses in both wet and dry years.

2. The estimated natural flow at Lees Ferry is greater than the estimated unregulated inflow to Lake Powell, because natural flow is an estimation of what the river flow would be in the absence of humans, whereas unregulated inflow is the estimation of inflow in the absence of upstream reservoirs but with existing consumptive uses and losses.

3. Average annual consumptive uses and losses in the Colorado River, including reservoir evaporation and use in Mexico was 15.0 million af/yr between 2001-2020, based on Bureau of Reclamation reports. Between 2016 and 2020, consumptive uses and losses averaged 14.5 million af/yr.

4. Facilities authorized by or linked to the CRSP include Flaming Gorge, Navajo, Blue Mesa, Fontenelle, Morrow Point, and Crystal reservoirs.

5. $y = (1.5699 \times 10^{-22}) (x^{3.9677})$; $R^2 = 0.96$, where y is increase in basin reservoir storage during the snowmelt season and x is the annual natural flow at Lees Ferry in the same year.

6. $y = (4.7903 \times 10^7) (x^{-1.1806})$; $R^2 = 0.99$, where y is the proportion of the increase in storage consumed or otherwise lost in the subsequent nine months and x is the increase in storage during that snowmelt season.

Section: New Colorado River Guidelines are Only the Beginning

Source: <https://www.inkstain.net/2024/05/new-colorado-river-guidelines-are-only-the-beginning/>

Much attention is focused right now on rewriting Colorado River operating rules, to replace the soon-to-expire 2007 reservoir operating guidelines. But there is a growing frustration that the struggle to solve that relatively narrow problem “mass balance” problem (how much water, and where?) leaves out a range of incredibly important issues:

If a reminder of this reality were needed, it was provided in the majority of comments submitted as part of the EIS scoping process, summarized by Reclamation as urging a “more holistic view of the system to include the integrity and health of the river and its tributaries,” a perspective which, among other

things, includes a consideration of “environmental justice,” “preserving natural and cultural values,” and “acknowledging and incorporating the rights and authorities of all Basin sovereigns.”

That’s from a new policy brief from my friends and colleagues at the Colorado River Research Group, a collaborative of researchers across the basin whose mission is to provide “an independent, scientific voice for the future of the Colorado River.” The brief grew out of conversations among the group’s members about both the strengths, as well as the shortcomings, of the current process.

We are mindful that much of what CRRG has been advocating for is directly on the table in the various proposals now being considered for post-2026 river management:

New and improved reservoir management strategies are clearly needed and, at a minimum, will need to accomplish the following items discussed in earlier Colorado River Research Group (CRRG) publications:

- Address those rules that have institutionalized the structural deficit, largely at the expense of protecting (and recovering) reservoir storage;- Better integrate the latest science about climate change and ongoing basin aridification, as well as real-time seasonal weather data, into the planning and operational framework;
- Improve incentives, opportunities, and mechanisms for all water users to participate in conservation programs and other efforts promoting system flexibility and resilience; and,- Interface with new agreements with Mexico focused on protecting reservoir levels, habitat restoration, and shortage-sharing.

But there are so many other important issues left untouched by the P26 process (sorry, yes, some of us have started shortening it to “P26”) that the list we came up with among CRRG members is too long to blockquote here in a blog post – click through to read the white paper, it’s not too long.

What we advocate for in the paper is that the other issues not be lost in our rush to solve the mass balance problem.

Section: Game theory on the Colorado River: The prisoner’s dilemma

Source: <https://www.inkstain.net/2024/05/game-theory-on-the-colorado-river-the-prisoners-dilemma/>

All of the controversy relative to the utilization of the Colorado is hampering the fullest development of the stream. As Governor Pittman said, it is impossible to plan ahead with any assurance when there is such conflict. Solution of all these differences is admittedly not easy. The economic and social future of the various Colorado river states will be influenced to a very great extent by the decisions. Naturally, each state seeks the most advantageous settlement for itself. The danger is that too much emphasis on the selfish advantage of each state will do each more harm ... than each could possibly benefit through pressing its selfish viewpoint to the end.

Some time these conflicts will have to be settled—and the settlement, if it is to be effective and enduring, will have to be on a reasonably fair basis to all. Continuing delay in reaching settlement is more harmful than compromise.

Salt Lake Telegram, Feb. 2, 1948

Section: What Happens if there is no Agreement on Post-2026 Colorado River Operations?

Source:

<https://www.inkstain.net/2024/04/what-happens-if-there-is-no-agreement-on-post-2026-colorado-river-operations/>

By Eric Kuhn

Given how far apart the competing proposals from the Colorado River Upper and Lower Division States are, a legitimate question is – “what happens if we get to the summer of 2026 and there is still no agreement on the post-2026 operational guidelines?” Well, believe it or not, that is a question upon which the seven basin states and the Secretary of the Interior already agree. The Upper Division States made the following comment in their proposal:

“As of the date of this submittal, Reclamation has not disclosed the No Action Alternative for this EIS. The No Action Alternative must acknowledge that, upon expiration of the 2007 Interim Guidelines, the operating criteria for Lake Powell and Lake Mead will revert to the Long-Range Operating Criteria (LROC) used to model baseline conditions in the Final Environmental Impact Statement for the Interim Surplus Guidelines dated December 2000.”

The language quoted by the Upper Division States letter comes from the Record of Decision (RoD) for 2007 Interim Guidelines (Section 8. C.). This termination language was included with the consent of the states. So, the obvious questions are – if the operating criteria are to revert to the LROC, what are the LROC, and what specifically do they mean for the operation of Lakes Mead and Powell?

What is the LROC?

The long-range coordinated operating criteria or “LROC” are a requirement of the 1968 Colorado River Basin Project Act, the federal legislation that, among many other things, authorized the Central Arizona Project (CAP). Prior to the creation of the 2007 Interim Guidelines for the coordinated operation of Glen Canyon and Hoover dams, the LROC created the framework for determining how much water to release from Lake Powell to Lake Mead each year.

The 60s were a turbulent decade for the Colorado River Basin. In 1962 Flaming Gorge and Navajo Reservoirs began filling, then in 1963, Lake Powell began filling. To manage the filling of over 30 million acre-feet of vacant space (25 maf in Lake Powell), in 1962 the Secretary issued filling criteria. And, as luck would have it, the 1960s were relatively dry. The filling criteria were controversial in both basins. A significant concern was power generation. The Upper Basin wanted to fill the reservoirs as fast as possible so the Upper Basin fund would begin accruing revenues that would be used to subsidize irrigation projects under the 1956 Colorado Storage Project Act. The Lower Basin was concerned with the impact of power generation at Hoover Dam – the revenues from which were being used to repay the federal treasury for Hoover Dam and other projects on the lower river.

The Upper Division States became concerned that the Lower Division States would continually interfere with the storage of water in the Upper Basin Reservoirs using Article III(e) of the Colorado River Compact, which states:

“The States of the Upper Division shall not withhold water, and the States of the Lower Division shall not require the delivery of water, which cannot reasonably be applied to domestic and agricultural uses.”

The Upper Division States (via the UCRC) decided they needed to use the federal legislation then pending before Congress to authorize the CAP to clarify the rules under which water would be stored and released from Lake Powell and the other CRSP storage reservoirs. A critical question that needed to be answered was how much holdover storage the Upper Division States would need to meet their compact obligations under Articles III(c) and III(d) of the compact. This was a difficult question to answer given that the two basins had very different interpretations of these provisions, especially the

obligation off the Upper Division States to Mexico under the 1944 Treaty. Today the two basins still have very different interpretations of these provisions.

The specific language of Section 6 of the 1968 Act was negotiated by the states with input from Reclamation. With Congressman Aspinall (the House Interior and Insular Affairs Committee Chair from Western Colorado) insisting that operating language be included in any legislation, the UCRC had the better hand. Colorado's Felix Sparks (its CWCB Director and UCRC Commissioner) said "we wrote every word of Section 6."

The requirement for the LROC is included in Section 602(a):

"SEC. 602 (a) In order to comply with and carry out the provisions of the Colorado River Compact, the Upper Colorado River Basin Compact, and the Mexican Water Treaty, the Secretary shall propose criteria for the coordinated long-range operation of the reservoir constructed and operated under the authority of the Colorado River Storage Project Act, the Boulder Canyon Project Act, and the Boulder Canyon Project Adjustment Act. To effect in part the purposes expressed in this paragraph, the criteria shall make provision for the storage of water in storage units of the Colorado River storage project and releases of water from Lake Powell in the following listed order of priority:

(1) releases to supply one-half the deficiency described in article III (c) of the Colorado River Compact, if any such deficiency exists and is chargeable to the States of the Upper Division, but in any event such releases, if any, shall not be required in any year, that the Secretary makes the determination and issues the proclamation specified in section 202 of this Act;

(2) releases to comply with article III(d) of the Colorado River Compact, less such quantities of water delivered into the Colorado River below Lee Ferry to the credit of the States of the Upper Division from other sources; and

(3) storage of water not required for the releases specified in clauses (1) and (2) of this subsection to the extent that the Secretary, after consultation with the Upper Colorado River Commission and representatives of the three Lower Division States and taking into consideration all relevant factors (including, but not limited to, historic stream-flows, the most critical period of record, and probabilities of water supply), shall find this to be reasonably necessary to assure deliveries under clauses (1) and (2) without impairment of annual consumptive uses in the upper basin pursuant to the Colorado River Compact: Provided, That water not so required to be stored shall be released from Lake Powell: (i) to the extent it can be reasonably applied in the States of the Lower Division to the uses specified in Article III(e) of the Colorado River Compact, but no such releases shall be made when the active storage in Lake Powell is less than the active storage in Lake Mead, (ii) to maintain, as nearly as practicable, active storage in Lake Mead equal to the active storage in Lake Mead, but no such releases shall be made when the active storage in Lake Powell is less than the active storage in Lake Mead, (ii) to maintain, as nearly as practicable, active storage in Lake Mead equal to the active storage in Lake Powell, and (iii) to avoid anticipated spills from Lake Powell."

Section 602(b) required that the Secretary submit draft criteria to the states by January 1, 1970, and, after receipt of comments, adopt criteria by July 1, 1970.

The language of the first two priorities for releases from Glen Canyon Dam appears straight forward, but there are still implementation questions. The first priority is to release water to meet the obligation of the Upper Division States to Mexico, if any. Today, the Upper Division States believe they have no obligation to Mexico, but the Lower Division States believe it's at least 750,000 af per year. Thus, lacking either an agreement among the states or an interpretation of Article III(c) by the Supreme Court, this priority is unquantified. The second priority is to meet the Upper Division State's 75 maf every ten years non-depletion obligation under III(d). It's an average of 7.5 maf per year, but the language does not require a specific flow amount in any one year. To add uncertainty, the Upper Division States believe that if climate change, not Upper Basin depletions, is a cause of ten-year flows falling below 75 maf, there is no violation of Article III(d).

The third priority is more complicated (perhaps convoluted). It's designed to address the question of how much holdover storage the Upper Basin can keep in the Lake Powell and the other CRSP reservoirs. Although the language may be confusing, the intent is simpler. The Upper Basin may keep as much storage as the Secretary determines necessary to allow the States of the Upper Division to meet their compact obligations during the most critical drought of record. If there is more than that amount in storage, then they must share the extra water with the Lower Basin, but only to the extent necessary to equalize active storage in Lakes Mead and Powell. Thus, was borne the term "equalization."

After the 1968 Act passed, the Department of the Interior immediately began an effort to prepare a draft LROC. The effort involved a federal-state task force. It also involved the first comprehensive use of a computer to simulate numerous different reservoir operational scenarios (I wonder how many of us still remember the days of computer punch-card decks?). Interior officials quickly realized that the effort was going to be complicated and contentious. For Interior's version of the events, see Chapter VII of the 1978 version of the Hoover Dam Documents. The contentious issues were the annual release amount from Glen Danyon Dam and how to calculate the amount of allowable holdover storage (now commonly referred to as "602(a) storage"). Colorado's Felix Sparks put it this way: "Everything we do at Glen Canyon Dam is ultimately about the Colorado River Compact."

After numerous meetings and conferences, on June 9, 1970, Secretary Walter Hickel formally approved the first LROC. Much of the language parroted the 1968 Act provisions for the CRSP reservoirs and for Lake Mead, the language of the 1964 decree implementing the 1963 decision in *Arizona v. California*. For the Upper Division States, however, the Secretary made three decisions that they strongly opposed. First, the LROC included a provision for an annual "minimum objective release" from Glen Canyon Dam of 8.23 maf. The problem was that this number was derived as 7.5 maf plus 750 kaf (the LB's interpretation of the UB's Mexican Treaty obligation) less 20 kaf (the mean flow of the Paria River). The Secretary accepted recommendations from the UCRC for language clarifying that the minimum objective release was an "objective" not a "requirement" and that the Secretary was not interpreting the Compact. The second problem was that the Secretary rejected their recommendation that a probability rule curve be used to set the 602(a) level. Instead, the Secretary determined that the 602 (a) level would be set on an annual basis. Finally, with the adoption of the LROC, the UCRC recommended that the lake Powell filling criteria be terminated. The problem with the filling criteria from the Upper Division States perspective was that it required revenues from power generation at the CRSP units be used to keep the Lower Basin whole for the impact of filling Lake Powell on power generation at Hoover Dam. The Secretary rejected this recommendation and the filling criteria stayed in effect through 1980.

The rejection of its important recommendations left the UCRC members bitter. New Mexico State Engineer and UCRC Commissioner, Steve Reynolds, put in his plain-spoken language – "they crammed it down our throats" (expletive deleted).

So what happens if we continue to use the LROC?

So, the big question is – if there is no agreement on post-2026 operating guidelines between the basins and the Secretary determines that the annual operation of Lakes Mead and Powell for Water Year 2027 (and perhaps beyond) will be guided by the LROC, what does this mean for the reservoirs? For Lake Powell, the answer may be not much. It means an annual release of 8.23 maf (or perhaps 8.1 maf -taking into account the river inflows immediately below the dam). Depending on the runoff in 2025 and 2026, there's a reasonable chance that this would be the amount released under a continuation of the 2007 Interim Guidelines. The differences are that there would be no balancing of the storage levels in Lakes Mead and Powell as is allowed by the three balancing tiers of the 2007 Interim Guidelines and if the next two years are dry, there could be some debate over whether the Secretary has the authority to reduce releases below 8.23 maf – something that might be needed given the need to protect elevations below 3,500 feet at Lake Powell because of concerns about use of its outlet works.. I believe the answer to this question is clearly yes. In a June 2, 2005, letter, Secretary Norton wrote "the Department

retains the authority pursuant to applicable law and the Operating Criteria to adjust releases from Glen Canyon Dam to amounts less than 8.23 million acre-feet per year. Specifically, the Department transmitted the following statement to the Governors of each of the Colorado River Basin States on June 9, 1970: “...(T)he operating Criteria imposes no firm or fixed obligation the 8.23 million acre-feet be released each year from lake Powell. That quantity is stated as an “objective””

As for releases greater than 8.23 maf, under the authority referenced in the June 2, 2005, letter the Secretary might also have the authority to increase releases. The LROC, however, anticipated that the occasional releases greater than 8.23 maf would be made as equalization releases under the Section 602(a)(iii) third priority. The LROC provides that the Secretary determines the equalization or 602(a) storage level annually, but it appears that in the near to mid-term future, there is very little risk that the storage level in Lake Powell will be anywhere near a level that would require a large equalization release (as happened in 2011). Based on the table in the 2007 Interim Guidelines, the end of water year elevation of Lake Powell would have to exceed 3666' (100'+ higher, or 10+ maf more than the current level). Further, I would argue that the 3666' level is no longer current. It assumed that 1954-65 was the critical drought. The recent 2000-2022 drought is every bit as dry and twice as long as the 1954-65 drought, therefore, a better estimate of today's 602(a) level is “all of the available CRSP storage plus a lot more.” The Upper Division State's proposal, in fact, does not even include an equalization provision. Furthermore, in separate studies in 1969 and 2004, Reclamation concluded that in the future, the 602(a) level would exceed the available CRSP storage capacity. This is one of the reasons that in 1970, the Secretary rejected the rule curve. Reclamation modelling concluded that with future Upper Basin development, the result of applying the UCRC's proposed rule curve (98.4%) meant that all available CRSP storage was needed to protect the Upper Basin. For more information on “602(a),” see Reclamation's 2004 Environmental Assessment on this subject. The Upper Basin depletions have not achieved the level of development anticipated in either 1969 or 2004, but natural flows have been far less. From Lake Powell's perspective there is no difference between upstream depletions and climate change caused reductions in natural flow.

For Lake Mead, operating under the LROC may be more complicated. The “70R” criteria is a flood control strategy for Lake Mead. Given the current reservoir levels, the chances of encroaching on the flood control space in Lake Mead is very small. The most pressing problem would be that after the termination of the 2007 Interim Guidelines, there would be no “current” guidance on the specifics of when and how much for the shortage levels that would still need to be imposed on Lower Basin mainstem users in 2027 and beyond. Under the 1964 Decree, however, the Secretary has the clear authority, if not a mandate, to set shortages. I think it's likely that the Secretary and three Lower Division States would agree on Lake Mead deliveries. A more difficult question may be the level of the annual deliveries to Mexico. If there is no internal U.S. agreement on the post-2026 guidelines, will Mexico be interested in extending Minute 323?

Operating under the LROC might be manageable

The bottom line is that if there is no agreement among the Basin States for consensus post-2026 operating guidelines, it may not be the ideal outcome the states want but given the broad authority and flexibility the Secretary has under the 1970 LROC, the situation would be manageable. In fact, operating Lake Mead and Lake Powell under the flexibility provided by the 1922 Compact, Section 602 of the 1968 Act, and the 1970 LROC might provide opportunities for a more flexible management approach that can be designed to address a broader range of issues, including balancing recreation resources, and addressing environmental conditions in the Grand Canyon.

Section: Five Major Proposals for Post-2026 management of the Colorado River

Source: <https://www.inkstain.net/2024/04/five-major-proposals-for-post-2026-management-of-the-colorado-river/>

With the submission of two additional proposals last week, we now have five major proposals for post-2026 Colorado River management.

The folks at the Water and Tribes Initiative have helpfully organized them in a single place. (Click on the

“Proposed Alternatives for Post-2026 Operating Guidelines” bubble.)

Tribal Principles

A set of guiding principles proposed by 17 of the basin's sovereign indigenous communities. (click here)

Upper Basin Proposal

What the label says, you already know about this one. (click here)

Lower Basin Proposal

What the label says, you already know about this one. (press release, alternative, presentation)

Environmental NGOs

The “Big Seven” Colorado River Basin environmental groups (click here)

Lake Powell/Grand Canyon/Lake Mead Ecosystem Proposal

A proposal from Jack Schmidt, Eric Kuhn, and John Fleck suggesting ways to manage the storage and distribution of water to provide more flexibility for environmental and other non-water supply benefits. (click here)

Section: New Mexico and the Colorado River

Source: <https://www.inkstain.net/2024/03/new-mexico-and-the-colorado-river/>

A guest post from my Utton Center colleague and Colorado River research collaborator Rin Tara, staff attorney and water policy and governance analyst at the Utton Center, University of New Mexico School of Law.

By Rin Tara

Over the last year, I had the joy of working with John on a law review article addressing potential risks to New Mexico's San Juan-Chama Project water supply. The paper, titled Last Call: The Limitations of New Mexico's Existing Water Management Framework in the Face of Reduced Colorado River Water Deliveries, was published earlier this month in the Colorado Environmental Law Journal.

The purpose of the project was to dig into the law surrounding the New Mexico's San Juan-Chama Project, which imports Colorado River water into the Rio Grande Basin that has become vital to water supplies in Santa Fe, Albuquerque, and New Mexico's Middle Rio Grande Valley. Our goal was to understand the durability of New Mexico's annual Colorado River deliveries. The short summary of our learnings is that these deliveries are vulnerable to provisions built into the legislation. The sections below are pulled from the paper, which I (self-servingly) recommend everyone read in full.

In 1962, Congress passed an Act authorizing the construction of the Navajo Indian Irrigation Project and the beginning stages of the SJCP. 76 Stat. at 96. The Act states that:

[T]he Secretary shall operate the project so that there shall be no injury, impairment, or depletion of existing or future beneficial uses of water within the State of Colorado, the use of which is within the apportionment made to the State of Colorado by article III of the Upper Colorado River Basin compact, as provided by article IX of the Upper Colorado River Basin Compact and Article IX of the Rio Grande compact.

This passage references two sections of the UCRBC. Article III of the UCRBC establishes percentages of the annual Colorado River flow allocated to each Upper Basin State.¹⁵³ New Mexico, as noted earlier, is entitled to 11.25 percent of the Upper Colorado River Basin Colorado River allocation. Colorado, in contrast, receives 51.75 percent. Although the Upper Basin states are subject to significant Colorado River curtailment within the Upper Basin should they exceed their UCRBC percentage allocation, sometimes referred to as the “penalty box,” New Mexico may be forced into curtailments before exceeding their full 11.25 percent allocation. This Act is worded such that New Mexico is not only prohibited from exceeding its 11.25 percent cap on Colorado River water usage, but New Mexico may also be required to curtail its SJCP water usage if the SJCP water diversions interfere with Colorado receiving its full 51.75 percent of the Upper Colorado River allocation. This language is significant because it permits Colorado to require New Mexico to reduce its SJCP water use if Colorado determines that the SJCP diversions are impeding Colorado's ability to make full use of its Colorado River water.

We note

that there will never be a priority call for SJCP water because the SJCP allocations exist outside of the prior appropriation system. Instead, priority calls are possible for users whose SJCP allocation has been reduced so much that the water users' other water rights are insufficient to meet their needs. A loss of SJCP water would create a cascade of water shortages, since users would be drawing more completely on other sources, which would create water stress on allocated water that originates in the Rio Grande Basin.

As anyone who follows this blog knows,

the broad challenges of meeting possible Colorado River shortages are mirrored in New Mexico's water management obstacles. Just as prior appropriation does not serve the Colorado River Basin broadly, it does not serve New Mexico. New Mexico has a cultural history of water shortage sharing, which draws a strong contrast from many other western states like California. As water shortages in the West continue to worsen, shortage sharing will be a critical part of water management. While prior appropriation has been durable up until this point, we have no past data that reflects our current reality. If existing norms are no longer effective for managing water shortages, then existing legal structures will also be ineffective. Fortunately for New Mexico, within the state, there are flexible systems of water management that allow water users to adapt collaboratively to reduced water availability. Acequia water management presents a community-centric model for shortage sharing, repartimiento. Similarly, the MRGCD also exemplifies a collaborative water governance system that does not rely on priority calls. Instead of priority calls, the state can instead look to these water-sharing models that more accurately reflect the cultural values of the state and share the inevitable economic burdens of reduced Colorado River water deliveries.

Although these discussions tend to raise blood pressure, I believe they are worth having now, before water stress makes these conversations feel all but impossible. New Mexico is uniquely situated to handle these challenges with creativity and set a new standard for handling the continued Colorado River shortages, basin-wide.

Section: Counseling patience on the current Colorado River kerfuffles

Source: <https://www.inkstain.net/2024/03/counseling-patience-on-the-current-colorado-river-kerfuffles/>

Despite the Sturm und Drang of last week's competing proposals to the federal government for managing drought and climate change on the Colorado River, there's a lot to be hopeful about.

On their faces, the Upper Basin and Lower Basin proposals have a lot of “Water’s for fighting over after all!” vibe. But if you take them “seriously but not literally”, to borrow a meme from recent political rhetoric, it’s clear there is much to be hopeful about.

Here’s the part I do take both seriously and literally. New Mexico’s representative in all of this, my representative, Estevan Lopez, said this:

We look forward to working with our sister Lower Basin States to resolve differences in approach and create a 7-state consensus alternative.

The key understanding the gap between the “Water is for fighting over!” rhetoric of last week and Estevan’s comment is to remember two interlocking things about the two basin submissions to the U.S. Bureau of Reclamation.

Tell us Your Plan Before You’ve Had a Chance to Develop It

The first piece to understand is that we are in the midst of intense, difficult, and importantly closed door negotiations among the Colorado River Basin states. The negotiations have a long way to go. Asking the states to freeze and make public a position now puts them in a difficult position!

The reason for the need to make preliminary proposals public now, a couple of years before we need to finalize action on this stuff, is legitimate. The whiz kids at Reclamation need time to do the “what if” modeling, a key step in the administrative process of development a National Environmental Policy Act review, the so-called “Environmental Impact Statement”, or EIS. This can’t wait for the states to work out a deal. That’s the reason for the March 2024 it’s-not-a-deadline “please send us your ideas” deadline from Reclamation.

Given the remaining uncertainty, the states faced a dilemma – submit something close to a “best and final offer,” the place you hope to end up? Or submit a “tough initial negotiating position” – essentially your starting point.

The Lower Basin, with a longer history of interstate wrestling with water use reductions going back to the 2001 Interim Surplus Guidelines, submitted something closer to the former, reflecting the Lower Basin’s willingness to support the first tranche of needed cuts, but suggesting the Upper Basin should share in the second tranche if drought and climate change require us to dig deeper.

The Upper Basin, using substantially less water and operating largely independently in terms of their use of Colorado River water, don’t have the same experience in intrabasin negotiation. The Upper Basin submitted something like the latter. Suggesting that the entire burden of cuts fall on the Lower Basin is obviously not where we’re doing to end up, but it preserves a tough negotiating position.

Let’s Model It!

The second important thing to remember, and that should give us pause about getting too worked up about the specifics right now, is that the whole point of this exercise is to sketch out some options that can be modeled to help inform decisions. It’s impossible to look at these proposals right now – even if we wanted to treat them as real and serious plans – and say what effect they would have.

That’s the point, at its best, of the NEPA process. Its purpose is to inform decisions.

For example, I’ve stared hard at one of the key differences between the two proposals – using total storage in Mead and Powell as the benchmark for making decisions about how to make cuts, versus using a larger suite of reservoirs. Without doing the modeling, I’m not sure I understand their practical implications. Perhaps the states, with internal modeling capabilities, have already done this, but it’s not public, so let’s model it and talk about it publicly. That’s what NEPA is for!

My Upper Basin Sympathies

I must declare my allegiances here. As a resident of the Upper Basin, it has been frustrating over the last decade to watch as the Lower Basin sucked the reservoirs down, only tapping the brakes fitfully,

and never quite hard enough until the last few years.

But it is from that vantage point that I've worked hard with Upper Basin colleagues in recent years to make clear to my fellow residents of Wyoming, Utah, Colorado, and New Mexico the realities underlying some of our shared groupthink, because (to push a poker analogy to the breaking point), some of the things we treat as high cards – “the Lower Basin is overusing its compact allocation,” and “we routinely take shortages in dry years” – may not be.

Because, ultimately, the best way to act on my allegiance to the Upper Basin is to funnel it through the allegiance we all should have to the Colorado River Basin as a whole.

Section: Holding Course: February brings more of the same for basin storage

Source: <https://www.inkstain.net/2024/03/holding-course-february-brings-more-of-the-same-for-basin-storage/>

A guest post from Jack Schmidt, crossposted from Utah State University's Center for Colorado River Studies

By Jack Schmidt | March 8, 2024

Reservoir storage didn't change much in February 2024, demonstrating that the Basin's water managers and users have succeeded in retaining the bounty of last year's big runoff.

This month's assessment of Colorado River reservoir storage will be short and to-the-point. A month from now (April 1) the traditional snow accumulation season will end, and I'll provide a more comprehensive assessment of the status of the basin's water supply.

What happened in February?

Total basin water storage was 28 million af (acre feet) on 29 February 2024 (Fig. 1). Since mid-December 2023, basin storage progressively increased bit by bit — by 32,000 af in February. The small value of the increase matters little; what matters is that basin storage did not decrease and hasn't for the last 2.5 months. To date, reservoir storage is only 20% less than the peak storage at the end of the 2023 runoff season. This is a very small loss in relation to the rate at which the bounty of previous runoff years had been consumed. Nevertheless, conditions in the basin are comparable to conditions in early May 2021, a perilous situation.

Figure 1. Graph showing total basin reservoir storage and total storage in Lake Mead and Lake Powell since January 2021. This graph demonstrates the efficient retention of reservoir storage accomplished this year, the result of reductions in consumptive use and addition of winter precipitation. The black arrows indicate the last time reservoir storage matched current conditions.

Although basin storage has not changed, the distribution of storage within the basin has somewhat changed, with storage being systematically shifted downstream. The combined storage in federal units of the Colorado River Storage Project upstream from Lake Powell (Flaming Gorge, Navajo, Fontenelle, Blue Mesa, Morrow Point, and Crystal) peaked in mid-July 2023 at 5.8 million af, and steadily declined thereafter (Fig. 2). In February, the combined storage in these reservoirs declined an additional 79,000 af to 5.0 million af.

Figure 2. Graph showing total basin reservoir storage (solid blue line) and storage in the different parts of the watershed. Notably, storage in Lake Mead has steadily increased during the past 18 months, while storage in Lake Powell has fluctuated more. Storage in CRSP reservoirs upstream from Lake Powell has steadily decreased since summer 2023, but is larger today than during the previous 2 years.

Similarly, storage in Lake Powell peaked in mid-July 2023 at 9.7 million af, but was 7.9 million af at the end of February 2024. Storage in Lake Powell declined by 203,000 af in February. Conversely, storage

in Lake Mead reached its nadir in August 2022 at 7.0 million af and has steadily increased thereafter to 9.7 million af at the end of February. Storage in Lake Mead increased by 310,000 af in February with most of that water coming from Lake Powell releases. Today, Lake Mead has 1.8 million af more water than does Lake Powell. The combined storage in the two reservoirs was 17.7 million af on 29 February, 109,000 af more than at the end of January.

Let's hope for another good month of precipitation in March.

Section: Ignoring an Inconvenient Colorado River Basin Risk

Source: <https://www.inkstain.net/2024/02/ignoring-an-inconvenient-colorado-river-basin-risk/>

By John Fleck

It is agonizing to watch this, but here we are.

With efforts by the Colorado River Basin states to craft an agreement to share the river's water skidding, brakes screeching, toward a cliff, we appear on the brink of repeating the disastrous mistake the authors of the Colorado River Compact made a century ago: ignoring inconvenient truths about the risks we face, washing away genuine uncertainties with convenient talking points.

As Eric Kuhn carefully documented in a post here yesterday, there is once again a genuine risk that we will ignore inconvenient truths about a huge uncertainty in our understanding of how much water the river can offer us, and for whom. We are pretending that an uncertainty literally at the scale of millions of acre feet in how we measure and manage water does not exist.

A masterful Upper Colorado River Basin public relations blitz, led by the Colorado Water Conservation Board, would have us believe one set of numbers about the river's future, a set of numbers that has given Upper Basin water users comfort that they can sit tight and blame others for the river's woes.

But as Eric's analysis showed, there are hidden assumptions behind the Upper Basin's numbers – assumptions that hide a genuine and irreducible uncertainty. The uncertainty is irreducible because more than a century after the adoption of the Colorado River Compact, there is still no agreed upon definition of how to measure the use of water. As Eric wrote, these are questions “with enormous potential impacts on the allocation and distribution of the shrinking Colorado River – questions we have avoided dealing with by draining the Basin's reservoirs. We no longer have that option.”

Arithmetic and Law

Eric is a master of the arcane and wonky details of the interface between Colorado River law and hydrology, and I commend you to his analysis – it rewards a careful read. But Eric once described my role in our collaboration as “dewonkifying”, so let me try to put this in simpler terms.

The 1922 Colorado River Compact based its allocations on “beneficial consumptive use”. But the phrase was never defined, and the definitions ended up bitterly contested in the decades that followed. It remains undefined to this day. Or rather, there are two competing definitions that yield very different results.

Each definition makes intuitive sense, and at first glance they look puzzlingly similar. But at the scale of the Colorado River Basin they yield very different results that have become a critical piece of the current basin management debate.

Method A is based on the collective amount of water communities take from the river, minus the amount they return – “diversions less return flows.”

Method B is based on the ultimate impact of that use on the Colorado River downstream of the use – for the Upper Basin, for example, at Lee Ferry, or for Arizona at the confluence of the Gila and the Colorado near Yuma. This is the “stream depletion theory”.

Those might sound so similar that the differences are trivial. And at localized scales they are. But, as Eric explained in yesterday's post, with a classically Eric Kuhn working out of the mathematical details (I love collaborating with this guy – he shows his work!) at the scale of the Lower Colorado River Basin the differences amount to nearly 2 million acre feet of water.

Under Method A, Lower Basin use is more than 10.1 million acre feet per year, well above its Colorado River Compact allocation of 8.5 million acre feet. This is the methodology the Colorado Water Conservation Board staff used in its now-famous PowerPoint slide purporting to demonstrate that the Lower Basin is using more than its legally allotted share of the Colorado.

But under Method B, Lower Basin use is some 8.3 million acre feet – less than its Compact allocation. Importantly, Method B is the method adopted by the Upper Basin Compact, and therefore the method used in the Upper Basin's management of its share of the river.

Let's Be Honest About the Uncertainties

To be clear, Eric and I are not arguing in favor of A or B. We are arguing, as we did in our book *Science be Dammed* (we spent chunks of three chapters on this question), that the lack of an agreement over the definition of "beneficial consumptive use" remains a genuine and important unresolved uncertainty in the Law of the River, and our discussions of the future management of the Colorado River need to acknowledge that uncertainty, not pretend that it does not exist.

This is what I, as a stakeholder whose community depends on the Colorado River, expect of those leading the interstate effort – public honesty about the genuine risks and uncertainties we face.

Section: On the Colorado River, there are no Simple Disputes

Source: <https://www.inkstain.net/2024/02/on-the-colorado-river-there-are-no-simple-disputes/>

By Eric Kuhn

One of the commentors to our January 19th, 2024, blog post titled "Are We headed for the First Colorado River Compact Tripwire?" – John C. (who, by-the-way runs a very talented water resources engineering firm) raised several finer points to explore further:

The first point deals with obligations of each Basin to contribute water needed to meet U.S. obligations to Mexico under the 1944 Treaty. The second deals with the question of how to measure, and therefore manage, in the context of overall Colorado River Basin management, the use of tributary water in the Lower Basin. Both represent unresolved legal questions with enormous potential impacts on the allocation and distribution of the shrinking Colorado River – questions we have avoided dealing with by draining the Basin's reservoirs. We no longer have that option.

The two issues have been disputed for decades. They are, of course, totally inter-related, and when one peels back the layers of each, the problems get so complicated that the only real solution may be for the Basin's states and other stakeholders to ignore their past positions and grievances and negotiate a river management approach that works on the river we have today, even if that means changes to the foundation of the Law of the River, the 1922 Compact.

Mexico Obligation

The different interpretations of the Upper Basin's delivery obligations to Mexico under Article III(c) are well understood throughout the Basin. III(c) says that water for Mexico should be provided from "surplus". If there is no surplus, the Upper and Lower Basins must each provide half the necessary water. But there has never been agreement on what that language means in practice. This unresolved uncertainty has enormous implications for how much water is available to each basin in the future.

The Lower Division States take the position that there's no current "surplus," as defined by Article III (c), thus the Upper Division States must deliver at Lee Ferry 50% of however much water is required to be delivered under the Treaty. The annual delivery is normally 1.5 maf/year, but under either the "extraordinary drought" provision of the Treaty, or Minutes, it could be less. In 2023, it was 1.4 maf.

The Lower Division States' position would dictate an average annual delivery of 8.25 maf/year at Lee Ferry, 7.5 maf under Article III(d) + 750,000 af under Article III(c). The 8.25 maf includes an average contribution of 20,000 af/year from the Paria River and would be adjusted for the occasional annual delivery of less than 1.5 maf. Veterans of the negotiations that led to the 2007 Interim Guidelines will recall that Arizona's Herb Guenther always brought with him to the meetings a posterboard sign with "8.25" written on it.

Going back to the 1970 decision by the Secretary of the Interior to set the "minimum objective release" from Glen Canyon Dam at 8.23 maf/year, the Upper Division States have consistently taken the position that their annual obligation Mexico has never been formally defined and, whatever it is, it is not 750,000 af/year, every year. While they vigorously complained, they never chose to formally challenge the issue in court or in Congress, perhaps because they concluded that they couldn't show that any of their interests were injured. Today, based on post-2000 hydrology, that dynamic may have fundamentally changed.

The basic position of each basin has not changed. If anything, because the stakes are much higher, the positions have hardened. In a December 20, 2022, scoping letter to Reclamation, Arizona's Tom Buschatzke, and Ted Cooke, wrote: "Article III(d) and (c) prohibits the Upper Division States from depleting the flow of the river at Lee Ferry below a rolling 10-year aggregate of 75 maf plus one-half of the Mexico delivery obligation. With reduced releases from Glen Canyon Dam potentially analyzed under the SEIS, if the 10-year rolling aggregate falls below the required aggregate volume, the Upper Division States could be subject to a "Compact call" that would require a reduction in consumptive use in the Upper Basin. In footnote 1, they add "A "surplus" currently does not exist because natural flows in the Colorado River have not exceeded 16 maf in the past 10 years." In their August 15, 2023, scoping letter, the three Lower Division States write: "The Post-2026 EIS must analyze whether alternatives are consistent with the 1922 Colorado River Compact non-depletion obligations and delivery obligations to Mexico. Alternatives should include actions necessary to ensure compliance with such obligations."

The Upper Division States are equally adamant that because of overuse in the Lower Basin, they currently have no annual obligation to Mexico under Article III(c). Note that I used the term "Lower Basin" because under the 1922 Compact, the Lower Basin includes the upper Gila River in New Mexico, where uses are small, Kanab Creek in Utah and Arizona, again uses are small, and the Virgin River, shared by Nevada, Arizona, and Utah, where uses are not small. The Virgin River is the water supply for the rapidly growing St. George area. As a state, Utah consumes the second largest amount of Lower Basin tributary water, about 150,000 af/year, albeit much less than Arizona.

In recent public presentations, Colorado Commissioner Becky Mitchell has stated that the Lower Basin's total annual use, including tributaries and reservoir evaporation, were 10.5 maf in 2020, 10.8 maf in 2021, and 10.4 maf in 2022, far more than the Lower Basin's compact apportionment (7.5 maf under Article III(a) plus 1.0 maf under Article III(b)). Their position is that the Lower Basin's overuse is "surplus" water that must first be used to meet the 1944 Treaty obligations to Mexico before the Upper Division States have any obligation to Mexico, a position the Lower Division States do not agree with.

The situation is messy. As I explain below, Mitchell's 10-plus million acre foot calculation is based on analysis that contains a hidden assumption about the correct way to measure water use, an assumption at odds with the method the Upper Basin has traditionally used to measure its own water use. But when one peels back the layers, it's even messier. First, there is no agreement on whether the obligation of the Upper Division States to Mexico is calculated on an annual basis, a ten-year rolling aggregate basis, or something else (I'm in the something else camp). The predominant position the Upper Division States is, as mentioned in John's comments, that the since the 1944 treaty provides for an annual delivery to Mexico (which can change), therefore, the obligation of the Upper Division States

is determined annually. As can be seen by the language in the Arizona and Lower Division States letter, the Lower Basin states' position is based on a ten-year rolling aggregate. The logic of this position is that the last sentence of Article III(c) states: "whenever necessary the States of the Upper Division shall deliver at Lee Ferry water to supply one-half of the deficiency so recognized in addition to that provided in paragraph (d)." Paragraph (d) is a ten-year requirement; does it make sense to add an annual requirement to a ten-year requirement? Further, the data necessary to determine whether a deficiency exists (and thus the obligation of each basin) would not be available until well after the water year is over.

My reading of the reports of the compact commissioners, their Congressional testimony, and the minutes of both the 1922 and 1948 suggests a third possible alternative. The negotiators of the 1922 Compact, the 1944 Treaty, and the 1948 Compact considered the surplus to be the difference between the average long-term natural flow of the river at the international boundary and 16 million acre-feet, the aggregate of the apportionments made by Articles III(a) and III(b). During the negotiations of the 1948 Compact, Colorado's Royce Tipton and Arizona's Charles Carson laid out the logic. Based on the comprehensive hydrologic analysis conducted by Reclamation (Appendix I of the 1947 Comprehensive Report), the estimated long-term natural flow below the confluence of the Colorado and Gila Rivers was 17.7 maf/year. Subtracting 16 maf left an average surplus of 1.7 maf, 1.5 maf for Mexico and about 200 kaf for surplus uses within the United States. Upper Division State officials argued that with this hydrology there was no deficiency. Colorado's Tipton and Clifford Stone (its 1948 Compact Commissioner), however, did acknowledge that the location of the surplus was an important factor.

With their 1940s understanding of the river hydrology, the Upper Division States did not want the deficiency calculated either on an annual basis or a ten-year running average. They understood that in both cases, they would be required to deliver more water to Mexico than using the long-term average. Under an annual determination, there would be many years (~50%) when there was a deficiency. Under the ten-year rolling average, there would be long periods when it would be below 16 maf/year and there would be a deficiency (the 1930s drought period for example) but provided the long-term average was more than 17.5 maf/year, there would be no deficiency. Tipton also made the point that in the future, the construction of additional storage reservoirs (like Lake Powell) would effectively "equate" the river. Today's problem with this approach is that in the 1940s, the basin's water managers assumed a level of "stationarity" (future river flows can be predicted by what happened in the past) that because of climate change does not exist today.

An additional problem we have today is the calculation of the deficiency is based on the natural flow at the international border with Mexico, not Lee Ferry. Note that the Arizona letter states, "natural flows in the Colorado River have not exceeded 16 maf/year." That statement is very likely true, but there are no data to back it up. Unlike Lee Ferry, there are no recent calculations of annual natural flows at the international border. There is no Colorado River system natural flow database. The existing database includes natural flows to Lee's Ferry, but downstream to the Imperial Diversion Dam, Reclamation acknowledges that they have little confidence in these data and much of it is not based on reconstructed natural flows. Importantly, the existing natural flow database does not include the Gila River system.

The last widely published estimate of natural flows at the international border was completed by the Bureau of Reclamation nearly eighty years ago (Appendix I). These data were used during the 1944 Treaty ratification hearings and, to a lesser extent, by the 1948 Compact negotiators. Appendix I shows an average natural flow at the border of 17.72 maf/year. At Lee Ferry, it was 16.41 maf/year (based on 1898-1943). Assuming a similar relationship between the flows today, from 2000-2023 the estimated natural flow at the border, including the Gila, would be approximately 13.5 maf/year ($12.44/16.41 \times 17.72$). Note, because of climate change, which appears to have a greater impact on the southern tributaries of the Colorado River system, the relationship may no longer be reasonable.

Updating the natural flow database to include the Gila River has been suggested by Upper Division State officials, but Arizona has historically objected. Clearly this would not be an easy task and there

would be large uncertainties, especially estimating with any certainty natural losses on the Colorado River below Hoover Dam and on the Gila River from the Phoenix area to Yuma under today's climatic conditions.

Lower Basin Tributary Use

The Gila River is also the central subject of John C.'s second comment that Arizona, California, and Nevada should be concerned that including mainstem consumptive uses, tributary consumptive uses (including those in Utah and New Mexico), reservoir evaporation, and maybe system losses, the Lower Basin total consumptive uses exceed its compact apportionment of 8.5 maf/year. Alternatively, Lower Basin tributary consumptive uses far exceed 1.0 million acre-feet – if one believes Article III(b) was intended to only cover Lower Basin tributaries. The problem with this argument is that there is no 1922 Compact definition of “beneficial consumptive use” (the “commodity” the Compact apportions). This is especially important for the Lower Basin tributaries.

There are different interpretations among the states and between the basins of how “beneficial consumptive use” should be defined and therefore measured. Each has a strong legal argument in its favor. But their approaches result in vastly different numbers, and as my co-author John Fleck has written, “There’s not enough water for all the lawyers to be right.”

Article VI of 1948 Compact defines and provides a method measuring compact apportionments for the Upper Basin. The 1964 decree in *Arizona v. California* defines how to measure the mainstem apportionments made to the Lower Division States under the 1928 Boulder Canyon Project Act. Since the 1963 decision did not interpret the 1922 Compact and there is no Lower Basin Sub-Compact, there is no accepted or defined method for measuring 1922 Compact apportionments on the Lower Basin tributaries (and arguably the entire Lower Basin). The methods used by the 1948 Compact and the 1964 Decree are very different. The 1964 Decree uses the concept of “diversions minus return flows.” It comes from the language of the Boulder Canyon Project Act. Stream losses and reservoir evaporation from Hoover Dam to the points of diversion are not considered a use, but rather a limitation on the available supply.

The 1948 Compact approach for defining and measuring consumptive use under Article VI is based on what was referred to as the “stream depletion” theory. Consumptive uses for the Upper Basin and for the individual states are measured as the net impact of man-made depletions on the natural (AKA virgin) flow of the Colorado River at Lee Ferry using the “input-output” method. The 1948 Compact gives the UCRC the authority to change the method by unanimous approval. The UCRC has instructed its staff to identify an alternative to the input-output method, so this may happen soon. Article VI was one of the most debated and carefully written articles in the 1948 Compact. The negotiators had a clear objective in mind. They wanted to carefully define how to measure compact apportionments so that water made available for consumption through “salvage by use” would not count as compact apportioned use.

The issue of salvaged water was a major dispute among the basin states in the 1940s and 1950s. Simply put, salvaged water is water made available for use by the reduction of natural losses caused by the development of the river. The best example of salvage by use is Arizona's Gila River. As the Gila River and its two major tributaries, the Salt and Verde Rivers, leave the rim country upstream of Phoenix, they have a combined average natural flow of over 2.0 maf/year (2.3 according to the 1947 report). As the river flows to its confluence at Yuma, in its natural state it loses about 1.0 maf/per year. By diverting and using the entirety of Gila River's waters upstream, these losses are eliminated. Thus, Arizona can consume a million af/year more water than what the Gila River contributes to the natural flow of the Colorado River system. Under the “stream depletion” theory, which was favored by Arizona and the Upper Division States, Arizona is only charged for a million acre-feet of 1922 Compact apportionment. Under California's “diversions minus return flows” theory, also favored by Nevada, all 2+ maf/year of Arizona's Gila River consumptive use would be charged as 1922 Compact apportionment.

The Upper Basin adopted the stream depletion theory during an era when the states were competing for every acre-foot possible. They thought could benefit by 400,000 -600,000 af per year. Simply put, in years when the water was physically available, the Upper Basin could consume 7.9 – 8.1 maf/year while only depleting the natural flow of the river at Lee Ferry by 7.5 maf/year (their compact apportionment). The negotiators never contemplated that 75 years later, the water available to the Upper Basin would be far less than 7.5 maf.

During the negotiations of the 1948 Upper Basin Compact, the decision to use the stream depletion theory was thoroughly debated. Wyoming's legal advisor, W. J. Wehrli, warned the other states that using this definition would benefit the Lower Basin far more than the Upper Basin. Additionally, he noted that it could reduce the amount of surplus water under Article III(c), potentially increasing the obligation of the Upper Division States to Mexico. Wyoming ultimately fell in line and agreed to the Article VI definition. During the Congressional debate over the authorization of the Central Arizona Project, Upper Division State officials (primarily Tipton and Stone) testified in favor of the stream depletion theory, arguing that the negotiators of the 1922 Compact intended this method to measure apportionments. Note, the compact does not include a definition of "beneficial consumptive use," they made their case based on an analysis of the minutes and the use of the term "depleted" in Article III(d).

The Upper Division States make a rhetorically powerful public argument that in the face of climate change, overuse in the Lower Basin is the central problem in the Colorado River Basin that must be solved to reach a sustainable future where water use, and the available supply are in balance. The argument is that when mainstem uses, reservoir evaporation, system conveyance losses, and tributary consumptive uses (in Arizona, Nevada, New Mexico, and Utah) are added together, the Lower Basin's total use exceeds ten maf/year, perhaps as high as eleven maf/year. The Lower Basin's compact apportionment, however, is only 8.5 maf/year (perhaps 7.5 maf/year depending on how Article III(b) is interpreted), resulting in an overuse of at least 1.5 maf/year. Such an overuse has major policy implications:

- If the Lower Basin is overusing its compact entitlement, why should the Upper Basin, which is using far less than its apportionment, reduce its uses to help bring the system into balance?
- Should the Lower Basin's overuse be considered "surplus" water under Article III(c)? If so, should this surplus water be delivered to Mexico before the Upper Division States have any obligation to deliver half of the deficiency to Mexico?
- These two questions have implications for Mexico, if the Lower Basin is overusing its compact apportionment, why should Mexico reduce its annual use?

The problem with the Upper Basin's argument is that it's based on the diversions less return flow theory. If the Lower Basin's uses are calculated based on the stream depletion theory – the methodology the Upper Basin adopted in its own 1948 Upper Basin Compact, uses may not be greater than 8.5 maf/year.

| Stream Depletion | Diversions less Return Flows | |

| Mainstem uses (a) | 6.6 maf/year | 6.6 maf/year |

| Reservoir Evaporation (b) | 0.5 maf/year | 0.86 maf/year |

| System Losses (c) | 0 | 0.45 maf/year |

| Tributary Uses (d) | 1.2 maf/year | 2.2 maf/year |

| Total | 8.3 maf/year | 10.1 maf/year |

Explanation:

a) Mainstem uses (deliveries from Hoover Dam) are very similar under both theories. Almost all mainstem uses are either fully consumptive or located low in the system. The average mainstem use by the Lower Division States over the last five years (2019-2023) is approximately 6.6 maf/year.

b) Under diversions less return flows the 0.86 maf/year is the average evaporation (2017-2021) from the December 2023 Reclamation study. Under the stream depletion theory, reservoir evaporation is calculated as surface evaporation less the natural losses in the inundation area that would have occurred had the reservoir not been built. This is how evaporation on Lake Powell is calculated. I estimated natural losses as .36 maf/year.

c) The 0.45 maf/year is from the December 2023 Reclamation study. Under the stream depletion theory, system losses are offset by salvaged water. As a practical matter, we have no idea how much salvaged water is currently generated on the mainstem below Hoover Dam, but we do know that the channel is much smaller today than it was before development. In 1945 during the Mexican Treaty ratification hearings, Colorado's Royce Tipton estimated the number to be 400,000 af/year. I'll assume they offset.

d) Under diversion less return flows, tributary uses are 2.2 maf/year, the latest data from the 2001-2005 Consumptive Uses and Losses and Report. Under the stream depletion theory, it's reduced by 1.0 maf/year, the estimated salvage on the Gila River from the 1947 study. Because of climate change, losses today may be greater, but no data are available.

Thus, using the diversions less return flows theory, the Lower Basin is clearly using more than 8.5 maf/year, but under Upper Basin's own stream depletion theory, it is not. Since the 1922 Compact neither defines nor prescribes a way to measure "beneficial consumptive use," the basic question – "is the Lower Basin overusing its compact apportionment?" simply cannot be answered.

These disputes also point to the fundamental flaw with the states' talking point that the 1922 Compact, the 1948 Upper Basin Compact, and the Mexican Treaty will serve as the foundation of the post-2026 operating rules. There is no agreement on what they say or mean. As the states continue their discussions with the goal of agreeing on a state proposal, they need to consider addressing the disputed compact issues in a straight-forward manner. Leaving these critical uncertainties for future generations to handle (like we have in the past) is no longer possible and asking the Supreme Court for a resolution will likely make matters worse, not better.

Section: No Change in Reservoir Storage ... and That's Good News

Source: <https://www.inkstain.net/2024/02/no-change-in-reservoir-storage-and-thats-good-news/>

By Jack Schmidt | February 12, 2024 (cross-posted from the Center for Colorado River Studies)

Nothing really changed in Colorado River Basin reservoir storage during January 2024. That is really good news as the basin prepares for the upcoming irrigation season.

1. Total basin water storage did not significantly change during January 2024 (Fig. 1, blue line). In fact, total basin storage increased a tiny bit—34,000 af (acre feet)—and was 28.0 million af on 31 January. Conditions in the basin remain comparable to conditions in spring 2021. There is no reason so stop our efforts to conserve reservoir storage!

Figure 1. Graph showing total basin storage in 46 reservoirs in the Colorado River basin between 1 January 2021 and 31 January 2024 (blue line). The black arrow points to the previous time when reservoir storage was at this amount. Total storage in Lake Mead and Lake Powell for the same period is shown by the orange line. Compiled from data downloaded at https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html

2. Although the total storage has not changed, storage in specific reservoirs has been adjusted. Storage in CRSP reservoirs upstream from Lake Powell, including Fontenelle (Fig 2, green line) decreased by 109,000 af and has been decreasing since mid-summer 2023. Storage in Lake Powell (Fig. 2, gold line) decreased by 303,000 af. There was no significant change in storage in other Upper

Basin reservoirs (Fig. 2., black line). In contrast, storage in Lake Mead (Fig. 2, purple line) increased by 368,000 af. On 31 January, active storage in Lake Mead exceeded active storage in Lake Powell by 1.27 maf; Lake Mead held 9.41 million af and Lake Powell held 8.14 million af. Storage in Lake Mead has slowly been increasing since mid-summer 2022.

Figure 2. Graph showing active reservoir storage in different parts of the Colorado River basin between 1 January 2021 and 31 January 2024. The category “other Upper Basin reservoirs” includes Granby, Dillon, McPhee, Strawberry, Starvation, Nighthorse, and 30 others. The category “CRSP reservoirs upstream from Lake Powell, and Fontenelle” includes Blue Mesa, Morrow Point, Crystal, Fontenelle, Flaming Gorge, and Navajo.

3. Basin reservoir storage continues to be depleted at an unprecedented low rate in comparison to the past decade (Fig. 3), as is also the case for the combined storage of Lake Mead and Lake Powell (Fig. 4). On 31 January 2024, the amount of basin storage gained by the 2023 snowmelt runoff had been reduced by 20%.

Figure 3. Graph showing the rate of reduction in basin-wide reservoir storage in each of the past ten years. The reduction in storage in 2023/2024 has been at a much slower rate than in other years. Each year that plots lower than 2023 (blue line) on this graph reflects a higher rate of loss in storage than the current year.

Figure 4. Graph showing the rate of reduction in the combined storage in Lake Mead and Lake Powell in each of the past ten years. The reduction in storage in 2023/24 has been slower than in any other recent year. Every year plots lower than 2023 (blue line) reflecting a higher rate of loss in storage than in this year.

Section: A dry forecast for the Colorado River Basin. A note on policy implications.

Source: <https://www.inkstain.net/2024/02/a-dry-forecast-for-the-colorado-river-basin-a-note-on-policy-implications/>

The UC Merced Evaporative Demand Drought Index (EDDI) points to drying over the Colorado River Basin over the next four weeks.

EDDI is a new experimental tool that offers potential for tracking quickly emerging drought conditions by analyzing the evaporative demand of the atmosphere. It combines how moist things are with how hot and dry the atmosphere is – how much moisture that “thirsty” atmosphere will be sucking up out of soil and plants. (More here.) I’ve started tracking it as an add-on to watching the Colorado River Basin Forecast Center’s runoff forecasts. The forecast is already “meh” (~1maf below average). EDDI suggests we should expect it to get worse.

A Compact Tripwire in the headlights

A reminder of something Erik Kuhn and I wrote about last month:

The Bureau of Reclamation’s January 2024 “Most Probable” 24-month study forecasts that annual releases from Glen Canyon Dam for both Water Years 2025 and 2026 will be 7.48 million acre-feet per year (maf). If this happens, the ten-year total flow at Lee Ferry for the 2017-2026 period will drop to about 83.0 maf, only about 500,000 acre-feet above 82.5 million acre-feet, the first 1922 Compact hydrology “tripwire.”

That line – 82.5 maf feet of Lee Ferry deliveries over a ten year period – has become a dividing line between two contending interpretations of the most important unresolved question in the century-old Colorado River Compact: How much water must the Upper Basin deliver to the Lower Basin? What happens if it doesn’t?

A plea – we need to be having a public conversation about this. The current Upper Basin-Lower Basin negotiations over the Post-Tripwire Colorado River Management Regime appear to have wandered into the Grimpen Mire:

“It’s gone!” said he. “The mire has him. Two in two days, and many more, perhaps, for they get in the way of going there in the dry weather, and never know the difference until the mire has them in its clutches. It’s a bad place, the great Grimpen Mire.”

– Arthur Conan Doyle, *The Hound of the Baskervilles*

We keep euphemistically calling them the “Post-2026 Guidelines”, but let’s be real about this.

We are talking here about the Post-Tripwire Colorado River Management Regime.

It could snow a lot between now and summer and bail us out again this year. But we can’t look away from the Tripwire.

Section: In Colorado’s San Luis Valley, paying for the water they use

Source: <https://www.inkstain.net/2024/02/in-colorados-san-luis-valley-paying-for-the-water-they-use/>

Folks in Colorado’s San Luis Valley are engaged in a bold experiment in western water management – charging farmers for the water they use. Jerd Smith explains:

A new rule approved by the area’s largest irrigation district, known as Subdistrict 1, and the Alamosa-based Rio Grande Water Conservation District, sets fees charged to pump water from a severely depleted underground aquifer at \$500 an acre-foot, up from \$150 an acre foot. The new program could begin as early as 2026 if the fees survive a court challenge.

The challenge in the valley is that, with climate change inexorably chomping at the Rio Grande, and the groundwater used to replace the river’s dwindling irrigation supplies, there simply isn’t enough water to keep farming all the acreage they’ve got up there.

The valley is operating under the same two constraints that we see up and down the river – less water flowing in, and requirements established in the Rio Grande Compact to pass some of what does come in to folks downstream – Colorado can’t use it all, but must pass some water along to water users in central New Mexico. Those of us in central New Mexico’s “Middle Rio Grande” (the stretch from Cochiti through Albuquerque to Socorro) get to use some, but must pass some of on to farmers in Southern New Mexico. Under the deal now pending before the U.S. Supreme Court, the southern New Mexican’s (the Elephant Butte Irrigation District and Las Cruces area) must then pass some water across the border to people in Texas and Mexico.

Paying to Reduce Use: Private v. Public Goods

In each of those stretches – Colorado, central New Mexico, and southern New Mexico – we face the challenge of reducing use in order to meet downstream obligations.

In New Mexico, our approach to problems like this has been to treat the water as a private good, and pay its users to not use the water. This year, for example, a pipeline of money from the federal government, through the state, to our local water agency, the Middle Rio Grande Conservancy District, is paying irrigators \$700 an acre to not irrigate.

The approach in the San Luis Valley is different. There, farmers who want to pump groundwater (recognizing that groundwater and surface water are an interconnected part of a single system, and that as river flow declines farmers have been pumping groundwater to replace it) have to pay for it. If you want to pump more, you have to pay more. And as it gets scarcer, the price needs to go up.

The legal terminology involving the notion of property rights here is tricky, but as a practical matter this suggests two very different approaches. In New Mexico, we are treating the water as the irrigators property, and paying them to forego its use. In Colorado, they're treating it as public property, and requiring them to pay if they want to use it.

The Coasian Solution

Students of the Berrens-Fleck Lab will recognize this as a version of the classic problem of assigning the property right, as laid out by Ronald Coase in his classic 1960 paper *The Problem of Social Cost*. Overuse of water in a climate change-constrained system is a classic "externality" – a burden pushed off onto others, rather than the people who get to benefit from the use of water.

Coase's answer – "assign the property right!" – has made his paper one of the most-cited papers in the history of papers, and won him a Nobel prize. Coase's argument is that by assigning the property right, and starting from that point to figure out who pays and how much to solve the problem, we can converge on solutions. You can either make the people being harmed pay to stop the harm, or the people causing the harm pay to stop the harm.

We can, for example, require the factory polluting our river pay the cost of installing pollution control equipment. Or we can make the folks downstream, or the community as a whole, pay. Either way will work. The question of which approach we take is an ethical and political question.

Colorado has chosen (or at least is trying to chose – this'll end up in court) one approach. New Mexico has chosen another.

Cartoon Coase

This is a cartoon of Coase's argument. In the paper (which is a terrific read) he's making a more nuanced argument involving transaction costs. In both the New Mexico and Colorado cases, the cost of setting up the payment system makes actually carrying out the policies we need super hard. But the cartoon helps frame our approach to western water management challenges more broadly.

The Colorado example – charge more to use water! – is rare. In the Lower Colorado River right now, we're paying farmers, through their agricultural districts, giant container ships full of money to reduce their use – the New Mexico approach. We're treating the water as their property, and paying them not to use it. This is an ethical and political (and possible legal?) choice.

But the key difference between the New Mexico/Lower Colorado approach and the classic Coasian cartoon is who's doing the paying. In both cases, at least for now, we're using Other People's Money (OPM), via the recently passed Large Container Ships Full of Money Act (LCSFMA). Those of us in the West have somehow worked a racket where folks in Maine and Georgia and elsewhere are paying to bail us out of our mess. (To be fair, I'm sure we're bailing them out in some way too.)

The processes by which we have to figure out how to move all this money and water around – to pay people to not use water, or to charge them for the water they use – are a great example of the power of the deeper insights in Coase's 1960 paper. Working out the ways things don't match up to Cartoon Coase is where the real value of the intellectual framework is found.

Sources and Methods

Two huge thanks. First, to Daniel Rothberg, whose Western Water Notes alerted me to the issue. And to Jerd Smith and the Colorado Sun, for supporting and publishing the great water journalism we all need to understand these issues. If you can, I'd encourage you to contribute to one or the other or both, to support the fundamental underlying knowledge base we all need to move forward on climate change and western water issues.

Section: Are We Headed for the First Colorado River Compact Tripwire?

Source: <https://www.inkstain.net/2024/01/are-we-headed-for-the-first-colorado-river-compact-tripwire/>

By Eric Kuhn and John Fleck

The Bureau of Reclamation's January 2024 "Most Probable" 24-month study forecasts that annual releases from Glen Canyon Dam for both Water Years 2025 and 2026 will be 7.48 million acre-feet per year (maf). If this happens, the ten-year total flow at Lee Ferry for the 2017-2026 period will drop to about 83.0 maf, only about 500,000 acre-feet above 82.5 million acre-feet, the first 1922 Compact hydrology "tripwire."

That line – 82.5 maf feet of Lee Ferry deliveries over a ten year period – has become a dividing line between two contending interpretations of the most important unresolved question in the century-old Colorado River Compact: How much water must the Upper Basin deliver to the Lower Basin? What happens if it doesn't?

The consequences of triggering the tripwire, which might happen in 2027, are significant. In the worst case scenario, it could plunge the basin into Supreme Court litigation over the interpretation of the 1922 Compact, which could result in a forced curtailment of post-compact water uses in the Upper Basin. Or, alternately, if the Basin States are willing to settle their long-term disputed issues or implement basic changes to the Law of the River via the renegotiations of the post-2026 operating rules, it could be a "non-event." This is one of the fundamental issues facing the states as they meet to develop their basin-state alternative.

The 82.5 maf tripwire is based on the 1922 Compact's two flow-related requirements at Lee Ferry; Article III(d) requires the four Upper Division States to not cause the progressive ten-year flow at Lee Ferry to be depleted below 75 maf. Additionally, Article III(c) provides that if there is not sufficient surplus water to meet the annual water delivery requirements of the 1944 Mexican Treaty, normally 1.5 maf, then each basin must provide half of the deficiency (the required annual delivery minus the available surplus). The Upper Division States must deliver their share of the deficiency at Lee Ferry in addition to their obligations under Article III(d).

The Upper Division and Lower Division States have never agreed on the meaning and interpretation of Article III(c). There have been numerous papers on the disputed issues by both compact scholars and practitioners and Article III(c) has never been interpreted by the U.S. Supreme Court. Suffice it to say that the Lower Division States believe that the Upper Division States must deliver at Lee Ferry a total of 82.5 maf every ten years (75 maf + 10 x 750,000), but the Upper Division States believe that they currently have no obligation to Mexico, so the number is at most 75 million. There are of course, nuances. The Lower Division States have suggested that the Upper Division States might also need to cover transit losses between Lee Ferry and Mexico and the Upper Division States have most recently suggested that if climate change, not Upper Basin depletions, is causing the ten-year flow to fall below 75 million, then their article III(d) non-depletion obligation must be appropriately adjusted. Further, if pursuant to either the extraordinary drought provision or a treaty minute, the required annual delivery to Mexico is less than 1.5 maf, then, even with no surplus, the Upper Division's 50% share would be less than 750,000 acre-feet.

One should recognize that the annual releases from the Glen Canyon Dam and the annual flow at Lee Ferry (the compact point) are not the same. Between the dam and Lee Ferry, the river gains flow from groundwater accretions (in part due to leakage around the dam) and from the Paria River. These gains can vary from about 30,000 acre-feet to over 300,000 acre-feet annually. The ten-year Lee Ferry flow for 2014-2023 was approximately 86.1 maf.

That amount – 86.1 maf – might seem like a safe cushion. But because it is a ten-year moving total, we are about to drop out years with big releases (9 million acre feet) and replace them with years with just

7.48 maf. At the end of 2024, because both 2014 (the year that drops out) and 2024 (the year that is added in) are 7.48 maf years, the ten-year flow will stay about the same. The way the ten-year flow calculation works is next year, 2015 will drop out and 2025 will be added in, and so on, but here is the problem; From 2015 through 2019, the 2007 Interim Guidelines dictated an annual release of 9 maf per year. With accretions, flows at Lee Ferry averaged about 9.18 maf per year (source: UCRC 74th Annual Report). Thus, if 2025 and 2026 are 7.48 maf years, the ten-year flow will lose about 1.5 maf/year making the total about 83 maf for the 2017-2026 period. Because the 2007 Interim Guidelines expire, we don't know what the annual release will be in 2027, but if it's less than about 8.5 maf, because 2017 was a 9 maf year, the ten-year Lee Ferry flow could drop below the tripwire – 82.5 maf (with two more 9 maf years, 2018 and 2019, in the pipeline).

We recognize that the 24-month studies don't predict the future. They are a planning and management tool. It's plausible that by 2027, a series of wet years could result in a ten-year flow that is much higher than 82.5 maf, something that is, in our view, unlikely. But as we sit here in January 2024, the 24-month study is the only planning tool we've got. It would behoove us to pay attention to what it is telling us.

If a future 24-month study projects that ten-year flows will fall below 82.5 maf, it will be a big deal for the basin. What provision of the Law of the River will control annual releases from Glen Canyon Dam – the post-2026 Operating Guidelines, the Lower Division's interpretation of the 1922 Compact (82.5 maf), or the Upper Division's interpretation of the 1922 Compact (75 maf or less)? If the Lower Division States agree to a ten-year flow target of less than 82.5 maf, are they effectively surrendering to the Upper Division States? If the Upper Division States agree to a flow target of 82.5 maf, are they effectively surrendering to the Lower Division States? If the Upper Basin states agree to either 82.5 maf or some smaller compromise delivery target and the hydrology remains bad, how will Wyoming, Colorado, New Mexico, and Utah – our states – approach the required water use reductions? What if there is not enough water in storage in Lake Powell (and the other CRSP reservoirs) to release sufficient water to bring the ten-year flows to 82.5 maf? Will the Lower Division insist that the UCRC implement a curtailment of post-compact uses in the Upper Basin. If the UCRC refuses to do so, will that plunge the Basin into litigation?

At the recent 2023 CRWUA meeting, representatives of the Lower Division States stepped up and made it clear they own the “structural deficit,” and the conference was buzzing with talk of the innovative system approach the Lower Division has put on the table. This is great news, but as Colorado's Royce Tipton concluded sixty years ago, the “structural deficit” is not a single number. It's a range that depends on the interpretation of the Lee Ferry obligations of the Upper Division States under the 1922 Compact. If the average annual flow requirement is 8.25 maf/year, (which Tipton referred to as “fictional”) he calculated the deficit to be about 1.2 maf/year. Today we believe it's about 1.4 -1.5 maf/year (Tipton's assumptions about system losses were probably too low and perhaps his Lee Ferry to Lake Mead inflow assumptions too high). If, however, the average annual flow requirement is 7.5 maf/year, he calculated the deficit to be about 2 maf/year, today maybe 2.2 maf/year. This difference is huge, especially for the Central Arizona Project, the junior user on the Lower basin mainstem.

In the past decades, water managers in the Colorado River Basin have made accomplishments that even two decades ago were considered out of reach: The 2019 Drought Contingency Plans, the shortage sharing agreements with Mexico, the increased recognition of the rights of the Basin's Native American communities are just a few. The Lower Division's recent pronouncement that they own the structural deficit is another major step forward, but it has a fundamental flaw. Representatives of all seven Basin states continue to stubbornly insist that the Law of the River, and specifically, the 1922 Colorado River Compact, will serve as the “foundation” of the post-2026 operating guidelines. The flaw is that is that we don't know if this foundation is based on a ten-year Lee Ferry flow of 82.5 maf, or 75 maf, or something different.

Put simply, the states agree that the Law of the River has to be the basis of what we do, but don't agree on what the Law of the River actually says. Without an agreement on this fundamental issue, calling the 1922 Compact a foundation is nothing more than self-delusional wishful thinking. Now that the Lower

Division States have agreed to own the structural deficit, is the next step for all seven states and the federal government to openly acknowledge that given the impacts of climate change on the river, the 1922 Compact's overallocation of water and its disputed Lee Ferry flow provisions are core problems for the basin, not the foundation? Finding a sustainable future, without litigation, will require accepting and acknowledging the basic problems we face, not avoiding them.

Section: Colorado River Basin Reservoir Storage at the End of 2023 – Holding On to What We Have

Source: <https://www.inkstain.net/2024/01/colorado-river-basin-reservoir-storage-at-the-end-of-2023-holding-on-to-what-we-have/>

By Jack Schmidt | January 9, 2024

There was not much loss in reservoir storage in the Colorado River basin in December 2023. Total storage in the basin's reservoirs only declined by 17,000 acre feet during the month, and the combined contents of Lake Mead and Lake Powell increased by 68,000 acre feet. At year's end, the basin's water users have only consumed 21% of the gain in storage caused by the large snowmelt of 2023.

Here are a few graphs depicting where we stand at the start of the new year.

1. The amount of water stored in the basin's reservoirs remains at an unprecedented low condition. On 31 December 2023, total basin storage was 28.0 million acre feet (af), of which 17.5 million af was in Lake Mead and Lake Powell (Fig. 1). The total amount of water stored in the basin is the same as it was in early May 2021. At that time, storage was less than at any other time in the 21st century, but we drained the reservoirs much more in the summer and fall of 2021 and 2022. The recovery of storage caused by the large runoff in 2023 provided some relief to the ongoing water-supply crisis, but water storage remains critically low.

This winter's snowfall has been meager so far, and Reclamation's January 1 prediction of unregulated inflow to Lake Powell is only 66% of average. Water managers are beginning to anticipate another summer of water-use restrictions. No one should think that last year's runoff eliminated the ongoing crisis.

Figure 1. Graph showing active water storage in 42 reservoirs in different parts of the Colorado River basin. Conditions at the end of December 2023 are comparable to conditions in early May 2021, indicated by the black arrows. Data downloaded at https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html.

2. Most of the basin's water storage is in Lake Mead and Lake Powell (Fig. 2). Releases from Lake Powell and reductions in Lower Basin water use were sufficiently large that there was significant recovery of storage in Lake Mead. At the end of December, storage in Lake Mead (9.05 million acre feet) exceeded storage in Lake Powell (8.44 million acre feet) by approximately 600,000 acre feet. The difference in storage between the two reservoirs is much less than during the previous two years when more water was stored in Lake Mead.

Figure 2. Graph showing water storage since January 2021. Note that storage in Lake Mead was significantly greater than in Lake Powell in 2021 and 2022. Large spring runoff in 2023 was captured in Lake Powell, and some of that accumulated inflow was subsequently released to Lake Mead. The rate of reduction in storage in reservoirs upstream from Lake Powell significantly slowed after mid-fall 2023. The category "other Upper Basin reservoirs" includes Strawberry, Granby, McPhee, Dillon, Starvation, Nighthorse, and smaller reservoirs. Water storage in Lake Mohave and Lake Havasu remains nearly constant. Note that the vertical axis is an arithmetic scale that has a break. Data downloaded at https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html.

3. The rate of loss in reservoir storage this year remains low relative to the rate of loss in previous years (Fig. 3), especially the rate of decline of the combined storage in Lake Mead and Lake Powell. (Fig. 4) The basin's water managers are doing a good job of reducing use and conserving water in reservoirs. Reclamation's estimate of probable consumptive water use in the Lower Basin in 2023, issued 31 December 2023, is 5.78 million acre feet, nearly 900,000 acre feet less than Lower Basin consumptive use in 2022. Will that degree of water conservation be enough? That depends on how much snowmelt occurs this spring.

Figure 3. Graph showing the rate of reduction in basin-wide reservoir storage in each of the past ten years. The reduction in storage has been at a much slower rate than in other years. Each year that plots lower than 2023 on this graph reflects a higher rate of loss in storage than this year.

Figure 4. Graph showing the rate of reduction in the combined storage in Lake Mead and Lake Powell in each of the past ten years. The reduction in storage has been slower than in any other recent year. Each year plotting lower than 2023 on this graph reflects a higher rate of loss in storage than in this year.

Acknowledgement: Eric Kuhn and John Fleck provided helpful suggestions that improved this posting.

Section: Colorado River stuff roundup

Source: <https://www.inkstain.net/2024/01/colorado-rivers-stuff-roundup/>

Peering out across the Colorado River landscape for 2024, a few things I've been reading to help catch y'all up:

Paying to conserve

Annie Snider had a terrific story in late November about the role of federal cash in the short term solutions to the Colorado River's challenges that needs to echo outside the usual Politico-reading crowd.

Big \$\$\$ flowing from the federal government have been crucial to shoring up the reservoirs by paying people to leave water in them, but what does that do to the prospects of long run solutions that require permanent water use reductions:

The gusher of federal money is likely to make a broader, long-term deal to save the West's most important river more expensive.

"It's all a grand experiment," said Kathryn Sorensen, a former head of Phoenix's water department, who noted that Democrats' Inflation Reduction Act funding is effectively creating a new market for water, with a new, higher price. "This market, especially one with a premium [price], might create some perverse incentives."

Using hydrology, rather than reservoir levels, to determine how much water people use

Luke Runyon, now at the Water Desk (wait, what?) included this in his "what to watch for this year on the river" piece following last month's Colorado River Water Users Association:

One more idea from the Las Vegas conference that's still largely conceptual, but is gaining some interest from those in power, is to use annual measures of basic hydrology — like snowpack levels and streamflows — to determine how much water ends up being delivered to the basin's varied users. It sounds simple: only use what nature provides.

But that idea flies in the face of the river's foundational governing document, the Colorado River Compact, which put fixed volumes of water use on paper, regardless of whether it was a dry or wet year. For now, the idea seems to be more of a talking point than a specific policy proposal, and we will see if proponents can turn it into something Lower Basin users can get behind.

The current management regime is based on reservoir levels, rather than the underlying system hydrology. My collaborators Jack Schmidt, Eric Kuhn, and I argued in our 2022 NEPA comments for the hydrology approach rather than the reservoir level approach. We are not alone. It makes a lot of sense, though, as Luke notes, no one's make a specific policy proposal (we just wave our arms in our comments and basically say "This'd be great, someone should figure out how to do it!").

Which leads me to....

Modeling and a shared understanding

Kevin Wheeler, Terry Fulp, and Roberto Salmon have a super-important new paper on the use of modeling to create a shared understanding of the resources of the Colorado River basin:

As participants directly involved in recent agreements seeking to address this challenge, we describe how a critical basin-wide lens has been developed through information transparency, including the evolution and voluntary exchange of computational models. To achieve greater sustainability, further resolve and commitment from water managers in the United States and Mexico is necessary to ensure the required enhancement of the knowledge systems that will be used to inform extremely difficult and critical decisions.

This touches on a crucial point in negotiating these big, complex, shared resource management agreements – the importance of having a shared understanding of the resource. This is one of the key threads through my book *Water is For Fighting Over*, which drew for this point on the work of Elinor Ostrom.

When I wrote the following passage eight years ago, I was very specifically writing about the work that Wheeler, Fulp, Salmon, and others had done in negotiating a landmark U.S.-Mexico water management agreement known as Minute 319.

Much of the negotiations that followed hinged on the development of a shared understanding of a seemingly straightforward problem: how to account for the water moving through the Colorado River System from one nation to the other. But to deal with the issue required doing something that had previously been beyond reach: extending "the network" of US water managers into Mexico. One of the network's key tools is a sophisticated computer model of the basin known as the Colorado River Simulation System. The Bureau of Reclamation supported an entire research team based at the University of Colorado in Boulder to develop and maintain it, and its calculations are used to make decisions about how much, where, and when, to release water.

Now Wheeler, Fulp, and Salmon have written for us a guide. I hope to return to this for a longer piece (see below, re business model), but in the meantime just read what they wrote, please, that's your homework.

Busking as a business model

I've got no business model here, just an extremely part time University gig that ensures me library privileges (important!) and a semi-formal institutional structure to provide routine access to a bunch of really smart people (very important!).

But it occurs to me that I'm kinda like a busker, somewhere between my failed model of performing journalism in bars (this did not work out) and putting out a guitar case seeded with a few fives. Thanks to my faithful readers who provided the fives, here's the guitar case.

**Section: Closing in on a post-2026 Colorado River management deal
(some terms and conditions may apply)**

Source: <https://www.inkstain.net/2023/12/closing-in-on-a-post-2026-colorado-river-management-deal-some-terms-and-conditions-may-apply/>

The news out of last week's Colorado River Water Users Association is that, behind the scenes, a deal is taking shape with the potential to bring Colorado River Basin water use into balance with water supply.

The deal would eliminate the "structural deficit", and creates a framework for a compromise over the Upper Basin's Lee Ferry delivery/non-depletion obligation.

This is huge. But so are the caveats – in terms of both the challenges remaining for a deal, and the definition of the problem we are trying to solve.

The U.S. Lower Basin states – California, Arizona, and Nevada – have converged on a set of numbers to permanently reduce their use on a year-in, year-out basis by a minimum of 1.25 million acre feet per year, eliminating the "structural deficit" – the year-in, year-out gap between inflows and outflows that has drained Lake Mead over the last two decades.

California and Arizona seem to have found a path to a compromise (the details of which have not been made public) after the early-2023 cage match that seemed to place us on the path to interstate litigation, with six states arguing for sharing the pain and California insisting on a priority administration that would have largely placed the burden of the impact of climate change on Arizona.

If separate negotiations with Mexico lead to additional reductions south of the border (which is how this has played out in the last two rounds of basin scheming), total durable, permanent Lower Basin reductions on the order of 1.5 million acre feet a year appear to be within reach.

If more cuts than that are needed to balance the system, the Lower Basin states at CRWUA made it pointedly, publicly clear that they are asking the Upper Basin states to share in the additional pain.

Implicit in that final point is the opportunity for a version of what we used to call the "Grand Bargain" – a Lower Basin concession that the river's flow may not be sufficient to deliver 82.5 million acre feet per year. That would require even deeper cuts in the Lower Basin. To avoid interstate litigation over a Colorado River Compact delivery shortfall, the Lower Basin is offering a "Modest Bargain" of a sort – an Upper Basin contribution of water matching in some way (it's not clear in what proportion to the additional Lower Basin cuts) in return for the Lower Basin not wading into a legal fight over the meaning of Article III(d) of the Colorado River Compact.

To the extent that this moves from the meeting rooms and hallway conversations of CRWUA to public view, the seven states need to come together on something that can be put down in writing, publicly, by (I think) March in order for the Bureau of Reclamation wizards to begin the modeling work. So this is on a very fast track.

This is a very big deal, and very good news. But....

Some terms and conditions may apply

There are a bunch of caveats.

The final AZ-CA-NV split of the 1.25-ish million acre feet is not fixed, but it is close, converging on a set of numbers that make sense, respecting some of California's senior priority status, but not insisting on it as thoroughly as the state's proposal of last February.

Suffice to say that the remarkable Lower Basin use numbers this year – currently at 5.8 million acre feet for the three U.S. states, the lowest total U.S. draw on Lake Mead since the modern record-keeping regime began in 1964 – shows that cuts like this are totally doable without wrecking the economies of the three states.

If we don't have three-state numbers yet, we're a lot farther from figuring out how each of the three states will divvy up the cuts among their users. This will be hard. Well for two of the states, anyway,

Nevada just has one major user to do the divvying.

But will voluntary cuts of the scale needed be possible without the big inflow of federal cash that has helped so much this year? The precedent set by all the money sloshing around the Lower Basin right now poses a challenge.

And what of the Upper Basin's relentless "it's a Lower Basin over-use problem!" rhetoric over the last year. Now that the Lower Basin folks have basically said "Yup, and here's what we're gonna do about that", have the Upper Basin folks painted themselves into a corner that makes the broader compromise needed on the next steps that much harder?

Who's at the table right now?

All of this is predicated on a narrow definition of the problem we are trying to solve, which is basically a mass balance problem – figuring out how to match our use of water with the supply available, rather than over-using and draining the reservoirs. This is important! But it's not the only thing.

This is a process dominated by the economically and politically powerful current water users. If we have a collective action problem here, we also have a "collective agency" problem. It is a system under which "agency" – the power to influence outcomes – is tightly controlled and narrowly distributed.

What about interests other than the big water agencies and their representatives in state government? This is clearly a state-to-state conversation right now, heading toward a desire for a seven-state proposal come March. In the last two rounds of this – the 2007 Interim Guidelines and the Drought Contingency Plan – the states' proposals have carried the day.

What about the Colorado River Basin's 30 Tribal Sovereigns? It's not clear to me how their needs and interests are being incorporated into this process. In this regard, one is reminded of Neil Gorsuch's dissent in this year's Navajo decision, where he analogized to the tribe standing in line after line again and again at the DMV, only to be told that this isn't the right line. Then which is?

What about non-water-consuming environmental values, which have similarly had a hard time figuring out which process might be the right one? The states could, in theory, act on behalf of those non-water agency interests in the deal they're so furiously negotiating. Will they? Will the federal government step in and insist if the states don't?

There was hope as we headed into the negotiation of the post-2026 river management regime that broader interests and values would be represented. It will be interesting to see what else beyond a seven-state proposal gets consideration in the discussions to come.

A note on sources and methods: I spent last week resting, looking at art, watching falling snow, reading a book (actually several), and not going to CRWUA. My deep thanks to my many friends who attended and filled me in what they heard and saw. All errors are mine. (Also, is that a Cocker Spaniel in the picture, a couple of seats to the chair's left?)